



Ugo Roccamatysi

Propulsion System Integration for an Advanced Blended Wing Body,
Liquid Hydrogen Fuelled Airliner (BW-25)

Faculty of Engineering and Applied Sciences
School of Aerospace, Transport and Manufacturing

MSc
Academic Year: 2025 - 2026

Supervisor: Prof. Craig Lawson, Prof. John Fielding, Dr. Fakhre Ali and
Dr. Cristina Villena Muñoz
April 2026



Faculty of Engineering and Applied Sciences
School of Aerospace, Transport and Manufacturing

MSc

Academic Year 2025 - 2026

Ugo Roccamatysi

Propulsion System Integration for an Advanced Blended Wing Body,
Liquid Hydrogen Fuelled Airliner (BW-25)

Supervisor: Prof. Craig Lawson, Prof. John Fielding, Dr. Fakhre Ali and
Dr. Cristina Villena Muñoz

April 2026

This thesis is submitted in partial fulfilment of the requirements for
the degree of MSc

© Cranfield University 2026. All rights reserved. No part of this
publication may be reproduced without the written permission of the
copyright owner.

Academic integrity declaration

I declare that:

- the thesis submitted has been written by me alone.
- the thesis submitted has not been previously submitted to this university or any other.
- that all content, including primary and/or secondary data, is true to the best of my knowledge.
- that all quotations and references have been duly acknowledged according to the requirements of academic research.

I understand that to knowingly submit work in violation of the above statement will be considered by examiners as academic misconduct.

Abstract

This thesis was written as part of the MSc Aerospace Vehicle Design Group Design Project, focused on the conceptual and preliminary design of the BW-25, a high-capacity long-range Blended Wing Body (BWB) airliner designed for a 2050 entry into service using liquid hydrogen (LH_2) as its primary fuel. The author was responsible for the propulsion system integration, covering the installation of two Rolls-Royce Trent XWB-derived LH_2 -powered turbofans on the aircraft's upper surface.

The work follows a safety-first approach, beginning with the capture of EASA CS-25 and CS-E airworthiness requirements, a Functional Hazard Analysis, two Fault Tree Analyses covering uncontained rotor failure and uncommanded thrust reverser deployment, and a rotor burst assessment that validated the engine layout with respect to fuel tank protection and dual-engine loss probability. Engine mounting configuration, pylon height, and lateral positioning were then determined, leading to the aerodynamic design of a non-axisymmetric separate-flow nacelle incorporating a negatively scarfed inlet. Nacelle drag, thrust reverser design, noise mitigation strategy, and engine accessories sizing were subsequently addressed, and the results consolidated into an integrated propulsion system architecture combining the LH_2 thermal management, lubrication, power generation, and control subsystems.

The study demonstrates that pylon-mounted upper-surface installation of a large high-bypass ratio LH_2 turbofan on a BWB is conceptually feasible and compliant with applicable certification requirements, providing a sound basis for progression to a preliminary design phase.

Keywords:

Nacelle Design, Uncontained Engine Rotor Failure, Boundary Layer Ingestion, Engine Stall, Noise Reduction, Rolls-Royce Trent XWB, Nozzle, Air Inlet, Spread Angle, Thrust Reverser, Engine Starting, Drag Assessment, Propulsion Integration

Acknowledgements

I would like to sincerely thank Prof Craig Lawson, Prof. John Fielding, Dr. Fakhre Ali and Dr. Cristina Villena Muñoz for their guidance and support throughout this Group Design Project. Their experience and advice were extremely helpful and contributed greatly to my learning.

I would also like to thank my fellow team members involved in the BW-25 project. Working with such a committed and talented group was a real pleasure, and I am proud of what we achieved together.

I am grateful to the faculty and staff of the School of Aerospace, Transport and Manufacturing at Cranfield University for providing a supportive and stimulating environment throughout my MSc studies.

Finally, I would like to thank my family and loved ones for their constant encouragement, patience, and support.

I acknowledge the use of Claude Sonnet 4.6 (<https://claude.ai/new>) for linguistic refinement and structural clarification, as English is not my first language. No AI tools were used for technical analysis, calculations, or generation of results. All engineering judgement, data interpretation, and conclusions were developed by me.

Table of Contents

Academic integrity declaration	i
Abstract.....	ii
Acknowledgements	iii
List of Figures.....	vii
List of Tables	x
List of Equations	xi
List of Appendix Figures	xii
List of Appendix Tables	xiii
List of Abbreviations	xiv
1 Introduction and Objectives	1
1.1 The BW-25 concept.....	1
1.2 BW-25 : Group design project.....	3
1.3 Objectives and scope of the thesis.....	4
1.4 Propulsion system integration involvement and CAD team contribution	4
2 Design Basis and System Interfaces	6
2.1 Propulsion system integration interactions with other teams	6
2.2 BW-25 : Conceptual specifications.....	7
3 Literature review	9
3.1 Installation of the engines	9
3.1.1 Current types of installation	9
3.2 Nacelle.....	12
3.3 Intake.....	13
3.3.1 Air inlet cowl.....	13
3.3.2 Air inlet design considerations	14
3.4 Nozzle	17
3.5 Thrust reversers.....	20
3.6 Noise reduction	23
3.7 Engine accessories.....	27
4 Safety assessment	30
4.1 Initial airworthiness specifications	31
4.2 Functional Hazard Analysis	31
4.3 Fault tree analysis.....	32
4.4 Rotor burst analysis.....	33
4.4.1 Perform an analysis.....	33
4.4.2 Safety distance between the engines.....	39

5 Engine integration and nacelle design	44
5.1 Engine integration in the BW-25	44
5.1.1 Engine Mounting Configuration Selection	44
5.1.2 Pylon Height Determination	46
5.1.3 Longitudinal and Lateral Engine Positioning	48
5.2 Engine nacelle design	50
5.2.1 Introduction to engine nacelle design	50
5.2.2 Inlet dimensions	51
5.2.3 Inlet lip shape design	60
5.2.4 Nacelle profile design	62
5.2.5 Primary nozzle and plug design	68
5.2.6 Final design of the nacelle	69
5.3 Drag assessment	72
5.3.1 Forebody drag	73
5.3.2 Midbody drag	77
5.3.3 Afterbody drag	78
5.3.4 Drag estimation results	80
6 Thrust reverser system	83
6.1 Feasibility and Interest Study	84
6.1.1 Landing Performance	84
6.1.2 Safety	84
6.1.3 Reliability	85
6.1.4 Weight and Economics	85
6.1.5 Human Factors	86
6.1.6 Conclusions	86
6.2 Design choices	87
6.2.1 Reverser type	87
6.2.2 Deflection angle	87
6.2.3 Blocking half configuration	88
6.2.4 Reverse Thrust Force Estimation	90
6.3 Actuation System — E-TRAS	91
7 Noise Mitigation	93
7.1 Acoustic Liners	93
7.2 Chevrons — Not Retained	94
8 Engine accessories	96
8.1 Architecture Overview	96
8.2 Engine Starting	96
8.3 Oil System	98
8.4 Cooling via Liquid Hydrogen	101
8.5 Power Loads Summary	101

9 CAD Team.....	104
9.1 Summary	104
9.2 CAD Rules and Standards	104
9.3 Integration Process	104
9.4 Guidance and Support to Project Teams	105
9.5 Surface Modelling and Engine Integration.....	105
10 Integrated propulsion system architecture.....	107
11 Conclusion and Further Work	108
11.1 Conclusion.....	108
11.2 Further Work.....	109
References.....	111
Appendices	116

List of Figures

Figure 1-1: BWB from JetZero's [4]	2
Figure 1-2 : Side view of the surface modelling of the BW-25	3
Figure 1-3 : Rear view of the surface modelling of the BW-25	4
Figure 2-1 : BW-25 General Arrangement	7
Figure 3-1 : BWB with pylon-mounted engines [7].....	10
Figure 3-2 : BWB with engines mounted on the upper surface [7].....	11
Figure 3-3 : NASA N3-X BWB with engines integrated through a S-duct [7]	11
Figure 3-4 : A350's nacelle [10]	13
Figure 3-5 : Air inlet cowl on A350 [10]	14
Figure 3-6 : Flow field for different operating conditions [13].....	16
Figure 3-7 : Nacelle and pylon schematic [15]	17
Figure 3-8 : Primary nozzle [10]	18
Figure 3-9 : Separate (a) and mixed (b) exhaust nozzle geometries [15]	19
Figure 3-10 : Thrust reverser on A350 [10]	20
Figure 3-11 : Cascade reversers [17]	21
Figure 3-12 : Multi-Door Reversers [17]	21
Figure 3-13 : Bucket reversers [17].....	22
Figure 3-14 : Clamshell reversers [17]	22
Figure 3-15 : Noise distribution around a turbofan [20]	24
Figure 3-16 : Noise absorbing materials and location [21]	25
Figure 3-17 : Chevron nozzle on the 787 [23]	26
Figure 3-18 : Negative Scarfed Intake.....	27
Figure 3-19 : 787 engine start system [25]	28
Figure 4-1 : Side view of a rotor burst on an Antonov 124 [31].....	33
Figure 4-2 : Fragments paths estimated by the AMC-20 [28]	34
Figure 4-3 : Fragments of the AN-124 rotor burst disc [31].....	35

Figure 4-4 : Rotor bursts of 15° plotted on the 2D plane of the aircraft	36
Figure 4-5 : Rotor bursts of 5° plotted on the 2D plane of the aircraft	36
Figure 4-6 : Rotor bursts of 3° plotted on the 2D plane of the aircraft	37
Figure 4-7 : 3D view of the rotor burst area for a 15° spread angle	38
Figure 4-8 : 3D view of the rotor burst area for a 5° spread angle	38
Figure 4-9 : 3D view of the rotor burst area for a 3° spread angle	39
Figure 4-10 : Rotor burst threat window [28]	40
Figure 4-11 : Front view of the engines	41
Figure 5-1 : Comparison of podded and BLI nacelle installations [33]	45
Figure 5-2 : Flow field in vicinity of nacelle in 2-D section at engine centre [6]	46
Figure 5-3 : Engine height on the BW-25	47
Figure 5-4 : Total pressure contours (Pa) on the fan inlet face of the outer turbofan in the low position (left) and high position (right) configurations: from top to bottom $\alpha = 11$ degrees [36]	48
Figure 5-5 : Engines position	50
Figure 5-6 : Subsonic inlet nomenclature [39]	52
Figure 5-7 : Typical ranges of key inlet design variables [39]	53
Figure 5-8 : Streamlines above the upper surface of the BW-25 at the engine level	54
Figure 5-9 : General N2A-EXTE Model arrangement drawing details [41]	54
Figure 5-10 : A320 scarf inlet for the Snecma – SILENCE(R) programme [42]	55
Figure 5-11 : Side view of the different forebody length for the TOP, SIDE and BOTTOM section	56
Figure 5-12 : Inlet lips	60
Figure 5-13 : Inlet lips profiles	62
Figure 5-14 : External contour profiles	64
Figure 5-15 : Afterbody key parameters	65
Figure 5-16 : Primary nozzle in blue	68
Figure 5-17 : Conical exhaust plug	69
Figure 5-18 : Engine nacelle CAD model	71
Figure 5-19 : Comparison between the nacelle designed and the A350's nacelle ...	71

Figure 5-20 : Subsonic afterbody drag relation [43]	79
Figure 6-1 : Cascade thrust reverser structure [49]	87
Figure 6-2 : Thrust reversers deflection angle	88
Figure 6-3 : Blocking half configuration illustration	89
Figure 6-4 : Block half thrust reversers on the C-17 [50]	89
Figure 6-5 : E-TRAS thrust reverser [51]	92

List of Tables

Table 2-1: Initial data useful for the integration of propulsion systems contained in the specification book	8
Table 4-1 : Example of FHA realized on the Loss of thrust on one engine	31
Table 4-2 : Fragments description	34
Table 4-3 : Dp and Risk factors [28]	41
Table 4-4 : Probability of one engine hitting the other in the event of a rotor burst	42
Table 4-5 : Probability to have a total loss of both engines due to a rotor burst.....	42
Table 5-1 : Key dimensions of the forebody.....	57
Table 5-2 : Inlet lip characteristics definition.....	61
Table 5-3 : External contour characteristics	64
Table 5-4 : Afterbody key characteristics calculation	67
Table 5-5 : Drag calculation parameters.....	72
Table 5-6 : Forebody drag calculation	77
Table 5-7 : Midbody drag calculation.....	78
Table 5-8 : Afterbody drag calculation	80
Table 5-9 : Total nacelle drag.....	80
Table 5-10 : Total nacelle drag for every altitude	82
Table 6-1 : Landing distances for the BW-25.....	84
Table 6-2 : Thrust reversers force.....	90
Table 8-1 : Properties of the Mobil Jet Oil II	98
Table 8-2 : Oil pump system characteristics	100
Table 8-3 : Nacelle power requirements	102

List of Equations

Equation 4.1 : Probability that a fragment hit a component [28]	40
Equation 5.1 : Boundary layer thickness correlation for a flat thin plate [38]	47
Equation 5.2 : Total pressure recovery at zero flight speed [39]	58
Equation 5.3 : Mach drag rise number correlation [43]	59
Equation 5.4 : Elliptical geometry for the intake lips	61
Equation 5.5 : Forebody geometry equation [45]	63
Equation 5.6 : Drag rise Mach number correlation for the afterbody [43]	66
Equation 5.7 : Prandtl-Schlichting turbulent flat plate correlation.....	74
Equation 5.8 : Skin friction coefficient.....	74
Equation 5.9 : Drag coefficient at the ideal condition.....	74
Equation 5.10 : Spillage drag empirical relation	74
Equation 5.11 : Mass Flow Parameter definition	75
Equation 5.12 : Freestream tube capture area.....	75
Equation 5.13 : Mass Flow Ratio definition.....	75
Equation 5.14 : Forebody drag.....	75
Equation 5.15 : Critical Mass Flow Ratio	76
Equation 5.16 : Skin friction coefficient of the midbody	78
Equation 5.17 : Midbody drag.....	78
Equation 5.18 : Skin friction coefficient of the afterbody	79
Equation 5.19 : Afterbody drag	80
Equation 8.1 : Moment of inertia of a hollow cylinder	97
Equation 8.2 : Angular acceleration for the engine start.....	97
Equation 8.3 : Mean angular velocity during the acceleration phase.....	97
Equation 8.4 : Power required for the engine start.....	97
Equation 8.5 : Required oil mass flow for the oil pumps	99
Equation 8.6 : Pump power.....	99

List of Appendix Figures

Figure_Apx 1 : UERF Fault tree analysis 137

Figure_Apx 2 : Uncommanded thrust reversers deployment in flight Fault tree
analysis 138

Figure_Apx 3 : Nacelle drag calculator python script 154

Figure_Apx 4 : CAD drawings 156

Figure_Apx 5 : Final architecture 158

List of Appendix Tables

Table_Apx 1 : Propulsion system integration requirements capture.....	116
Table_Apx 2 : Functional Hazard Analysis of propulsion systems	127
Table_Apx 3 : Intake lip coordinates	139
Table_Apx 4 : Top external contour coordinates.....	140
Table_Apx 5 : Side external contour coordinates	142
Table_Apx 6 : Bottom external contour coordinates.....	144
Table_Apx 7 : Engine performances	146

List of Abbreviations

AMC	Acceptable Means of Compliance
APU	Auxiliary Power Unit
BLI	Boundary Layer Ingestion
BWB	Blended Wing Body
CAD	Computer Aided Design
CDR	Critical Design Review
CFD	Computational Fluid Dynamics
CS	Certification Specifications (CS-25, CS-E)
DLIM	Double-sided Linear Induction Motor
EASA	European Union Aviation Safety Agency
ECS	Environmental Control System
EECS	Electronic Engine Control System
EHA	Electro-Hydrostatic Actuator
EMI	Electromagnetic Interference
E-TRAS	Electrical Thrust Reverser Actuation System
ETOPS	Extended-range Twin-engine Operations Performance Standards
FADEC	Full Authority Digital Engine Control
FHA	Functional Hazard Analysis
FOD	Foreign Object Debris
FTA	Fault Tree Analysis
GH ₂	Gaseous Hydrogen
HIRF	High Intensity Radiated Field
HLFC	Hybrid Laminar Flow Control
HP	High Pressure (spool)
HPC	High Pressure Compressor
HPT	High Pressure Turbine
ICAO	International Civil Aviation Organization
IP	Intermediate Pressure
IPC	Intermediate Pressure Compressor
IPS	Ice Protection System

IPT	Intermediate Pressure Turbine
ISA	International Standard Atmosphere
JASC	Joint Aircraft System/Component (code)
LDR	Landing Distance Required
LH ₂	Liquid Dihydrogen
LP	Low Pressure
LPT	Low Pressure Turbine
LTO	Landing and Take-Off (cycle)
MFP	Mass Flow Parameter
MFR	Mass Flow Ratio
NACA	National Advisory Committee for Aeronautics
NASA	National Aeronautics and Space Administration
NPR	Nozzle Pressure Ratio
OEI	One Engine Inoperative
SFC	Specific Fuel Consumption
SLS	Sea Level Static
TR	Thrust Reverser
UERF	Uncontained Engine Rotor Failure
UHBR	Ultra-High Bypass Ratio
V ₁	Take-off decision speed
VFSG	Variable Frequency Starter-Generator
VR	Rotation speed
VSV	Variable Stator Vane
WOW	Weight on Wheels

1 Introduction and Objectives

1.1 The BW-25 concept

There will continue to be an increased demand for efficient transportation methods for a significant number of passengers over long distances around the world due to the ongoing growth of commercial aviation globally [1]. The aerospace industry has developed many new technologies to improve the efficiency of existing conventional passenger aircraft but conventional designs have reached the limits of how much they can be improved aerodynamically and in terms of propulsion. All current conventional passenger aircraft utilize the "tube-and-wing" design concept consisting of a tubular fuselage, high-aspect-ratio wings, and a low-aspect-ratio vertical stabilizer located aft of the horizontal stabilizer. It is unlikely that the tube-and-wing design configuration can be improved significantly to achieve further improvements in efficiency.

In the marketplace, there remains a strong demand for larger capacity aircraft with lower direct operating costs, as well as the growing concern for environmental issues has led to the need for aircraft that produce fewer gaseous emissions, make less noise, and allow for the use of alternative fuels [2]. Therefore, the inability to develop solutions to address these demands using only incremental changes to conventional aircraft configurations necessitates developing more revolutionary approaches to aircraft design.

Throughout history, the aerospace industry has been hesitant to move away from established technology. However, the development of new operational requirements and the advancement of the knowledge base in the underlying sciences of flight provides a rational justification for the aerospace industry to evaluate alternative aircraft configurations. One such design alternative is the Blended Wing Body (BWB) aircraft, which integrates the wing and fuselage into a single lifting surface [3]. While this configuration typically eliminates the conventional vertical tail to minimize weight and aerodynamic drag, it does not strictly remove the need for directional stability. Instead, yaw control is provided

through wing planform effects, split drag rudders, differential thrust, and active flight control systems. Furthermore, this architecture allows for significant additional internal volume by forming a thicker midsection.



Figure 1-1: BWB from JetZero's [4]

This design configuration has a number of potential advantages. For example, it can offer a lower wetted surface area relative to the volume of payload, better structural efficiency because of the way the loads are distributed, more flexibility in terms of how payloads can be packaged, and potentially better performance on very-high-capacity transportation missions. The BWB configuration also has the potential to provide an advantage when integrating propulsion systems and could make it easier to apply hybrid laminar flow control (HLFC) technologies.

In spite of the apparent merits of the BWB configuration, the disadvantages and practical realities that would limit the use of the BWB configuration are poorly understood. Consequently, the ability to assess the overall performance, cost-effectiveness and operational feasibility of this design configuration continues to be problematic. Therefore, identifying the benefits and the constraints of the BWB configuration will be important.

The objective of the project related to this study is to conduct a thorough examination of the Blended Wing Body configuration in order to determine whether

the BWB configuration has the potential to become a future generation passenger aircraft. There have long been interests in flying wing and blended body configurations but the development of new technologies and design methods past few decades have brought those ideas back into focus. With a planned service entry date of approximately 2050, the BWB configuration allows for the possibility of incorporating a variety of advanced systems, such as liquid hydrogen fuelled turbofan engines, and large scale laminar flow control systems, in order to achieve significant increases in the efficiency and environmental performance of the aircraft. Throughout the Group Design Project, the design configuration developed by the team was referred to as the BW-25 [5].

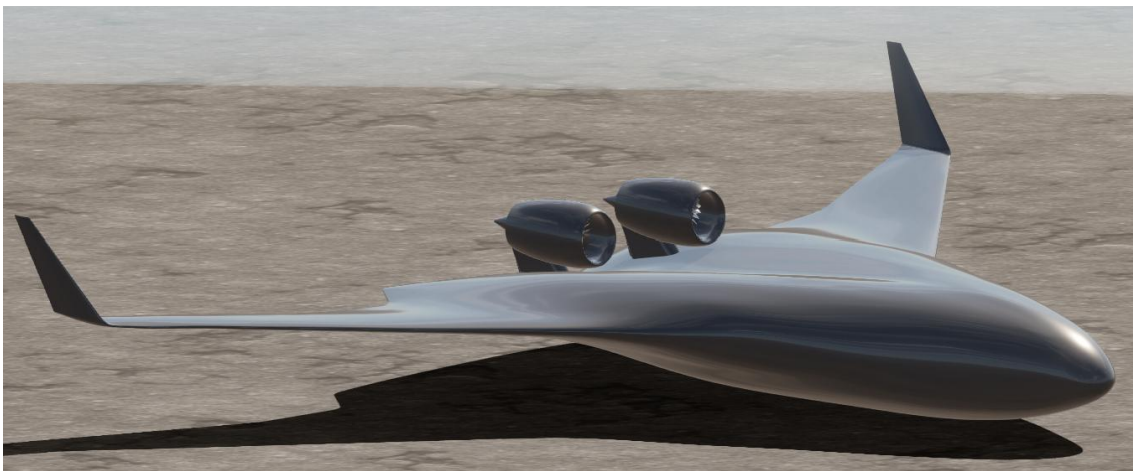


Figure 1-2 : Side view of the surface modelling of the BW-25

1.2 BW-25 : Group design project

The Aerospace Vehicle Design course for September intake 2025 includes a Group Design Project where approximately sixty students worked together as one design team to develop the BW-25 from the conceptual design phase through to the preliminary and detailed design phases, culminating in a Critical Design Review (CDR). The project required effective communication and organisation across a large team to maintain design integrity. For this reason, the team was organised into three primary groups: Systems, Avionics, and Structures. In addition to their primary tasks, secondary responsibilities were assigned to students on a voluntary

basis. The author of this thesis was assigned the primary task of propulsion system integration, and also contributed to the CAD team as a secondary responsibility.

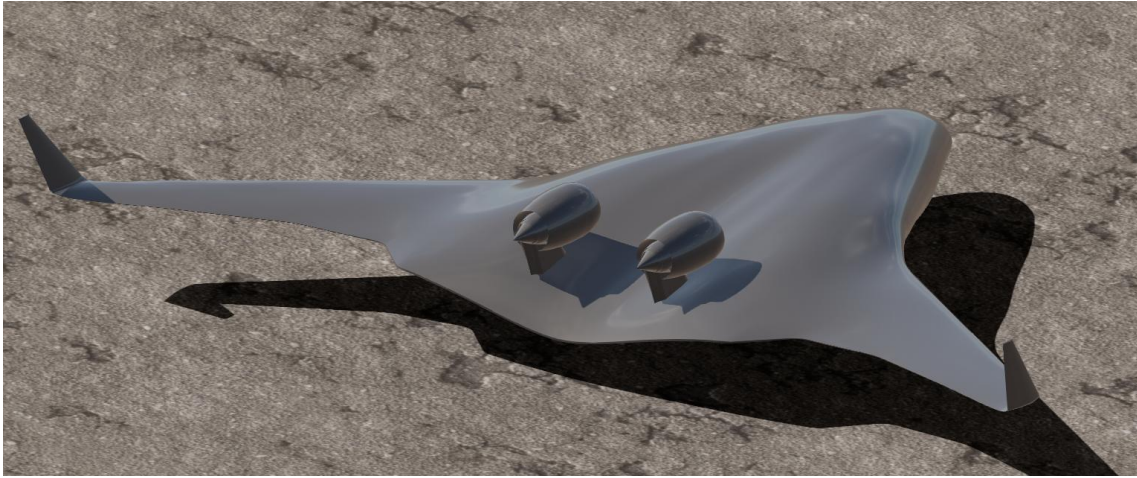


Figure 1-3 : Rear view of the surface modelling of the BW-25

1.3 Objectives and scope of the thesis

The main goal of the thesis is to present a unified conceptual integration of the propulsion systems for the BW-25 aircraft, using the Rolls-Royce Trent XWB LH₂-fuelled engines mounted on the upper surface of the Blended Wing Body. The objective is to provide a safe and reliable method of installing the engines on the airframe while addressing the aerodynamic, acoustic, and system-level constraints associated with the installation.

The thesis focuses on the conceptual and preliminary design-level integration of the propulsion systems. The scope therefore covers the engine mounting location on the airframe, the design of the engine nacelle, inlet, nozzle, and thrust reversers, the acoustic treatment for noise reduction and the connection of the liquid hydrogen fuel system to the engines. The key interdependencies with other disciplines that shaped this integration work are identified and described in Section 2.1.

1.4 Propulsion system integration involvement and CAD team contribution

The author's primary responsibility in the BW-25 project was to integrate the Rolls-Royce Trent XWB liquid hydrogen engines onto the upper surface of the Blended

Wing Body. The first step was to determine the engine placement on the airframe, a decision driven by rotor burst constraints and the need to ensure clean airflow ingestion into the engines. From this starting point, the nacelle, inlet, and nozzle were designed in accordance with the air mass flow specifications provided by the propulsion performance responsible Guillaume Fleury.

Noise reduction was addressed through the incorporation of acoustic insulation within the nacelle, and chevron technology was evaluated as an additional noise attenuation solution, although it was ultimately not retained in the final design. The thrust reverser system was also designed, with a cascade configuration selected following a comparative assessment of several options including more recent technologies. Coordination with the fuel system team (Daniella Barnett: Transfer and feed and Rohit Suchitra Mugunthan: Tanks and gauging) was also carried out to enable the integration of their liquid hydrogen cooling system within the engine.

The author also supported the CAD team by contributing to the surface modelling of the aircraft and integrating the propulsion system components into the overall assembly, ensuring consistency across the model throughout the design iteration.

2 Design Basis and System Interfaces

This section defines the framework within which the propulsion system integration work was conducted. It covers the key interfaces with other disciplines, as well as the initial specifications that constrained the design.

2.1 Propulsion system integration interactions with other teams

This section outlines the key interdependencies and interfaces between propulsion system integration and other disciplines throughout the BW-25 project. The most central interface was with propulsion system performance, where collaborative efforts focused on determining engine parameters for the nacelle and the thrust reverser design. Regarding the fuel system, the interface covered the liquid hydrogen cooling system architecture, fuel temperature and pressure conditions at the combustor chamber inlet, and rotor burst failure scenarios with the associated protection measures for sensitive components. Discussions with the secondary power and APU discipline focused on identifying the energy sources and amount of power required throughout the flight to sustain engine and nacelle functions. For the structural team, the key inputs provided for pylon design were the engine height and weight. No specific interface was required with the IPS discipline, as anti-icing provisions were integrated directly within the inlet cowl. Similarly, no joint work was carried out with the HLFC discipline, although HLFC influences inlet airflow conditions during cruise.

2.2 BW-25 : Conceptual specifications

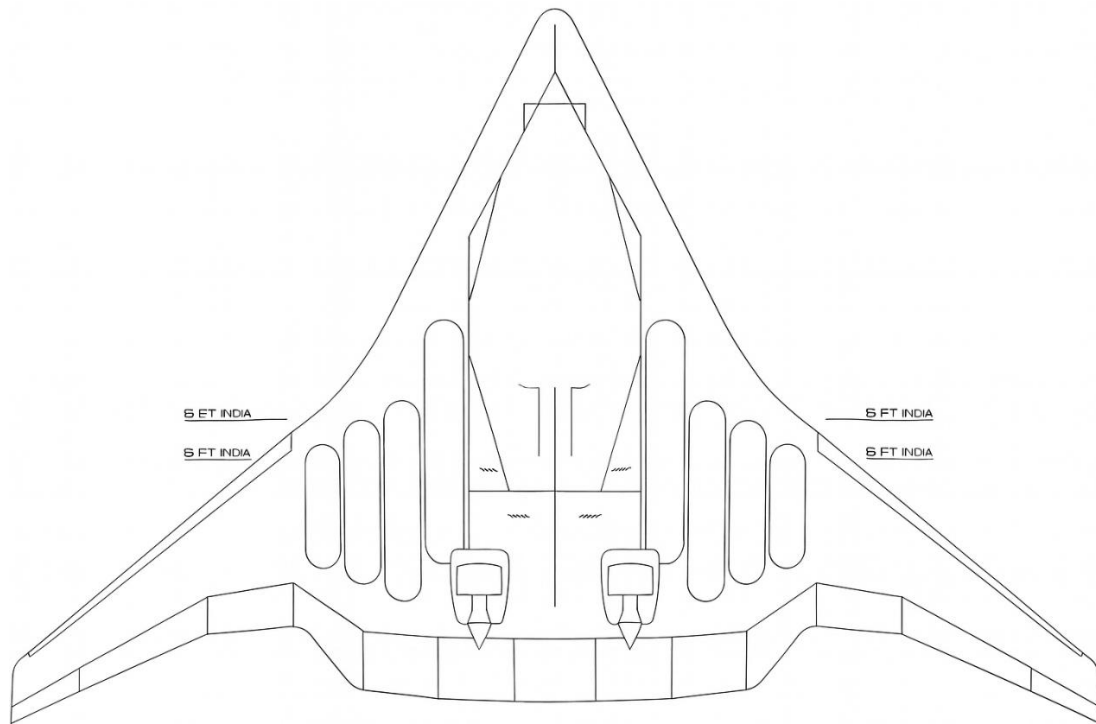


Figure 2-1 : BW-25 General Arrangement

The specification sheet provides several pieces of information related to the integration of propulsion systems, which the author of the thesis must follow to guide their work. These include:

- Anticipating the requirements for 2050 in terms of airport noise level restrictions.
- Ensuring that the technologies used are consistent with 2050 standards.
- Two conventional turbofan engines with a high bypass ratio will be used.
- The engine model will be a derivative of the Rolls-Royce Trent XWB.
- The engines will be powered by H_2 .
- The engines must be mounted on the upper surface of the aircraft.
- Several installation architectures must be studied and lead to the choice that maximizes the overall benefits of the aircraft.

- It should be possible to significantly reduce the noise emitted by the aircraft by correctly choosing the engine position.
- Attention must be paid to the case of uncontained engine rotor failure.

Table 2-1: Initial data useful for the integration of propulsion systems contained in the specification book

Parameter	Value
Lateral distance of engine centreline from aircraft centreline	5.0 m
Location of engine CG from datum	41.0 m
Fan diameter	3.0 m
Engine length	5.812 m
One engine weight	7400 kg
One {nacelle+pylon}	2,295.87 kg
Height of the engines from the XY plane	4.25 m

3 Literature review

3.1 Installation of the engines

3.1.1 Current types of installation

The Blended Wing Body configuration offers a wide range of engine installation options, each with distinct aerodynamic, acoustic, and structural implications. Unlike conventional tube-and-wing aircraft, the BWB's geometry creates opportunities to significantly reduce engine noise at ground level by using the aircraft structure as an acoustic barrier between the engines and the ground. However, the choice of installation architecture involves trade-offs that the existing literature does not fully resolve, particularly in the context of a large commercial hydrogen-fuelled BWB.

A conventional under-wing pylon-mounted installation is generally considered unsuitable for a BWB. The absence of a conventional high wing means that under-wing mounting would expose the engines and pylons directly to the freestream flow, generating significant aerodynamic drag without the acoustic shielding benefit that makes upper-surface mounting attractive.

Pylon-mounted engines on the upper surface represent the most extensively studied adaptation for a BWB, and benefit from the largest body of experimental and numerical data [6]. The engines are mounted above the fuselage on pylons, keeping them away from the boundary layer and ensuring relatively clean, undistorted airflow into the inlet. This arrangement also provides good FOD protection and facilitates maintenance access. However, the pylons and nacelles remain exposed to the freestream, generating non-negligible drag. Furthermore, the literature reveals a lack of consensus on the optimal pylon height and engine lateral positioning. Pylon height directly governs the trade-off between boundary layer ingestion — which increases with proximity to the surface — and structural complexity, which grows with pylon length. No universally accepted design guidelines exist for this trade-off in the context of upper-surface BWB installations.

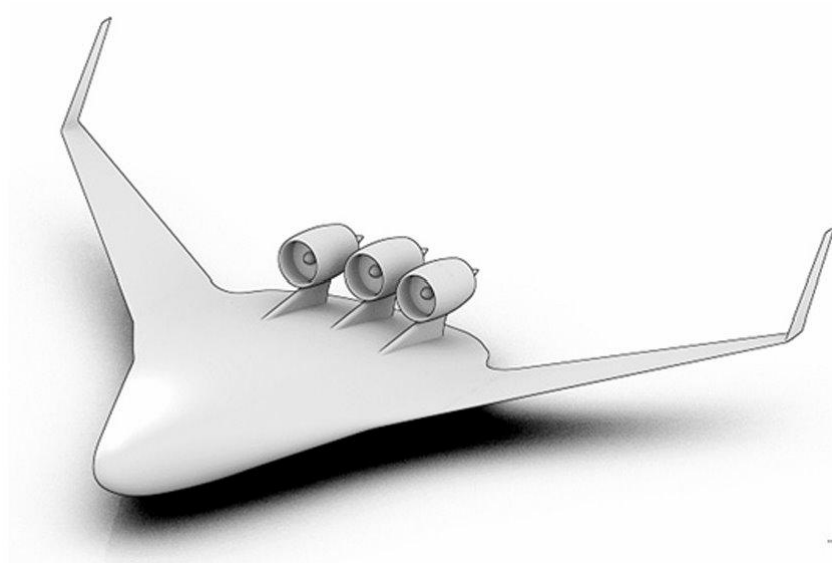


Figure 3-1 : BWB with pylon-mounted engines [7]

Fuselage surface-mounted engines eliminate conventional pylons, significantly reducing drag and improving acoustic shielding, as the fuselage mass provides a more effective noise barrier between the engines and the ground. This configuration also results in a lower thrust moment arm relative to the aircraft centre of gravity. However, the proximity to the fuselage surface means the engines ingest a portion of the boundary layer that has developed over the upper surface, leading to inlet flow distortion [8]. This Boundary Layer Ingestion (BLI) effect can reduce inlet total pressure recovery and potentially cause fan instability if not carefully managed. The interaction between the ingested low-momentum flow and fan face distortion has been studied primarily for axisymmetric or idealised geometries, making the

application of existing BLI correlations unreliable for the highly three-dimensional flow field over a real BWB upper surface.

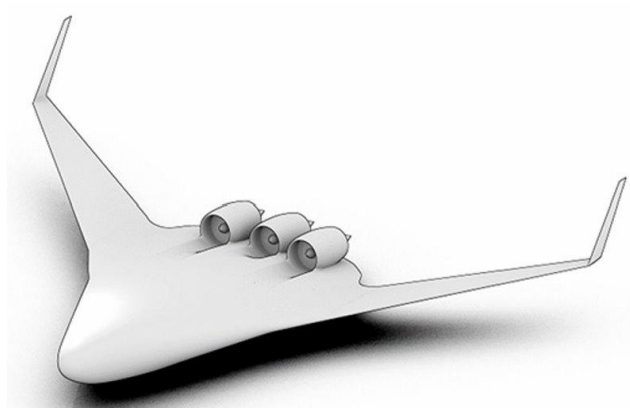


Figure 3-2 : BWB with engines mounted on the upper surface [7]

Embedded engines with S-duct inlets are considered the most aerodynamically promising configuration, eliminating external nacelle drag almost entirely, maximising BLI efficiency, and providing the greatest degree of noise suppression by fully enclosing the engines within the structure [9]. The NASA N3-X concept is a notable example of this approach. Nevertheless, the S-duct configuration imposes significant internal volume requirements, and the curved inlet geometry introduces pressure losses and flow distortion that must be carefully controlled. This configuration is therefore difficult to reconcile with the internal volume constraints of a commercial BWB carrying large cryogenic fuel tanks.

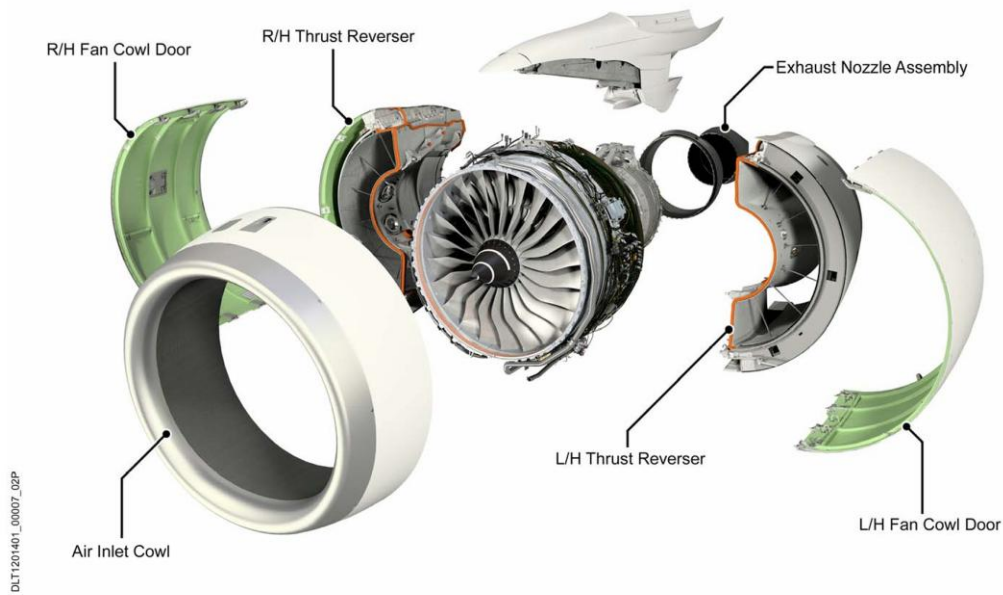


Figure 3-3 : NASA N3-X BWB with engines integrated through a S-duct [7]

Beyond the choice of mounting architecture, two cross-cutting design parameters receive insufficient attention in the available literature. The longitudinal positioning of the engines along the BWB chord affects the pitching moment balance, the fuel line routing, and the acoustic shielding geometry — yet few studies quantify this effect systematically for upper-surface configurations. Additionally, the lateral separation between engines in a twin-engine BWB introduces a specific constraint related to uncontained rotor burst, where regulatory requirements impose minimum separation distances that, combined with one-engine-inoperative yaw control requirements, impose boundaries on engine positioning that are not often explicitly addressed in the BWB engine location literature.

3.2 Nacelle

The nacelle has a major influence on the overall performance of the propulsion system, primarily due to its protection of the engine while capturing the correct amount of airflow to produce the required thrust. Furthermore, the design of the nacelle will have a significant impact on the aerodynamics of the aircraft and how efficiently thrust can be produced from the exhaust flow at the rear of the nacelle [10]. The nacelle characteristics are therefore highly correlated to the engine type and size. Hence, the Rolls-Royce Trent XWB, which was used as a reference for this project, is one of the world's largest turbofan engines that is currently in service, with an external diameter of about three meters [11]. The Trent XWB is only installed on the Airbus A350, and the nacelle of the Airbus A350 exemplifies some of the key issues when installing very large high-bypass-ratio turbofan engines into commercial aircraft in order to maintain both acceptable aerodynamic performance and satisfactory structural and systems integration.



POWERPLANT MAJOR ASSEMBLIES

Figure 3-4 : A350's nacelle [10]

The main components of the nacelle that are discussed in subsequent sections are the air inlet cowl, thrust reversers and nozzles.

However, this example refers to a nacelle mounted under the wings, whereas the BW-25 configuration requires the engines to be installed on the upper surface of the fuselage. This arrangement helps reduce the drag produced by the engines. The literature highlights new challenges such as air intake distortion in the engines, boundary layer ingestion, and more difficult access due to the engine position, which can affect the aircraft's performance, stability, and maintainability [12].

3.3 Intake

3.3.1 Air inlet cowl

The air intake cowl is a primary component of the engine nacelle, serving as the critical interface between the external flow and the engine. Its primary objective is to deliver airflow into the engine with minimal distortion across the entire flight envelope. Structurally, it comprises an inlet lip, inner barrel, outer barrel, and bulkhead components that stiffen the cowl and enable attachment to the engine fan case. The inner barrel typically incorporates acoustic liners to attenuate fan noise,

while the outer barrel defines the external aerodynamic profile of the nacelle. Material selections include metallic alloys and composite structures optimised for strength, weight, and longevity.

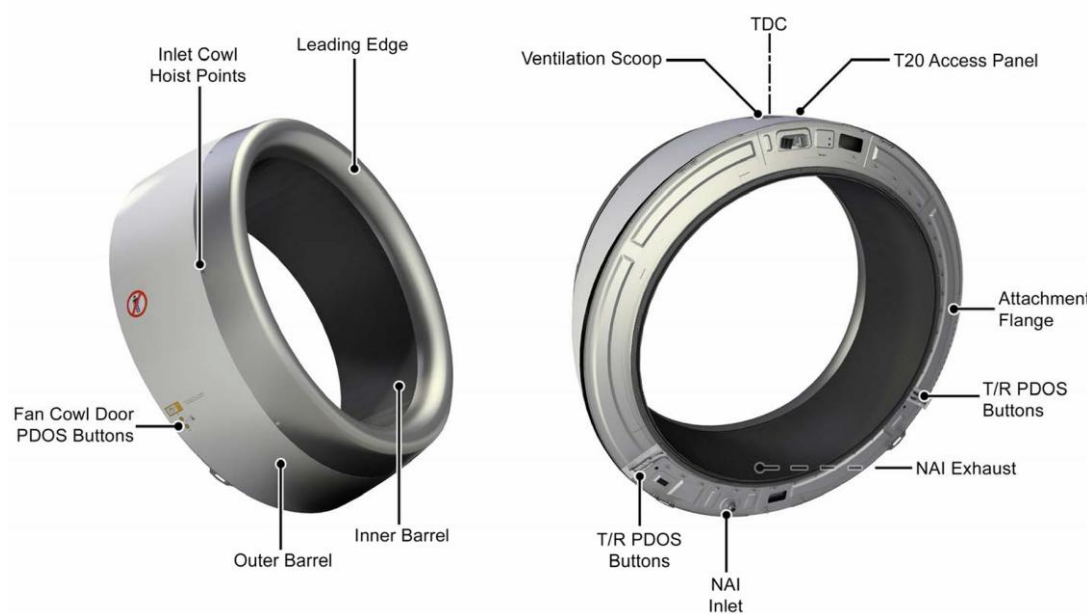


Figure 3-5 : Air inlet cowl on A350 [10]

The inlet cowl must also integrate anti-icing provisions to prevent ice accumulation on the lip, meet bird strike and FOD resistance requirements, and provide maintenance access. Most conventional thermal anti-icing systems rely on engine bleed air, which is not available on the BW-25 given its more electric architecture. Alternative electric anti-icing solutions exist but are less mature and introduce additional power demands on the electrical generation system. The inlet cowl design is therefore a multidisciplinary compromise between aerodynamics, structures, systems integration, and certification requirements — a balance that becomes more complex in the context of an upper-surface mounted engine on a BWB.

3.3.2 Air inlet design considerations

The aerodynamic behaviour of a turbofan inlet varies considerably across the flight envelope, making its design an inherently compromise-driven process. Two major drag components must be considered. Ram drag arises from the momentum

extracted from the freestream as air is captured and decelerated into the inlet — it is an unavoidable consequence of mass flow ingestion and scales directly with engine demand. Spillage drag occurs when the inlet captures less air than the approaching streamtube, causing excess flow to spill around the lip [13]. At cruise, where the inlet operates at a mass flow ratio between 0.70 and 0.80, spillage drag becomes significant and is strongly influenced by lip geometry and local flow acceleration around the highlight. Minimizing spillage drag requires keeping local Mach numbers below sonic on both the fan cowl and within the inlet duct — if the local Mach number becomes too high, shock waves develop, causing wave drag, boundary layer growth, total pressure loss, and potentially shock-induced flow separation. The inlet throat area and forebody curvature radius are therefore critical parameters for cruise performance.

During low-speed operations such as take-off and climb, the inlet operates at mass flow ratios above one, with the fastest flow velocities located inside the duct rather than on the external nacelle surface. At high angles of attack, the stagnation point migrates below the inlet lip, forcing the flow to accelerate sharply around the lower lip and generating large adverse pressure gradients that increase the risk of flow separation and inlet distortion. Crosswind conditions introduce a further level of complexity by shifting the stagnation point laterally, creating two distinct separation mechanisms: excessive diffusion near the highlight at low fan speeds, and shock-induced separation on the windward lip at higher fan speeds.

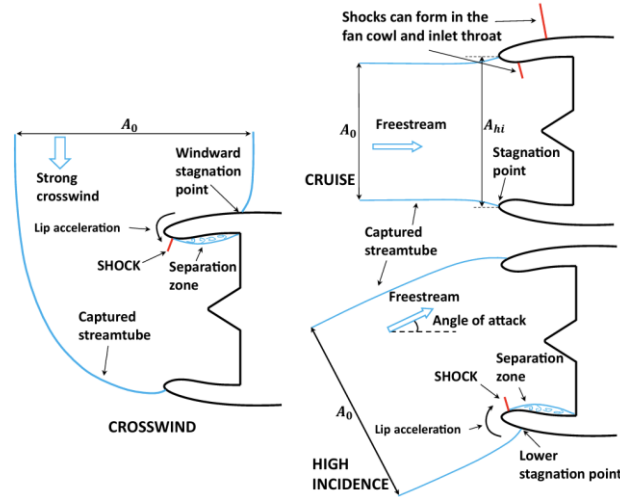


Figure 3-6 : Flow field for different operating conditions [13]

These competing requirements — a sharper lip for cruise drag minimisation versus a thicker, more rounded lip for low-speed flow attachment — create a fundamental design trade-off that is well documented for conventional under-wing installations. However, its resolution becomes more complex for upper-surface mounted configurations on a BWB, where the inlet is exposed to a non-uniform oncoming flow field and a boundary layer that has already developed over the length of the upper surface. This boundary layer carries a momentum deficit that further degrades total pressure recovery at the fan face, independently of the angle of attack effects described above [14].

High levels of total pressure distortion at the fan face are known to reduce the stall margin of the fan and compressor stages, potentially triggering rotating stall or surge. This represents a key design driver that must be addressed through careful inlet lip sizing, adequate inlet length definition, and verification of sufficient stall margin across the full flight envelope in coordination with the engine performance discipline.

Finally, the interaction between the bypass exhaust jet and the surrounding flow contributes to overall nacelle drag through boat-tail drag, which arises from adverse pressure gradients along the aft fan cowl when the nozzle is choked. The boat-tail angle of the rear nacelle geometry must therefore be carefully selected to minimize this drag component — an aspect discussed further in Section 3.4.

3.4 Nozzle

The exhaust nozzles are positioned at the rear of the engine and provide the means to control the direction and velocity of both the core and bypass exhaust streams. Their fundamental purpose is to generate thrust by expanding and accelerating exhaust gases at pressures greater than ambient. Nozzle geometry has a significant impact on engine performance and efficiency, as it governs the mass flow rate through the engine across the entire flight envelope and directly influences compressor stall margins at low speeds. The exhaust nozzles therefore represent a significant design trade-off between aerodynamic performance, noise reduction, thermal protection of rear structural components, and stable engine operation.

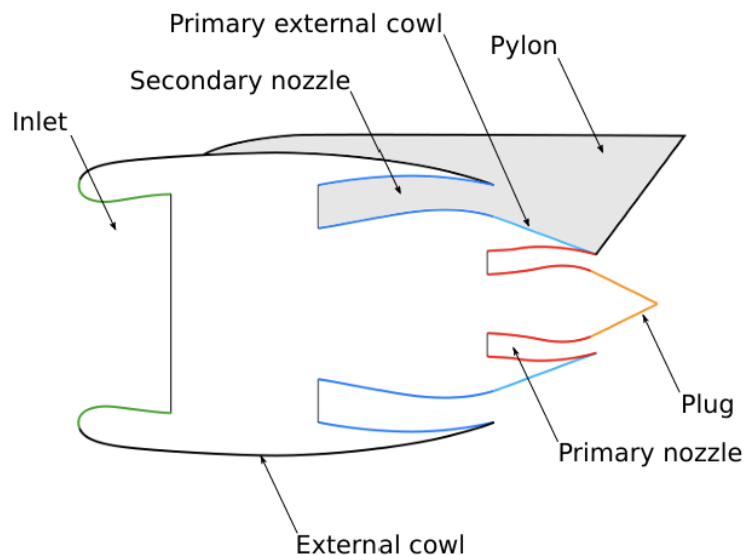


Figure 3-7 : Nacelle and pylon schematic [15]

Turbofan engines use two primary nozzle configurations. Separate-flow nozzles use two distinct annular exits for the core and bypass flows and are the standard solution for high bypass ratio turbofans due to their low structural weight and mechanical simplicity. However, the hot core exhaust in a separate-flow configuration is a major source of jet mixing noise, as jet noise is highly dependent on exhaust temperature. Mixed-flow nozzles address this by combining core and bypass flows through a mixer before exhaust, lowering the effective jet temperature and velocity and thereby reducing jet noise while also improving propulsive

efficiency [15]. These benefits come at the cost of increased weight, greater design complexity, and friction losses introduced by the mixer. For high bypass ratio engines, the weight and complexity penalty of the mixer typically outweighs the acoustic benefit in conventional installations. For BWB upper-surface configurations, the airframe already provides significant acoustic shielding from below, which partially offsets the jet noise penalty of a separate-flow nozzle and further reduces the relative benefit of a mixed-flow solution. However, this reasoning must be tempered by the increasingly stringent noise targets expected for a 2050 entry into service — if future ICAO regulations impose noise limits that cannot be met through shielding and liner treatment alone, the acoustic contribution of a mixed-flow nozzle may become relevant even on a high bypass ratio engine. The existing literature does not resolve this trade-off for BWB configurations, and the optimal nozzle architecture will ultimately depend on the margin between the achieved noise levels and the certification limits applicable at the time of entry into service.

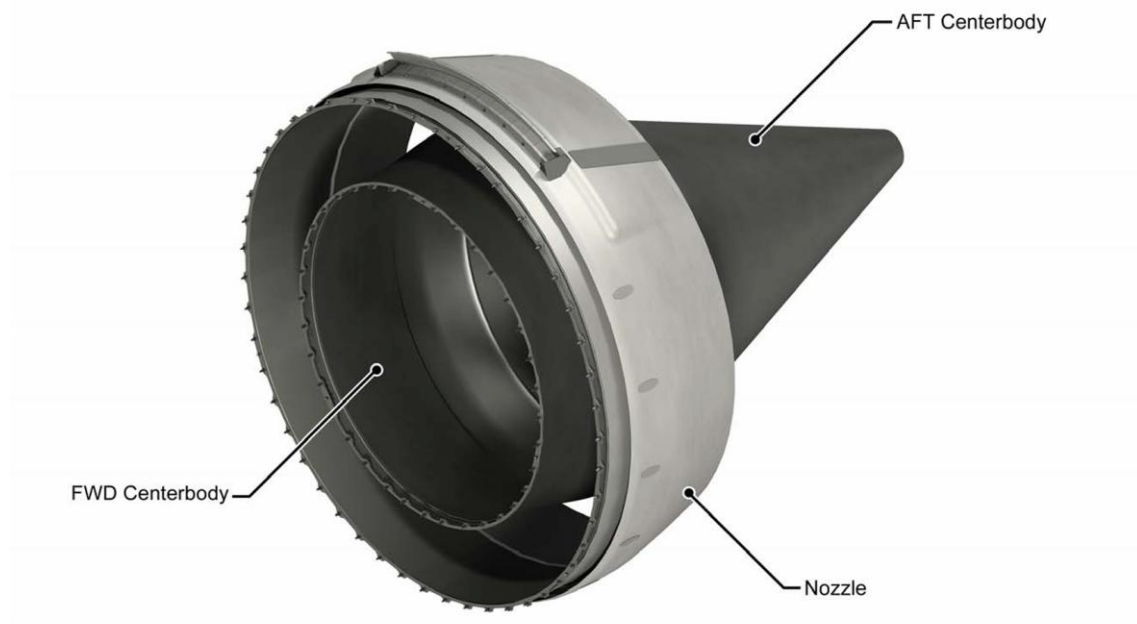


Figure 3-8 : Primary nozzle [10]

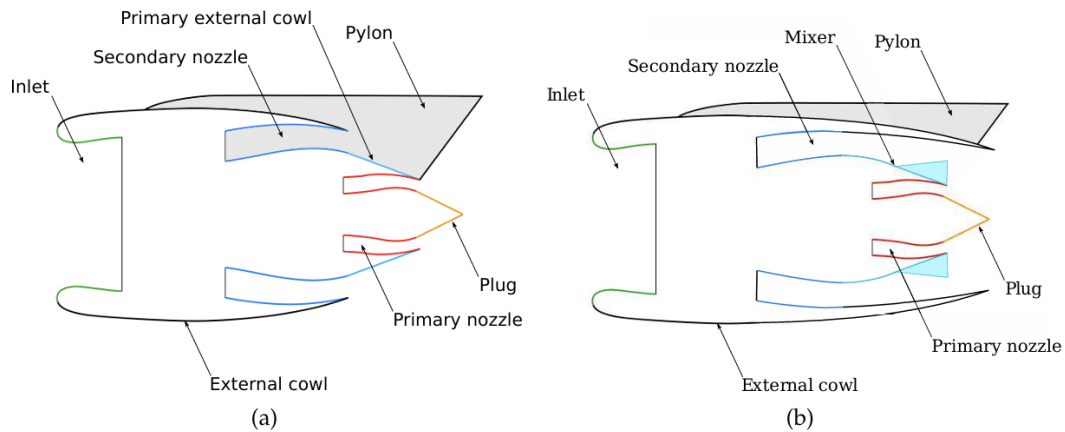


Figure 3-9 : Separate (a) and mixed (b) exhaust nozzle geometries [15]

Boat-tail drag represents a significant nozzle-related drag component that warrants particular attention. It arises from the pressure forces acting on the aft-facing external surface of the nacelle as the flow expands around the nozzle exit. For large high bypass ratio engines, the nozzle exit diameter is substantial, making the boat-tail area and its drag contribution non-negligible. The literature on boat-tail drag optimisation is predominantly based on under-wing nacelle geometries, where the afterbody is exposed to a relatively undisturbed freestream. For upper-surface mounted configurations on a BWB, the proximity of the nozzle exit to the airframe upper surface modifies the local flow field around the afterbody, potentially altering the pressure distribution and the magnitude of boat-tail drag in ways that are not well captured by existing correlations [16].

Finally, the use of liquid hydrogen introduces exhaust characteristics that differ from kerosene-fuelled engines and that have received limited attention in the nozzle design literature. LH_2 combustion produces a higher adiabatic flame temperature and a larger proportion of water vapour in the exhaust stream. The higher water vapour content modifies the specific heat capacity of the exhaust gas, affecting the Nozzle Pressure Ratio (NPR) required to achieve a given thrust level and influencing the acoustic signature of the jet. The implications of these differences for nozzle geometry sizing and acoustic treatment remain insufficiently addressed in the open literature, representing a knowledge gap of direct relevance to the BW-25 integration.

3.5 Thrust reversers

Thrust reversers redirect the engine exhaust flow forward to generate a braking force that assists wheel braking in reducing landing roll distance. They also provide increased operational safety during adverse weather conditions or on runways with limited length and reduce mechanical stress on brakes and landing gear. Modern thrust reverser systems are activated only when weight-on-wheels is confirmed and engine speed is at idle, preventing accidental airborne deployment. The amount of reverse thrust produced can be modulated in real time based on pilot input and flight conditions through electronic control systems.

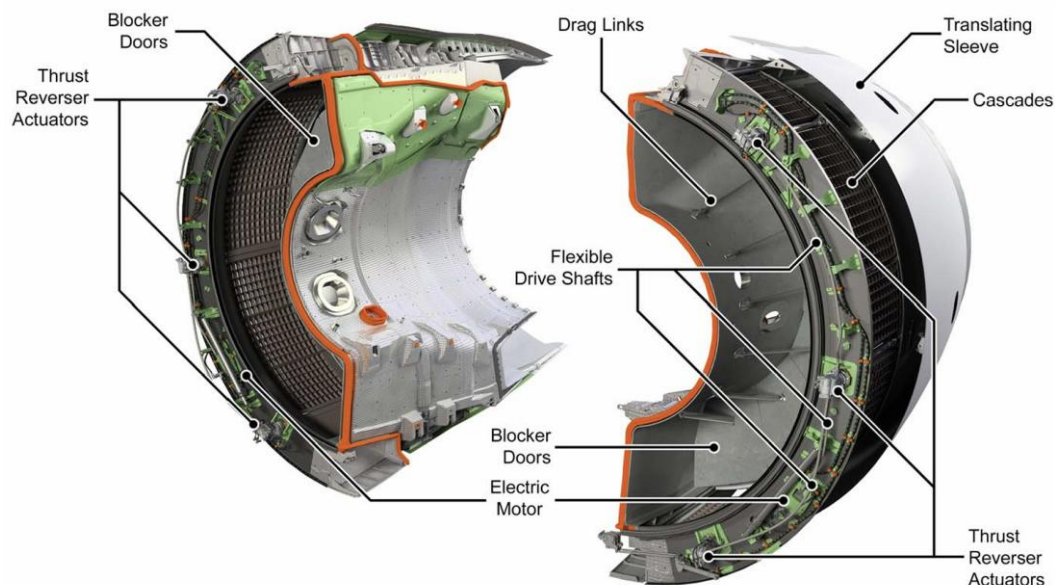


Figure 3-10 : Thrust reverser on A350 [10]

Four main thrust reverser architectures exist for turbofan engines, each suited to different bypass ratio and nacelle configurations. Cascade reversers are the most common solution for high bypass ratio commercial turbofans. A translating outer sleeve travels aft to expose fixed cascade vanes surrounding the bypass duct, while blocker doors deflect the bypass airflow through the vanes and redirect it forward. This configuration provides efficient reverse thrust with minimal interference with the engine core and is the natural reference for the BW-25 given its high bypass ratio engine. Multi-door reversers use a series of petals distributed around the bypass duct circumference that pivot into the airflow when deployed, offering good load

distribution and redundancy, and are particularly suited to engines with constrained nacelle volumes. Bucket-type reversers consist of two large doors that swing into the core exhaust path, but their volume scales rapidly with nozzle diameter, making them impractical for large high bypass ratio engines and confining their use to regional jets and military aircraft. Clamshell reversers operate similarly to bucket-type systems but incorporate cascade vanes for better airflow management and are primarily applied to low bypass or mixed exhaust engines.

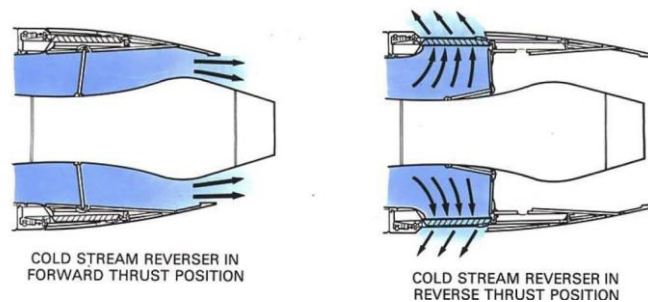


Figure 3-11 : Cascade reversers [17]

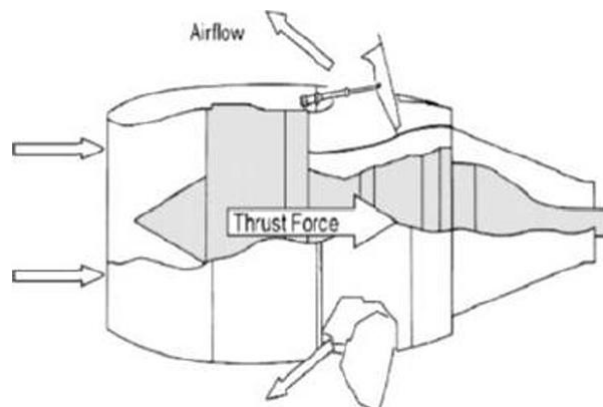


Figure 3-12 : Multi-Door Reversers [17]

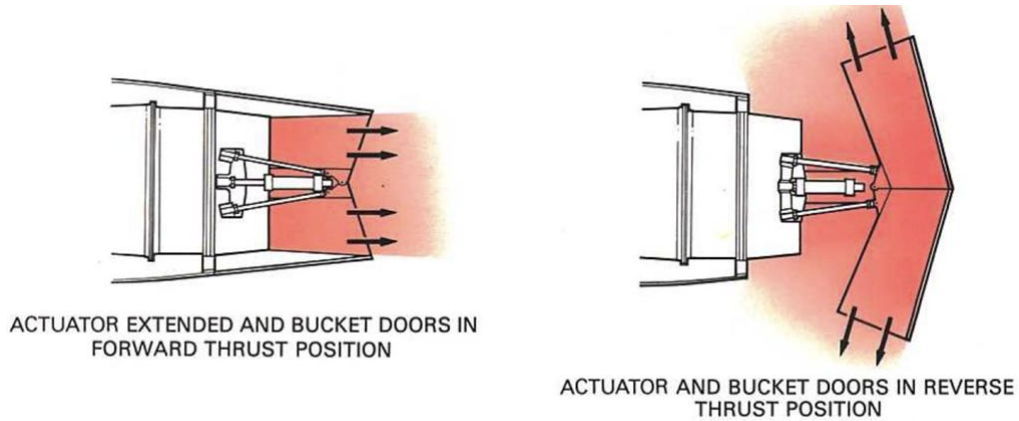


Figure 3-13 : Bucket reversers [17]

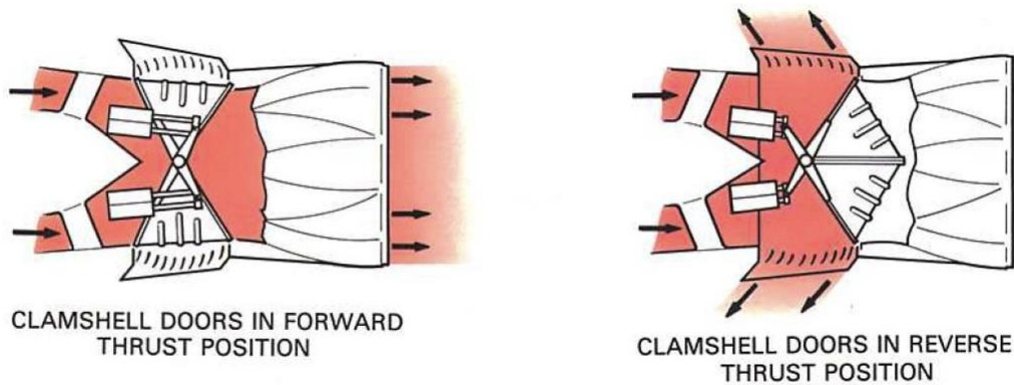


Figure 3-14 : Clamshell reversers [17]

For the BW-25, the upper-surface engine mounting introduces specific constraints on thrust reverser design that have received some attention in the literature. Studies on BWB configurations have identified two principal concerns associated with upper-surface thrust reverser deployment: the risk of contaminated air reingestion by the adjacent engine due to the proximity of the reversed flow discharge to the opposite inlet, and an adverse effect on the aircraft pitching moment during the landing roll, which is particularly sensitive on a BWB without a conventional tail [18]. These findings highlight the importance of engine lateral separation and thrust reverser deflection angle as design parameters that must be jointly optimised to mitigate both reingestion risk and longitudinal control degradation during deceleration.

3.6 Noise reduction

The BW-25, designed for a 2050 entry into service, must comply with increasingly stringent airport noise regulations and minimise its acoustic footprint during the Landing and Take-Off (LTO) cycle. International noise certification standards, defined under ICAO Annex 16 [19], impose strict limits at three reference points around the airport: sideline, flyover, and approach. Future regulations are expected to tighten these limits significantly, making noise reduction a key design driver for next-generation aircraft.

The dominant noise sources in a turbofan installation are fan noise, which propagates both forward through the inlet and rearward through the bypass duct and dominates during approach, and jet mixing noise, generated by the shear between the exhaust flow and the surrounding air, which dominates during take-off. The upper-surface engine mounting inherent to the BWB provides a significant degree of passive acoustic shielding by interposing the aircraft structure between the engines and the ground. However, the quantification of this shielding benefit remains subject to uncertainty — most studies are based on simplified geometric models or scaled experimental rigs that do not fully capture the complex diffraction and scattering phenomena associated with a real BWB planform. Shielding effectiveness varies considerably with frequency, engine lateral position, and observer angle, and existing studies disagree on the magnitude of the noise reduction achievable, particularly at sideline measurement points where the shielding geometry is less favourable. Passive shielding alone is therefore insufficient to guarantee compliance with 2050 noise targets, and active mitigation technologies must be incorporated into the nacelle design.

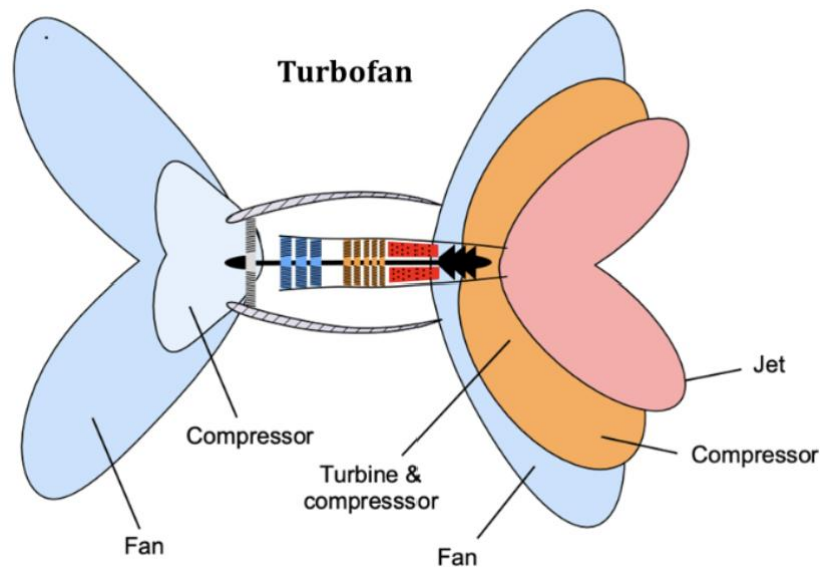


Figure 3-15 : Noise distribution around a turbofan [20]

Acoustic liners are among the most established noise mitigation solutions, consisting of honeycomb-structured panels with porous facing sheets that absorb acoustic energy through viscous losses within the cells. Their impedance must be tuned to the dominant frequency range of the targeted noise source. When integrated into the inlet cowl and fan duct, liners attenuate fan noise in both forward and rearward arcs, contributing directly to certification compliance [21]. However, the majority of liner design literature addresses conventional under-wing geometries. For upper-surface mounted engines, the acoustic propagation environment within the inlet and fan duct differs due to the modified refraction effects of the boundary layer flow over the BWB surface, and whether existing liner impedance design methodologies remain valid under these conditions is not clearly established. Advances in composite materials and adaptive liner technologies, where impedance can be actively controlled, are expected to further improve performance by 2050 and may partially address this gap.

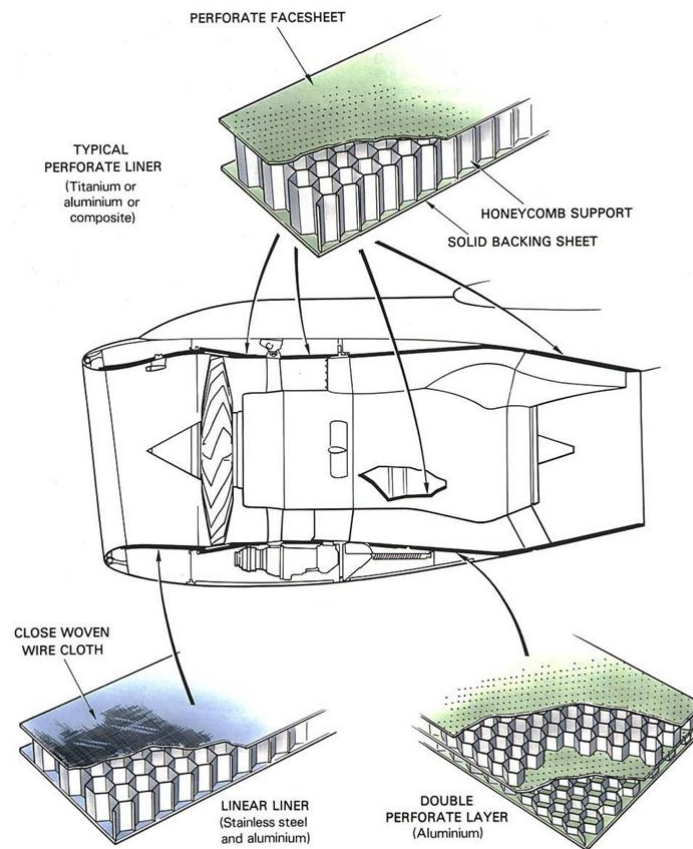


Figure 3-16 : Noise absorbing materials and location [21]

Chevron nozzles incorporate serrated trailing edges at the nozzle exit to promote mixing between bypass and core jet flows, generating streamwise vorticity that breaks down the large-scale turbulent structures responsible for low-frequency jet noise. However, increasing chevron immersion tends to raise high-frequency noise levels, requiring careful optimisation to balance the two contributions [22]. Chevrons also introduce a small but non-negligible thrust penalty. Given the stringent 2050 noise targets, chevrons were evaluated for the BW-25 but ultimately not retained, as the combination of upper-surface shielding and acoustic liner

treatment was considered sufficient to meet the noise objectives without incurring the associated aerodynamic penalty.



Figure 3-17 : Chevron nozzle on the 787 [23]

Negatively scarfed inlets feature an extended lower lip that tilts the inlet axis upward, redirecting intake noise toward the sky rather than the ground [24]. This technology has shown promising results in research programmes such as the European SILENCE project. However, the asymmetric inlet geometry introduces aerodynamic penalties that are particularly problematic for an upper-surface mounted engine: the extended lower lip increases local flow velocities on the upper inner surface, raising the risk of flow separation at high angles of attack, and the

asymmetric geometry increases structural loading on the upper lip. For the BW-25, this design could be interesting concerning the future noise regulation in 2050.

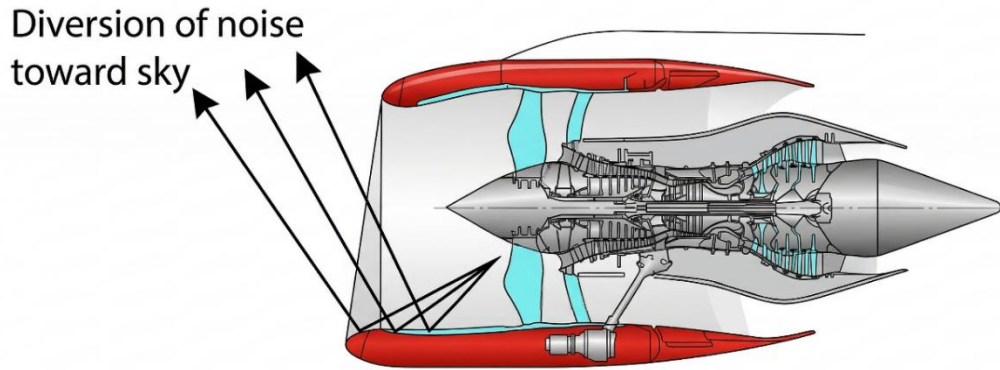


Figure 3-18 : Negative Scarfed Intake

Finally, the noise characteristics of hydrogen combustion introduce a dimension that is almost entirely absent from the current acoustic literature. LH_2 combustion produces a higher water vapour content in the exhaust, which modifies atmospheric absorption of high-frequency sound and could partially attenuate jet noise at ground level. However, the quantitative impact on certification noise levels has not been systematically studied for aviation applications, and it remains unclear whether this effect can be credited under current ICAO standards.

3.7 Engine accessories

Beyond the core propulsion components, a number of auxiliary systems must be accommodated within the nacelle envelope to ensure correct engine operation and to supply power to the aircraft. Although their detailed design falls outside the scope of this thesis, understanding their function and spatial requirements is necessary to make informed nacelle sizing decisions.

The BW-25 adopts a more electric architecture in which no bleed air is extracted from the engine core. All systems that would conventionally rely on pneumatic power are instead powered electrically, avoiding the efficiency penalty associated with diverting high-pressure, high-temperature air from the compression stages and contributing to reduced specific fuel consumption. Engine starting and electrical

power generation are handled by Variable Frequency Starter-Generators (VFSGs), an architecture already demonstrated on the Boeing 787. During engine start, electrical power supplied by the APU drives the VFSGs in motor mode, spinning the engine core via the accessory gearbox until self-sustaining speed is reached. Once running, the engine drives the gearbox, which operates the VFSGs as generators to supply electrical power to the aircraft systems [25]. The oil pumps are an exception to this fully electric philosophy and are instead shaft-driven through the accessory gearbox, as their operation is intrinsically linked to engine speed. Shaft power extraction is therefore limited to this engine-dedicated function, while all remaining accessories are electrically powered. The gearbox casing must be designed with robust sealing to prevent oil leakage within the nacelle fire zone, which is particularly important given the proximity of the LH₂ fuel system.

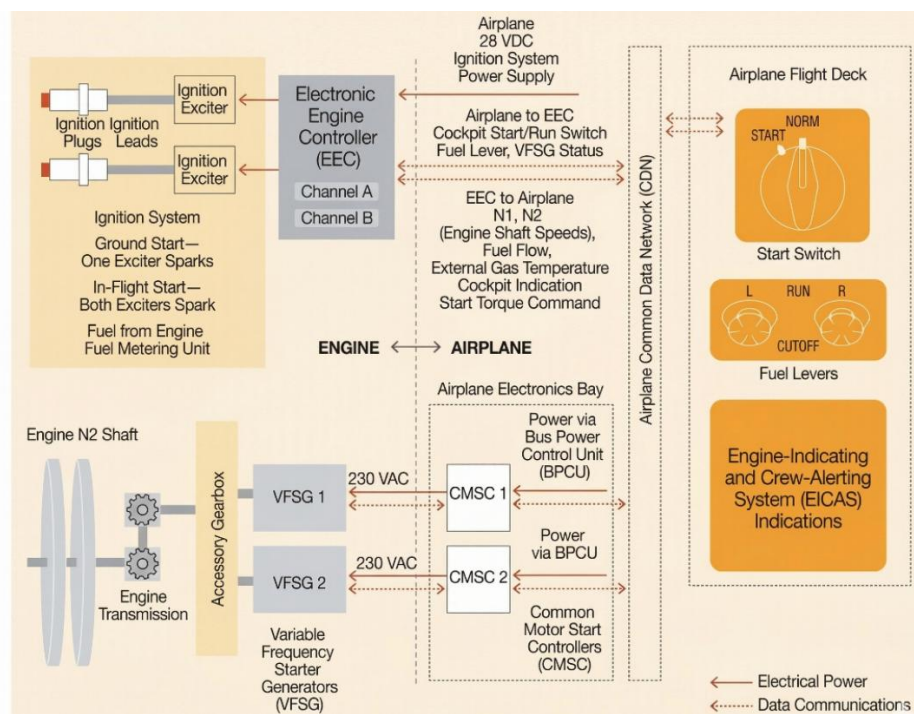


Figure 3-19 : 787 engine start system [25]

The VFSG architecture is well established on kerosene-fuelled platforms, but its behaviour in the thermal environment created by a cryogenic fuel system introduces questions that the literature does not yet address. The proximity of cryogenic LH₂ lines to the accessory gearbox and electrical components raises concerns about thermal fatigue, condensation, and potential embrittlement of materials not

qualified for cryogenic environments. These interactions have been studied in the context of space propulsion systems, but their translation to an aviation nacelle environment with its specific certification requirements remains largely unexplored.

The liquid hydrogen cooling system exploits the cryogenic properties of LH_2 as a heat sink through a series of heat exchangers integrated within the engine. The LH_2 is progressively warmed through successive exchange stages — cooling turbine components, the oil circuit, and the ECS supply air — before recovering additional thermal energy from the exhaust gases and reaching the combustor at the required injection temperature. This cascaded pre-conditioning architecture improves overall engine thermal management efficiency without requiring dedicated cooling systems. However, it introduces a degree of thermal coupling between engine systems that has limited precedent in commercial aviation — a failure or degradation in one heat exchanger stage affects the thermal state of subsequent stages and ultimately the fuel injection temperature. The fault propagation characteristics of this integrated thermal architecture, and its implications for system safety assessment under EASA CS-25 and CS-E, are not addressed in the available literature.

The engine oil system operates as a closed recirculatory loop in which oil is continuously pumped through the engine, filtered, cooled via the LH_2 heat exchanger, and returned to the reservoir, ensuring lubrication and cooling of all rotating components including bearings and gears.

Finally, the spatial integration of the accessory gearbox, VFSGs, and associated electrical routing within the nacelle envelope receives limited attention in published nacelle design methodologies. The elimination of bleed air ducting frees up some nacelle volume, but the addition of high-current electrical cabling, power electronics, and cryogenic fuel lines introduces new routing and thermal isolation requirements whose trade-offs have not been systematically studied for large high bypass ratio engines — a gap that is directly relevant to the BW-25 nacelle sizing process.

4 Safety assessment

The design of aircraft propulsion systems involves stringent safety considerations that must be addressed throughout the entire design process, not only at its completion. Identifying safety-critical and failure-prone elements early is essential to avoid costly redesign work and to ensure compliance with regulatory requirements from the outset.

For this reason, the safety assessment presented in this thesis follows a structured, front-loaded approach. It begins with the capture of all applicable airworthiness requirements from EASA's CS-25 and CS-E regulations, which form the regulatory framework within which every design decision must be justified. These requirements directly constrain the architectural and design choices made for the propulsion system.

Building on this foundation, a Functional Hazard Analysis was conducted to identify potential failure conditions and assess their consequences on the engine, the propulsion system, and the aircraft as a whole. The most severe failure cases identified through this process were then investigated further through a Fault Tree Analysis, which traces system-level failures back to their root causes at the component level.

In parallel, a Rotor Burst Analysis was carried out to evaluate the consequences of an uncontained engine rotor failure — one of the most extreme and potentially catastrophic failure modes. This analysis determined the trajectory and energy of high-speed debris released from a failed rotor and allowed hazard zones to be mapped onto the aircraft layout. These zone maps were shared with other design teams, including those responsible for fuel tanks and structural components, so that placement decisions could be made with full awareness of the associated risks.

Only after this complete safety framework was established did the nacelle design process proceed, ensuring that all architectural choices were informed by and compliant with the identified safety constraints.

4.1 Initial airworthiness specifications

The basis of this thesis work was to correctly capture all the legislative specifications regarding air safety related to the integration of propulsion systems, in order to make architectural and design choices that comply with the applicable requirements.

To achieve this, the very first task of the project was to gather all the requirements to gain an overall understanding of all aspects and constraints that need to be considered to carry out the project task. Of course, some restrictions were taken from the project specification booklet, but the majority of these restrictions come from EASA's CS-25 [26] and CS-E [27] regulations, along with their Acceptable Means of Compliance (AMC) [28].

The complete capture of these requirements was carried out in Appendix A.

4.2 Functional Hazard Analysis

The Functional Hazard Analysis (FHA) focuses on the main function of the system under study. Possible failure conditions that could affect this function are then identified [29]. The impact of each failure can vary, and its consequences on the aircraft and its occupants are assessed. This evaluation makes it possible to identify the most severe failure cases, which are given priority in the safety design process. For example, with the function of providing thrust generation, one of its failure conditions is "loss of thrust on one engine." The study of this failure condition can be developed as follows:

Function	Failure Condition	Flight Phase	Effect on Aeroplane/Crew/Occupants	Probability		Classification
				Qualitative	Quantitative	
Provide thrust generation	Loss of thrust on one engine	Take-Off, Climb, Cruise, Approach	Asymmetric thrust and increased drag leading to yaw, reduced climb performance and possible drift-down; high crew workload and noticeable discomfort for occupants.	Extremely Remote	1,00E-07	Hazardous

Table 4-1 : Example of FHA realized on the Loss of thrust on one engine

The detailed FHA is in the Appendix B.

4.3 Fault tree analysis

A Fault Tree Analysis (FTA) is a deductive reliability and safety technique to investigate how an undesirable event or system failure may occur. The initial step in conducting an FTA involves identifying the critical failure condition to be prevented from occurring, referred to as the top event, and then working down through the system to identify all possible combinations of lower level faults that could result in the occurrence of the top event [30]. The relationship between each of the lower level faults identified are shown by the use of logic symbols (AND gate, OR gate), indicating the way multiple faults may interact to produce the top event. Qualitative applications of FTA are to identify the minimum number of faults required to produce a given hazard. Quantitative applications involve estimating the probability of the top event based on the probability of the faults that make up the system. FTA is commonly applied within the Aerospace Industry and many other Safety Critical Industries to assist in establishing priorities for design improvement activities, assessing compliance with safety standards, and to assist decision making in reducing the potential for catastrophic failure.

Two detailed FTAs were completed as part of this work and are both contained in Appendix C. The first addresses an Uncontained Engine Rotor Failure (an extreme engine event resulting in a catastrophic loss of rotor integrity and release of high energy debris from the engine case, potentially causing damage to the aircraft structure, systems, or occupants). The second addresses the uncommanded in-flight deployment of the thrust reverser system. This FTA traces the top event down through two simultaneous conditions that must both be satisfied for the failure to occur: a complete failure of all locking functions (primary mechanical lock and secondary electrical lock) AND an uncommanded actuator motion reaching the deploy position. The use of an AND gate at the top level reflects the safety architecture of the E-TRAS system, in which the dual locking barrier and the command chain must fail independently and simultaneously.

4.4 Rotor burst analysis

A Rotor Burst occurs when rotating engine components such as turbine or compressor disks fail catastrophically, releasing high-speed, high-energy fragments capable of penetrating the engine case and damaging critical aircraft systems. A real-world example of such an event occurred on 13 November 2020, when an Antonov An-124 departing Novosibirsk experienced what appeared to be an uncontained failure of its No. 2 engine shortly after take-off. High-energy debris penetrated the fuselage and wing-root area, likely severing fuel lines, electrical wires and control cables, resulting in the loss of communication and electrical power. The aircraft was forced to return and landed without spoilers, thrust reversers, brakes or electrical power, overrunning the runway — though the crew survived [31].



Figure 4-1 : Side view of a rotor burst on an Antonov 124 [31]

4.4.1 Perform an analysis

The AMC-20 document defines a structured approach to assess the risks associated with rotor burst events, describing a method to predict the possible trajectories of high-energy fragments released from each rotor stage [28]. Four fragment

categories are defined, each characterised by its size, mass assumption, and spread angle:

Table 4-2 : Fragments description

Fragment type	Mass assumption	Spread angle
Single 1/3 disc fragment	1/3 of bladed disc	$\pm 3^\circ$
Intermediate fragment	1/30 of bladed disc	$\pm 5^\circ$
Small fragments (shrapnel)	Outer half of blade airfoil	$\pm 15^\circ$
Fan blade fragment	1/3 of blade airfoil section	$\pm 15^\circ$

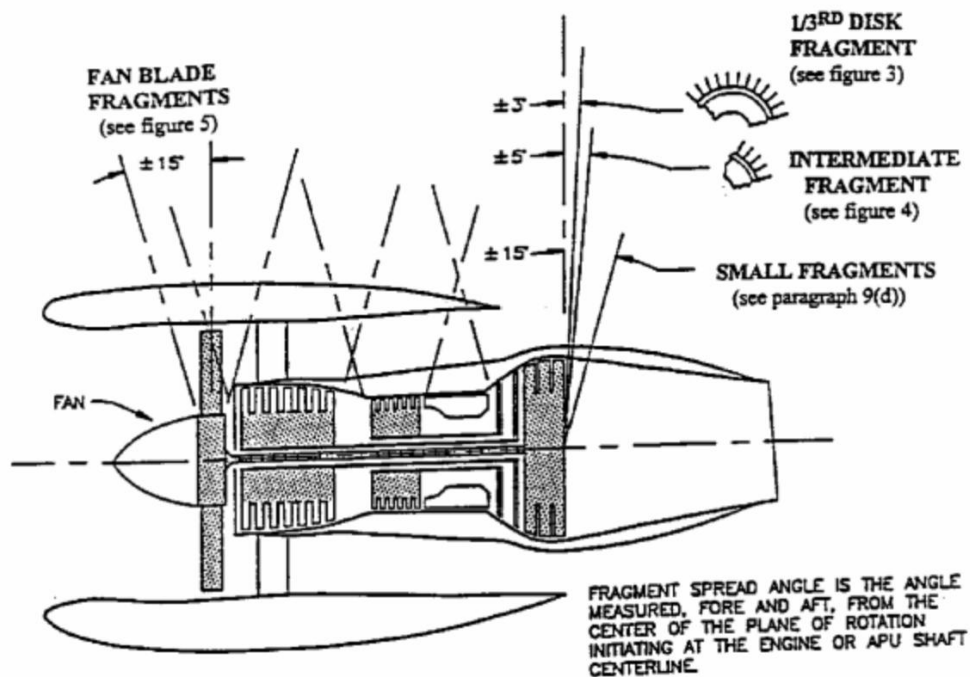


Figure 4-2 : Fragments paths estimated by the AMC-20 [28]

In all cases, only the translational kinetic energy is considered for energy calculations, based on the rim speed of the disc, while rotational energy is neglected. The one-third disc fragment represents the most severe case due to the large mass and concentrated spread angle. For small fragments and fan blade fragments, while primary structures are generally resistant to penetration, secondary structures such as fairings and control surfaces may be vulnerable to cumulative impacts.

Figure 48: Trent 900 IP turbine disc after overspeed burst test



Figure 4-3 : Fragments of the AN-124 rotor burst disc [31]

The author of the thesis created rotor burst areas based on the aircraft layout from the initial concept in the specification booklet. These areas were shared with other project teams, especially the team responsible for fuel tanks and systems, so they could take precautions and try to position the fuel tanks outside these zones.

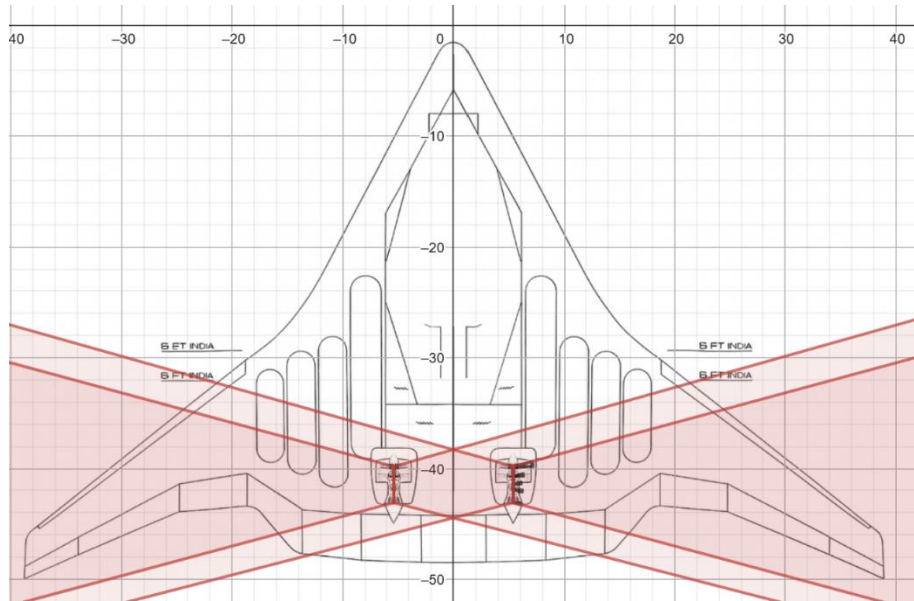


Figure 4-4 : Rotor bursts of 15° plotted on the 2D plane of the aircraft

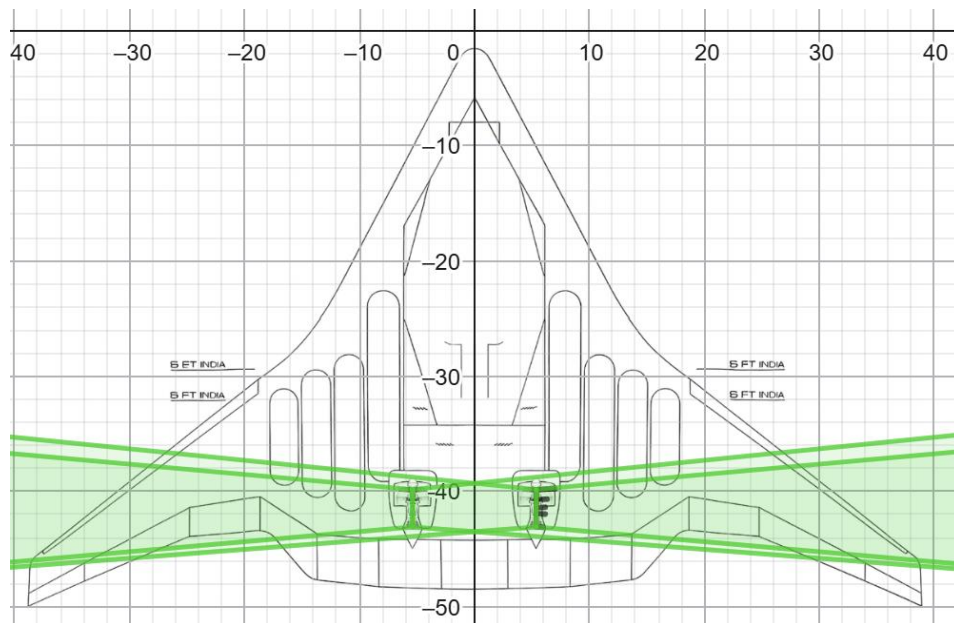


Figure 4-5 : Rotor bursts of 5° plotted on the 2D plane of the aircraft

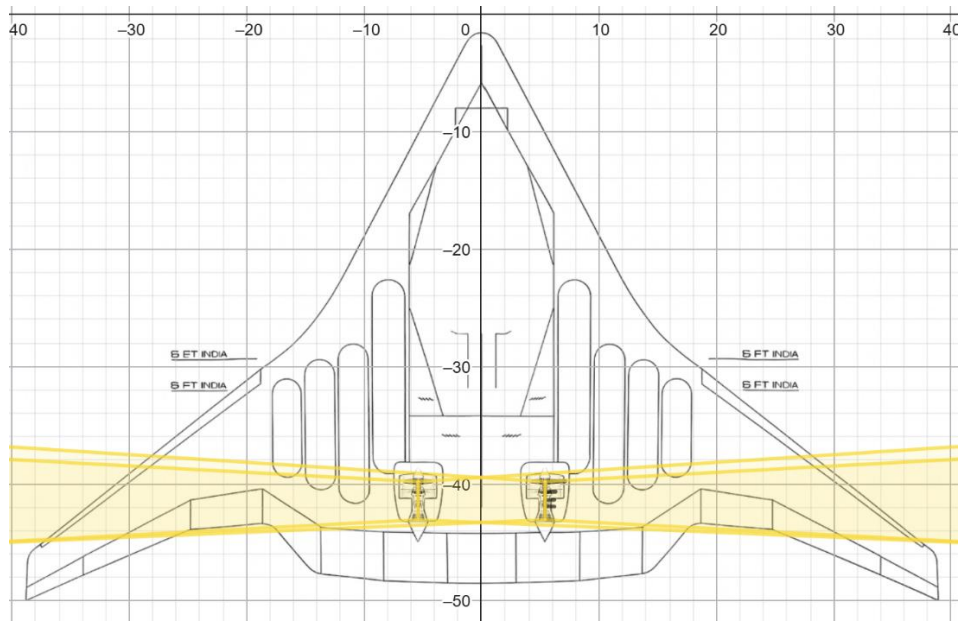


Figure 4-6 : Rotor bursts of 3° plotted on the 2D plane of the aircraft

These maps were distributed among other project team members to use when planning locations for their components or to add structural reinforcement in consideration of risk zones. The catastrophic case of a one-third disc fragment (with an assumed spread angle of 3°) has some of the fuel tanks falling inside the risk zones; since this case assumes a fragment with infinite energy, the fuel tanks would need to be repositioned outside these zones at the first stage of the project.

To give all affected teams a clear and shared understanding of the rotor burst risk zones, the analysis was also integrated into the final 3D model of the aircraft for all three spread angles ($\pm 15^\circ$, $\pm 5^\circ$ and $\pm 3^\circ$). This allowed each team to visualise how the engines are positioned relative to their own components, and to assess whether any parts of their systems fall within the hazard cones:

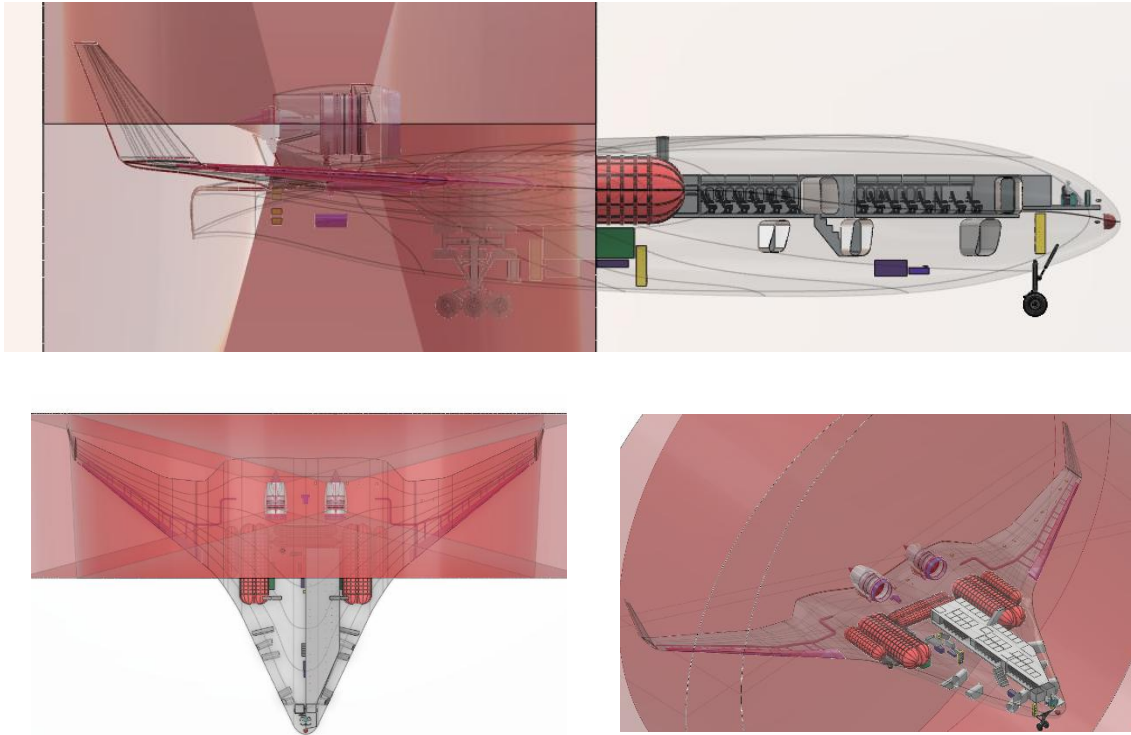


Figure 4-7 : 3D view of the rotor burst area for a 15° spread angle

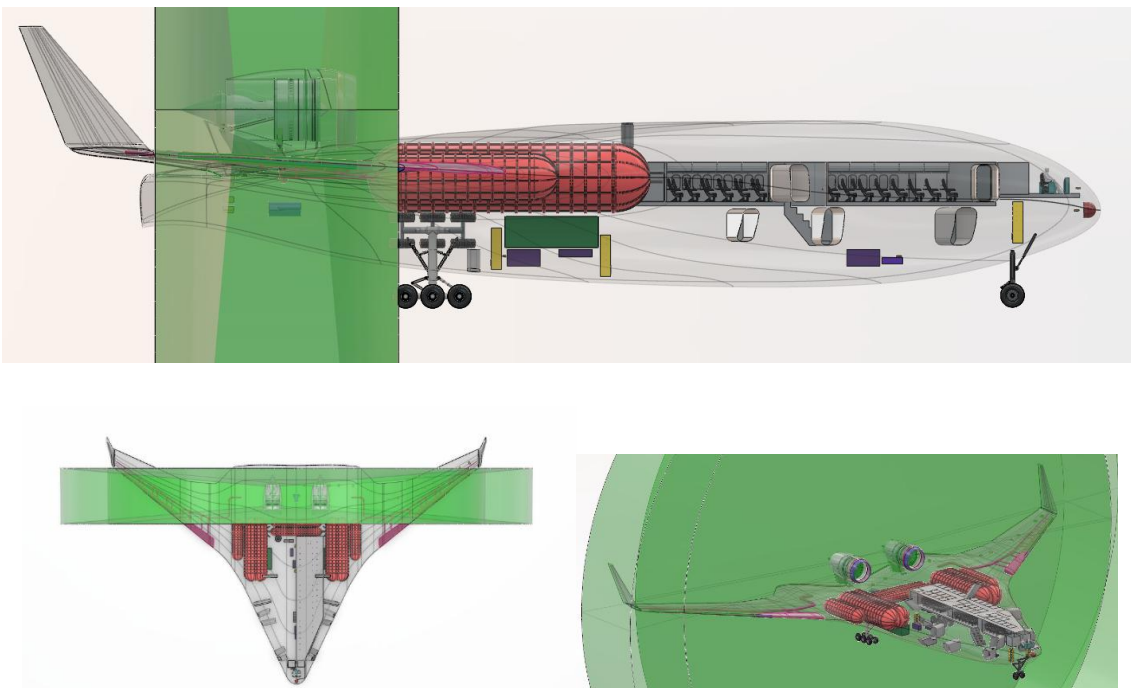


Figure 4-8 : 3D view of the rotor burst area for a 5° spread angle

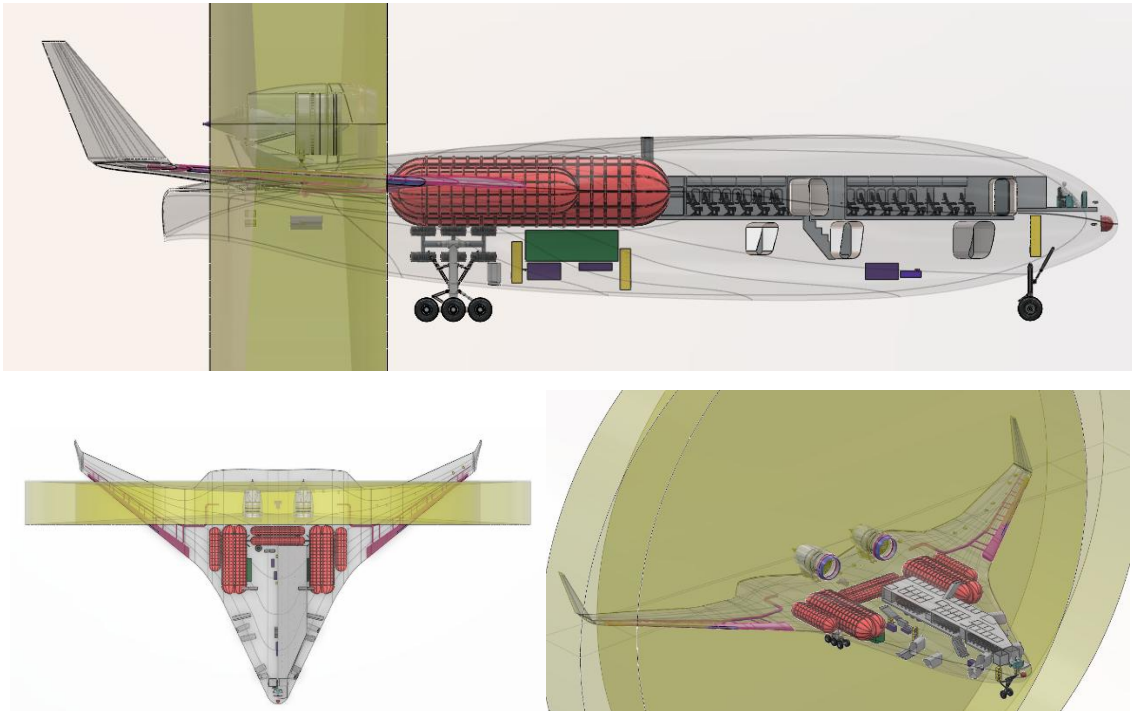


Figure 4-9 : 3D view of the rotor burst area for a 3° spread angle

For all three spread angles ($\pm 15^\circ$, $\pm 5^\circ$ and $\pm 3^\circ$), all fuel tanks are positioned outside the rotor burst hazard zones in the final aircraft configuration. Even in the worst-case scenario of a one-third disc fragment with a $\pm 3^\circ$ spread angle — which assumes infinite fragment energy — no fuel tank falls within the risk zone. This confirms that the engine and tank layout resulting from the design process satisfies the rotor burst safety requirements across all fragment categories defined by AMC-20.

4.4.2 Safety distance between the engines

The purpose of this paragraph is to determine whether, in the initial engine configuration (spacing of 10 m), the probability of losing both engines as a result of a rotor burst is lower than the target of 10^{-9} catastrophic accidents per flight hour set by EASA. Indeed, if one engine lost by a rotor failure hit with one of his fragment the second engine, there will be a total loss of the thrust of the engines leading to a fatal crash. To do this, AMC-20 provides a methodology for calculating the probabilities of a component being hit by a rotor burst. The author has considered here the worst case, namely a spread angle of 3° with infinite energy.

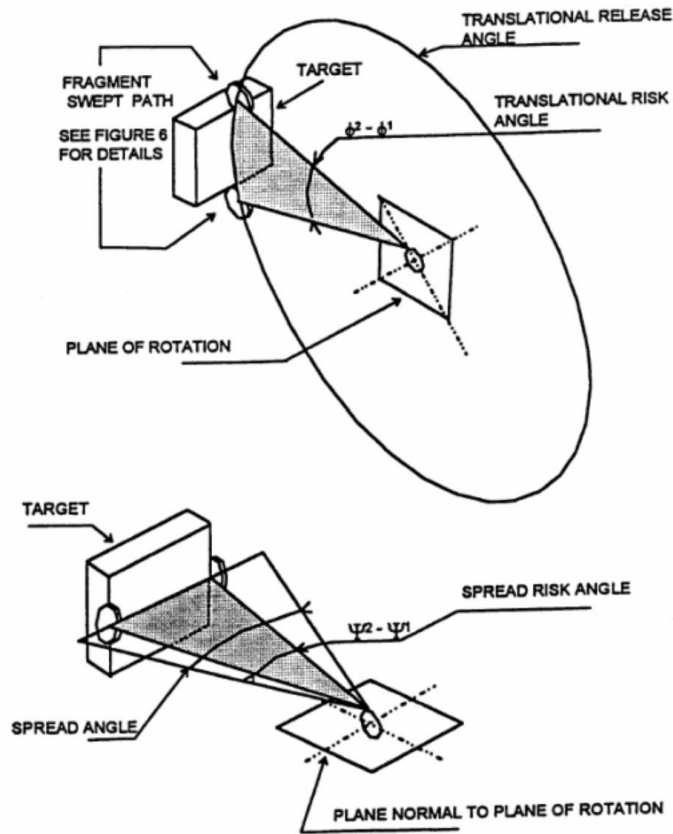


Figure 4-10 : Rotor burst threat window [28]

The formula given by AMC-20 for calculating the probability that a component is hit by a fragment is as follows:

$$P = \frac{\text{Degree of the translational risk angle}}{360} \times \frac{\text{Degree of the spread risk angle}}{\text{Degree of the spread angle}} \times D_p \times R$$

Equation 4.1 : Probability that a fragment hit a component [28]

Where D_p quantifies the probability that a rotor burst occurs during a given flight phase, and the Risk factor is the probability that fragments from a rotor burst reach and damage a critical element of the aircraft, taking into account geometry,

integration, and existing protections for that flight phase. AMC-20 provides a table of estimates for these factors.

Phase	Dp	Risk
Take-off before V1	0.35	1
T,O,-V1 to first power reduction	0.2	1
Climb	0.22	0.4
Cruise	0.14	0.2
Descent	0.03	0.4
Approach	0.02	0.4

Table 4-3 : Dp and Risk factors [28]

As Figure 4-9 : 3D view of the rotor burst area for a 3° spread angle shows, for a spread angle of 3°, the engine opposite the one experiencing the rotor burst is necessarily within the spread angle risk zone, giving $\frac{\text{Degree of the spread risk angle}}{\text{Degree of the spread angle}} = 1$. And by measuring geometrically, *Degree of the translational risk angle* = 15.90° in the initial aircraft configuration.

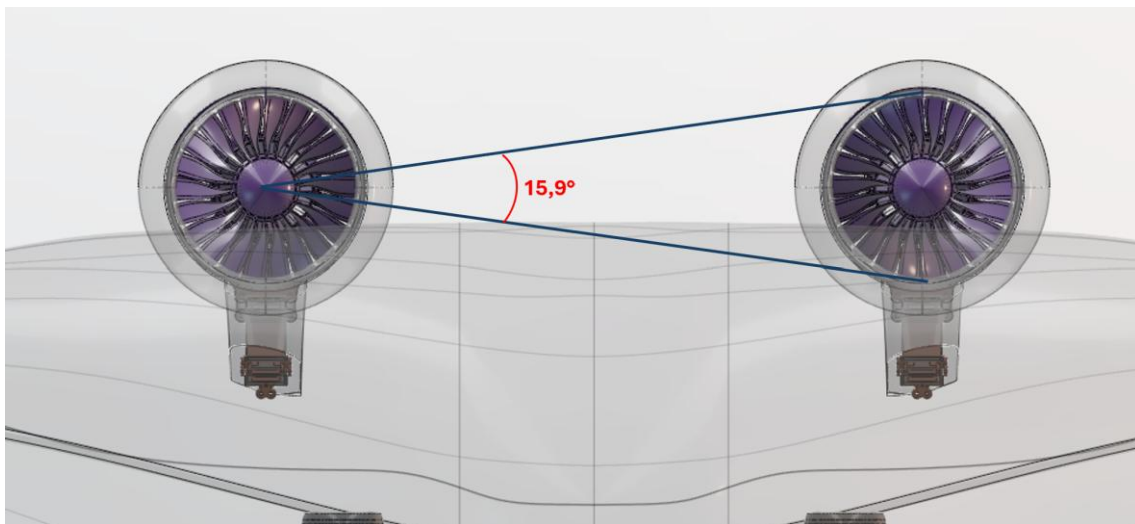


Figure 4-11 : Front view of the engines

The probability can therefore be calculated that a rotor burst fragment will hit the opposite engine if such an event occurs.

Table 4-4 : Probability of one engine hitting the other in the event of a rotor burst

Phase	Probability
Take-off before V1	0.01545833
T,O,-V1 to first power reduction	0.00883333
Climb	0.00388667
Cruise	0.00123667
Descent	0.00053
Approach	0.00035333

And knowing that in 2026, the A350 has accumulated more than 14 million flight hours, so the number of flight hours for the Rolls-Royce Trent XWB was 14 million [32]. And since its entry into service, the A350, which is the only aircraft equipped with these engines, has never experienced a rotor burst. The author therefore estimate a pessimistic probability of a rotor burst on the Trent XWB at 1 in 14 million, i.e. 7.14×10^{-8} per flight hour.

So the final probability to have the loss of both engines by a rotor burst is the product between the probability previously calculated and the estimate probability to have a failure conducting to a rotor burst:

Table 4-5 : Probability to have a total loss of both engines due to a rotor burst

Phase	Probability
Take-off before V1	1.10×10^{-9}
T,O,-V1 to first power reduction	6.31×10^{-10}
Climb	2.78×10^{-10}
Cruise	8.83×10^{-11}
Descent	3.78×10^{-11}
Approach	2.52×10^{-11}

The probability remains below or at the 10^{-9} threshold across all flight phases. The marginal exceedance for the take-off phase (1.10×10^{-9}) is considered acceptable given the highly conservative assumptions of the AMC-20 methodology, notably the assumption of infinite fragment energy and zero structural energy absorption. The

engine layout may change, but the distance of 10 meters between the two engines is an indication to keep in mind to remain within the requirements set by EASA.

5 Engine integration and nacelle design

5.1 Engine integration in the BW-25

5.1.1 Engine Mounting Configuration Selection

The choice of engine mounting configuration is a critical design decision for a Blended Wing Body aircraft, as the propulsion system is intrinsically linked to the aerodynamic and structural architecture of the airframe. Three principal configurations were considered for the BW-25: fuselage-embedded engines with Boundary Layer Ingestion (BLI), surface-mounted engines sitting directly on the upper fuselage without a pylon, and pylon-mounted engines elevated above the upper surface. The pylon-mounted configuration was ultimately selected for the reasons detailed below.

The embedded BLI configuration is theoretically attractive from a propulsive efficiency standpoint. BLI involves a fundamental trade-off: while ingesting the boundary layer reduces ram drag, this benefit is largely offset by the reduced total pressure recovery at the fan face, leading to an increase in Specific Fuel Consumption [7]. More critically for the present project, the non-uniform flow entering a BLI inlet generates fan face distortion that, if excessive, can push the fan beyond its stall margin, requiring either a costly fan rematch or active flow control. The severity of this distortion is highly sensitive to the angle of attack, and its assessment at the most demanding flight conditions — particularly take-off rotation — requires high-fidelity 3D CFD simulations to validate engine operability. Conducting such simulations to the standard required for a credible design decision was not feasible within the schedule of the present project. It should also be noted that the BW-11 reference configuration, which shares the same geometry with the BW-25, employs a distributed embedded propulsion system of a fundamentally different nature — an array of BLI fans integrated via S-ducts — and therefore provides no direct design heritage for an embedded conventional turbofan installation on the BW-25.

The surface-mounted configuration, in which the nacelle sits directly on the upper fuselage without a pylon, was also considered. While this arrangement minimizes

wetted area and eliminates the aerodynamic drag of the pylon itself, it introduces several difficulties. The engine intake would sit within the thick, low-energy boundary layer developing over the rear fuselage, particularly at high angles of attack during take-off rotation, generating significant inlet flow distortion [7]. Confirming that this distortion would remain within engine operability limits would again have required dedicated high-fidelity CFD analysis. Furthermore, this configuration removes available space from the compartment beneath the engine, complicates maintenance access, and presents complex structural attachment challenges to the fuselage skin without the structural depth provided by a pylon.

The pylon-mounted configuration was therefore selected for three converging reasons. First, the inability to confirm within the project timescale that distortion levels would remain within engine operability limits at high angles of attack made both the embedded and surface-mounted options too high a risk to adopt without further analysis. Second, the structural design group required the definition of a pylon as part of their own work, making the pylon-mounted arrangement a necessity for multidisciplinary consistency across the project. Third, pylon-mounted engines above the rear fuselage offer a well-understood approach in terms of maintenance access and propulsion-airframe integration, with no complex structural penetrations of the fuselage structure.

Benefit Potential Larger Than Task Order #7									
		Baseline Podded	TO # 7 Task 27	Max Benefit No AFC	Max Brnt With AFC				
		w/Short Duct				AR 0.86	TO # 19 AR 2	Canted	
Capture	PT1/PT0	1	0.974	0.984	0.979	0.986	0.976	0.985	
Ram Drag	ESFC (%)	0	-6.85	-5.14	-6.27	-4.7	-7.47	-4.97	
Inlet Recov	PT2/PT0	0.998	0.971	0.974	0.973	0.98	0.97	0.979	
Engine SFC	(%)	base	5.69	4.92	5.13	4.02	6.2	4.23	
Drag	(%)	base	1.29	-0.22	-1.93	-5.7	-5.9	-5.7	
Weight	(lbs)	base	16300	600	-6400	-4500	-4500	-4500	
Design TOGW	(lbs)	768200	796000	767500	746300	735930	737200	736900	
Block Fuel	(lbs)	249760	257500	248700	236000	224200	225360	225090	
delta fuel & CO2	(%)	base	3.1	-0.43	-5.5	-10.2	-9.8	-9.9	
3,000 nm Range: 70% Load Factor (68,795 lbs)									
TOGW	(lbs)	56600	589400	566100	553900	55200	552300	552300	
Fuel Burned	(lbs)	85800	88300	85500	81400	77620	77940	77890	
delta fuel & CO2	(%)	0	2.9	-0.4	-5.1	-9.6	-9.2	-9.3	

Figure 5-1 : Comparison of podded and BLI nacelle installations [33]

It should nevertheless be noted that even the podded pylon installation above a BWB body presents significant aerodynamic integration challenges. Several studies have documented strong shock waves and flow separation arising in the channel formed between the nacelle lower surface and the upper fuselage. Keller's CFD investigation of a hybrid wing body with podded UHBR engines showed that the initial engine integration led to a shock within the nacelle-body channel that increased cruise drag by 83% compared to the isolated components [6]. Similarly, Xin et al. identified a convergent-divergent channel effect as the root cause of these shock waves and flow separation and demonstrated that careful body surface contouring to maintain a quasi-parallel flow channel beneath the nacelle could reduce cruise drag by 8.7% [14]. These findings underline that, while the pylon-mounted configuration is the more conservative choice, its aerodynamic integration nonetheless requires careful attention to nacelle-airframe interference.

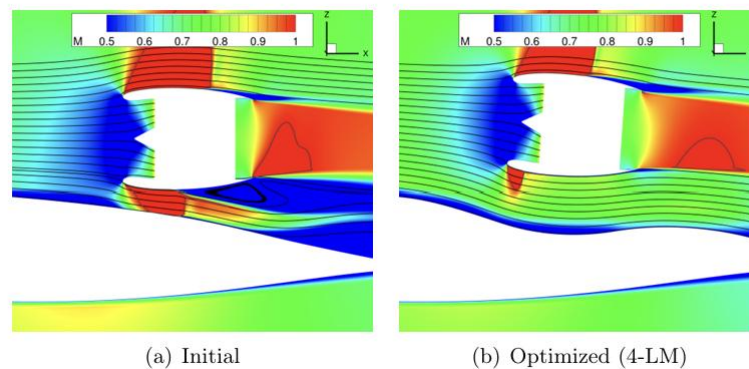


Figure 5-2 : Flow field in vicinity of nacelle in 2-D section at engine centre [6]

5.1.2 Pylon Height Determination

Once the pylon-mounted configuration was selected, a key parameter to determine was the pylon height — that is, the vertical clearance between the upper fuselage surface and the engine fan face. The primary driver for this dimension is the local boundary layer thickness at the engine inlet station: the engines must be positioned sufficiently above the surface to avoid ingesting the low-energy boundary layer flow, particularly at the most critical flight condition.

Existing boundary layer studies on the BW-11 configuration were conducted either at cruise conditions or at fixed low angles of attack and therefore do not capture the most demanding scenario for engine installation [12], [34], [35], [36], [37]. The critical design point was identified as the take-off rotation, which combines low speed, high angle of attack, and the longest boundary layer development run over the upper fuselage, resulting in the maximum boundary layer thickness at the engine station. The boundary layer thickness was therefore estimated analytically using the empirical turbulent flat plate correlation:

$$\frac{\delta}{x} \cong \frac{0.16}{(Re_x)^{\frac{1}{7}}}$$

Equation 5.1 : Boundary layer thickness correlation for a flat thin plate [38]

The engine station was located at $x = 36$ m from the nose of the aircraft. At rotation, the aircraft reaches a speed of $V_R = 61.7$ m/s, and the kinematic viscosity of air at sea level ISA conditions is $\nu = 1.51 \times 10^{-5}$ m²/s. The local Reynolds number at the engine station is therefore :

$$Re_x = \frac{V_R \cdot x}{\nu} = \frac{61.7 \times 36}{1.51 \times 10^{-5}} = 1.47 \times 10^8$$

Substituting into the boundary layer thickness correlation :

$$\delta = \frac{0.16 \times 36}{(1.47 \times 10^8)^{1/7}} = 39 \text{ cm}$$

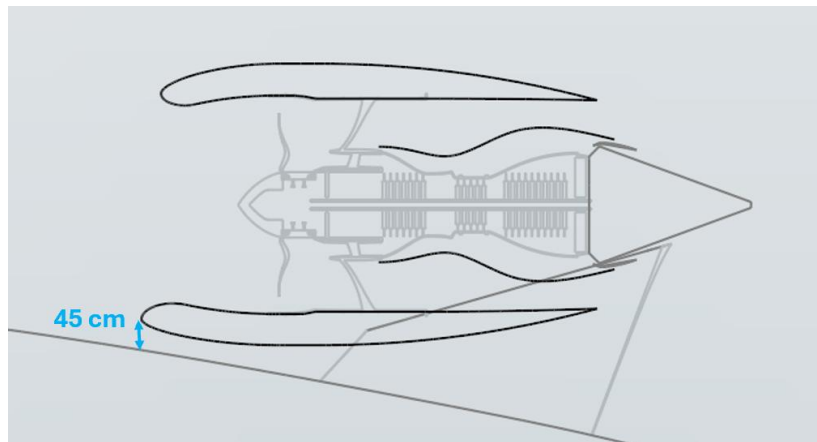


Figure 5-3 : Engine height on the BW-25

A pylon height of 45 cm was therefore selected, providing a 6 cm margin above the estimated boundary layer edge. This choice is supported by the CFD study of Luque Olano (2020) on the BW-11 with pod-mounted engines, which demonstrated that shifting the outer nacelle upwards by 40 cm significantly reduced fan face distortion and total pressure losses at high angles of attack, as illustrated in Figure 5-4. It is acknowledged that the flat plate assumption represents a simplification, as the actual boundary layer over the BWB upper surface is subject to favourable and adverse pressure gradients, curvature effects, and the influence of the high angle of attack at rotation. The analytical result should therefore be regarded as a first-order engineering estimate, with a higher-fidelity CFD characterisation recommended as future work.

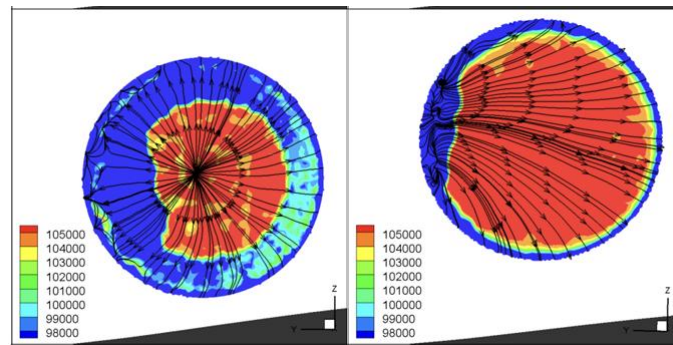


Figure 5-4 : Total pressure contours (Pa) on the fan inlet face of the outer turbofan in the low position (left) and high position (right) configurations: from top to bottom $\alpha = 11$ degrees [36]

5.1.3 Longitudinal and Lateral Engine Positioning

Beyond the pylon height, the longitudinal and lateral placement of the engines was also constrained by several competing design requirements.

In the longitudinal direction, the available installation space is tightly bounded fore and aft by structural and system constraints. Forward of the engine station, the fuel tanks occupy the internal volume. So placing the engine pylon base in this region would introduce an unacceptable rotor burst risk, as an uncontained engine failure could directly threaten the fuel system. Aft of the engine station, the trailing edge control surfaces — elevons — occupy the available planform area, leaving no structural attachment point for a pylon base without interfering with the control surface kinematics and hinge lines. These two constraints effectively fix the

longitudinal position of the engines within a narrow band on the upper rear fuselage. The engine longitudinal position was therefore set at 41 m from the datum, in accordance with the BW-25 green book specifications, reflecting the only location compatible with both the fuel tank boundary forward and the elevon structure aft.

In the lateral direction, the placement was constrained primarily by airworthiness considerations related to rotor burst containment. Moving the engines too close to the aircraft centreline would bring both engines within the debris dispersion zone of a single uncontained rotor failure, creating an unacceptable risk of losing both powerplants simultaneously — a scenario that certification requirements explicitly seek to prevent. Conversely, moving the engines further outboard towards the wing tip region offered no meaningful aerodynamic or structural benefit and would have introduced adverse pitching and rolling moment asymmetries in the event of an engine failure, compounding the already challenging stability and control characteristics of the BWB configuration. Furthermore, one of the key advantages of the rear-mounted engine configuration on a BWB is the noise shielding effect provided by the wide centerbody planform, which intercepts the downward radiation of fan and jet noise towards the ground. Moving the engines further outboard would progressively reduce this shielding effectiveness, as the engines would extend beyond the trailing edge of the centerbody and be increasingly exposed to direct noise propagation paths towards ground observers. The lateral engine spacing was therefore set at 10 m apart, in accordance with the BW-25 green book specifications, representing a balanced trade-off between rotor burst containment, engine-out controllability, and noise shielding effectiveness.

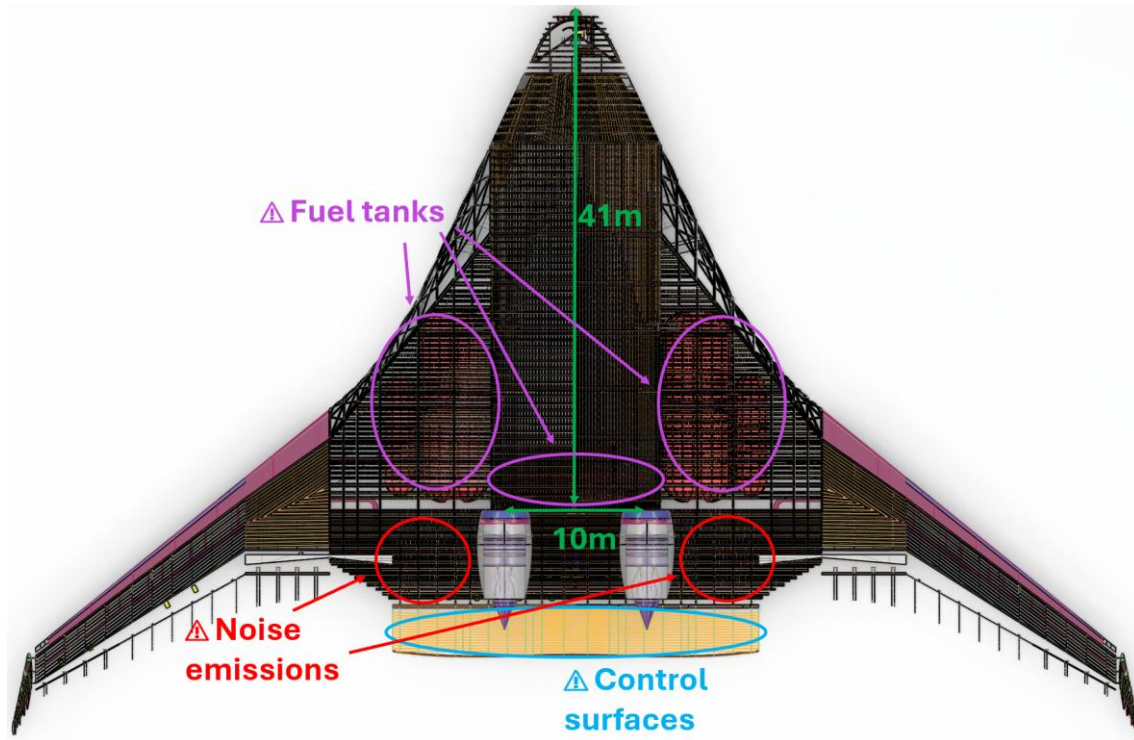


Figure 5-5 : Engines position

5.2 Engine nacelle design

5.2.1 Introduction to engine nacelle design

The nacelle is a multifunctional structure whose design must simultaneously satisfy aerodynamic, structural, acoustic, and safety requirements. Its primary role is to house and protect the engine while delivering airflow to the fan face with minimal distortion across the entire flight envelope. Beyond this core function, the nacelle defines the external aerodynamic profile of the propulsion system, contributing directly to the overall drag of the aircraft through its external skin friction, inlet spillage drag, and boat-tail drag at the nozzle exit. It also provides the structural interface between the engine and the pylon, integrates the fire protection system, and accommodates the acoustic treatment required to meet noise certification requirements. Furthermore, the nacelle must incorporate thrust reverser provisions, contain and direct the exhaust flow through appropriately designed nozzles, and ensure compliance with bird strike and FOD resistance requirements [10]. All of these functions must be achieved within a weight budget that does not

compromise the overall aircraft performance, making nacelle design an inherently iterative and multidisciplinary process.

For the BW-25, the more-electric architecture eliminates engine bleed air, which modifies the internal volume requirements of the nacelle by removing conventional pneumatic ducting while adding cryogenic fuel lines and high-current electrical routing, as outlined in Engine accessories Section 3.7.

Given the engine selected for the BW-25 — a derivative of the Rolls-Royce Trent XWB with a fan diameter of 3.0 m and an overall length of 5.812 m — a separate exhaust flow nacelle configuration was adopted. The aircraft is designed to cruise at Mach 0.80 at an altitude of 45,000 ft, which defines the primary aerodynamic design point for the nacelle external profile and inlet sizing. This choice is consistent with current trends in high bypass ratio commercial turbofan installations, where the structural simplicity and lower external drag of the separate-flow architecture outweigh the propulsive efficiency benefit of a mixed-flow solution, particularly given that the upper-surface installation already provides significant acoustic shielding that partially compensates for the jet noise penalty of the separate-flow configuration, as discussed in Section 3.4.

The present section focuses on the design of the air inlet cowl and the external nacelle profile, including drag estimation. The exhaust nozzle, thrust reverser system and acoustic treatment are each addressed in dedicated sections thereafter.

5.2.2 Inlet dimensions

The inlet is the forward section of the nacelle and a key role in the overall drag characteristics of the propulsion system. Its geometry must be carefully sized to deliver the required mass flow to the fan face with minimal distortion and pressure loss, while keeping spillage drag to a minimum across the flight envelope [39]. The key geometric parameters governing the inlet design are illustrated in Figure 5-6:

- D_{fan} : fan face diameter (3.0 m, from engine specifications)
- D_{th} : throat diameter, the minimum inner diameter of the inlet

- D_{hl} : highlight diameter, the outer diameter at the inlet lip leading edge
- D_{max} : maximum outer nacelle diameter
- L_f : nacelle forebody length

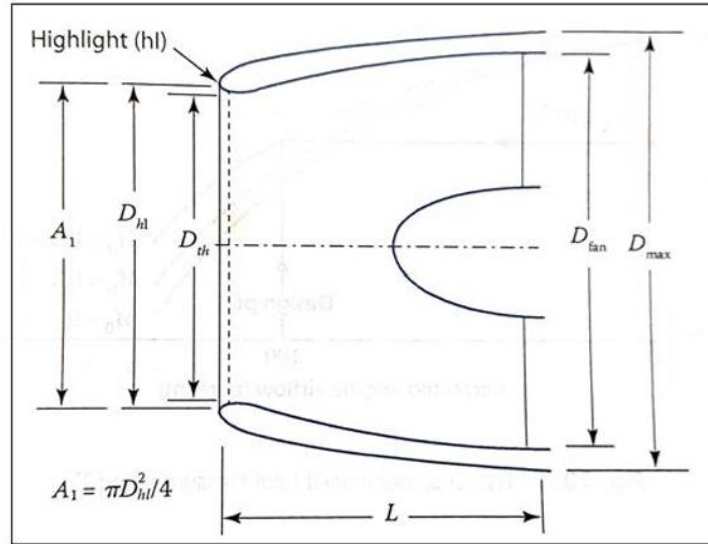


Figure 5-6 : Subsonic inlet nomenclature [39]

In practice, D_{th} , D_{hl} and D_{max} do not necessarily share the same proportions in all directions, and non-axisymmetric nacelle geometries are in fact commonplace in commercial aviation. Since nacelle design typically relies on CFD simulations to optimise the airflow and minimise drag across the various flight phases — an analysis that falls outside the scope of this conceptual design study — the sizing approach followed here is based on empirical correlations from Mattingly's *Aircraft Engine Design* [39], derived from a survey of nacelles developed over the past four decades and summarised in Figure 5-7. These correlations are particularly well-suited to the present work as they distinguish explicitly between the top, side, and bottom sections of the nacelle, reflecting distinct aerodynamic and structural drivers. From a structural standpoint, the lower nacelle must accommodate the accessory gearbox, requiring additional internal volume beneath the fan case. From an aerodynamic standpoint, at high angles of attack during take-off rotation, the stagnation point migrates toward the lower lip, creating a strong adverse pressure gradient that increases the risk of flow separation — a thicker, more rounded lower

lip delays this separation [40]. Crosswind conditions introduce a similar risk on the windward side, explaining why the side section is intermediate in thickness. Conversely, the top lip is kept thin to minimise wave drag at cruise, as an excessively thick upper lip would accelerate the local flow beyond sonic speed. For the BW-25, a non-axisymmetric geometry is therefore adopted, allowing the lower, upper and lateral sections to be tailored independently in accordance with these drivers.

Variable	Values		Key Influences
	Maximum	Minimum	
D_{hl}/D_{th}	1.13 Top 1.16 Side 1.20 Bottom	1.11 Top 1.12 Side 1.12 Bottom	Takeoff performance, low-speed angle of attack capability, and crosswind capability
D_{hl}/D_{max}	0.840 Top 0.865 Side 0.860, 0.775* Bottom	0.815 Top 0.820 Side 0.732 Bottom	High-speed cruise drag and engine-out performance
A_{fan}/A_{th}	1.24	1.16	Inlet diffuser recovery and influence on distortion
L/D_{fan}	0.85 ^{SMR} 0.65 ^{LR}	0.65 ^{SMR} 0.50 ^{LR}	Inlet diffuser recovery and influence on distortion

*Depends on gearbox location: core-mounted, fan-mounted, respectively
^{SMR}Short-to-medium-range twin-engine civil transport
^{LR}Long-range twin-engine civil transport

Figure 5-7 : Typical ranges of key inlet design variables [39]

5.2.2.1 Throat diameter D_{th}

The throat diameter is the primary sizing variable of the inlet. The ratio A_{fan}/A_{th} from Mattingly's correlations ranges from 1.16 to 1.24, corresponding to D_{th} values between 2.69 m and 2.79 m for a fan diameter of 3.0 m. A mid-range multiplier of 1.20 was selected, yielding:

$$D_{th} = \frac{D_{fan}}{\sqrt{1.20}} = \frac{3.0}{\sqrt{1.20}} = 2.74 \text{ m}$$

5.2.2.2 Nacelle forebody length L_f and scarf inlet

The forebody length ratio L_f/D_{fan} typically ranges from 0.50 to 0.65 for long-range twin-engine civil transport, and from 0.65 to 0.85 for short-to-medium range configurations. A mid-range value of 0.59 was selected, giving a baseline forebody length of:

$$L_f = 0.59 \times D_{fan} = 0.59 \times 3.0 = 1.77 \text{ m}$$

Unlike a conventional axisymmetric inlet where L_f is a single value, the BW-25 inlet incorporates a negatively scarfed geometry. As discussed in Section 3.6, a scarfed inlet redirects intake noise upward toward the sky, reducing ground-level noise exposure, and improves ingestion of the airflow sweeping over the fuselage of the aircraft.

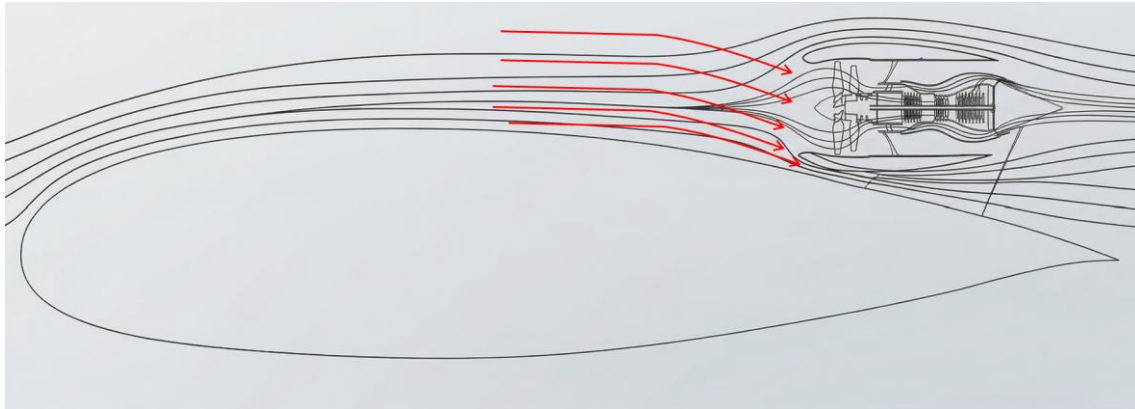


Figure 5-8 : Streamlines above the upper surface of the BW-25 at the engine level

An alternative approach would have been to apply a tilt angle to the engine itself to align it with the local streamlines above the fuselage for the aerodynamics considerations — a solution adopted in NASA's N2A-EXTE concept as shown in Figure 5-9 [41].

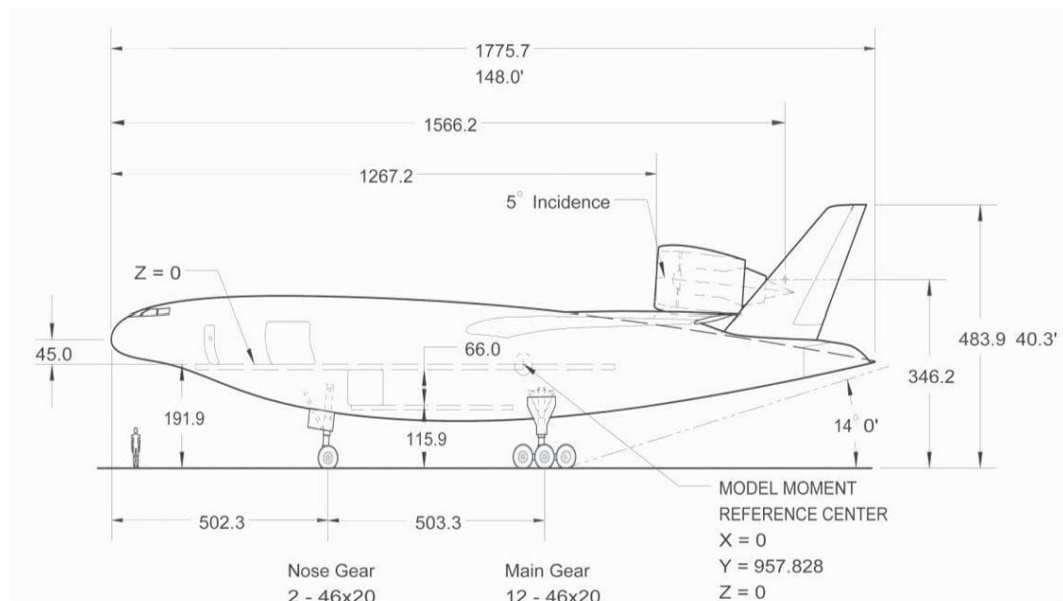


Figure 5-9 : General N2A-EXTE Model arrangement drawing details [41]

However, this option was rejected for the BW-25, as tilting the thrust line upward would further increase the already significant nose-down pitching moment generated by the aft-mounted engines relative to the aircraft centre of gravity. The scarfed inlet was therefore preferred, as it achieves improved flow ingestion and acoustic benefits without modifying the thrust vector orientation.

A negative scarf angle of 5° , consistent with values found in the literature for similar applications — including the A320 scarf inlet developed for the SNECMA SILENCE(R) programme shown in Figure 5-10 — has been chosen for the nacelle design of the BW-25.



Figure 5-10 : A320 scarf inlet for the Snecma – SILENCE(R) programme [42]

This angle results in three distinct forebody lengths for the top, side, and bottom sections, derived geometrically from the baseline L_f as follows:

$$L_{f, TOP} = L_f - \frac{D_{th}}{2} \tan(5^\circ) = 1.77 - \frac{2.74}{2} \times 0.0875 = 1.650 \text{ m}$$

$$L_{f, SIDE} = L_f = 1.784 \text{ m (fixed value)}$$

$$L_{f, BOTTOM} = L_f + \frac{D_{th}}{2} \tan(5^\circ) = 1.77 + \frac{2.74}{2} \times 0.0875 = 1.922 \text{ m}$$

These values are illustrated in Figure 5-11.

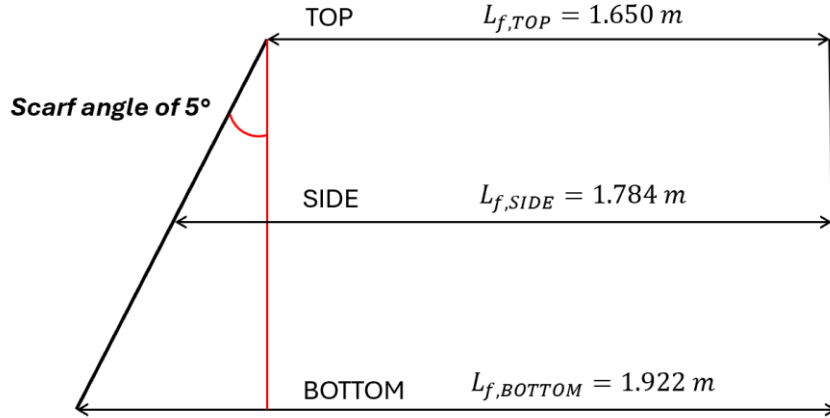


Figure 5-11 : Side view of the different forebody length for the TOP, SIDE and BOTTOM section

5.2.2.3 Highlight diameter D_{hl}

The highlight diameter is derived from the throat diameter using the ratio D_{hl}/D_{th} from Mattingly's correlations. Mid-range multipliers were selected independently for each section to reflect the non-axisymmetric geometry: 1.12 for the top, 1.14 for the side, and 1.16 for the bottom. The larger multiplier at the bottom is consistent with the aerodynamic and structural drivers described above, providing a more rounded lower lip for improved flow attachment at high angles of attack and sufficient internal volume for the accessory gearbox. This gives:

$$D_{hl,TOP} = 1.12 \times 2.74 = 3.069 \text{ m}$$

$$D_{hl,SIDE} = 1.14 \times 2.74 = 3.124 \text{ m}$$

$$D_{hl,BOTTOM} = 1.16 \times 2.74 = 3.178 \text{ m}$$

5.2.2.4 Maximum outer diameter D_{max}

The maximum outer nacelle diameter is determined from the highlight diameter using the ratio D_{hl}/D_{max} . Mid-range values of 0.828 were selected for the top and side sections, and 0.754 for the bottom section, consistent with the larger external profile required at the bottom to house the accessory gearbox:

$$D_{max,TOP} = \frac{D_{hl,TOP}}{0.828} = \frac{3.069}{0.828} = 3.706 \text{ m}$$

$$D_{max,SIDE} = \frac{D_{hl,SIDE}}{0.828} = \frac{3.124}{0.828} = 3.772 \text{ m}$$

$$D_{max,BOTTOM} = \frac{D_{hl,BOTTOM}}{0.754} = \frac{3.178}{0.754} = 4.215 \text{ m}$$

For simplicity in the subsequent nacelle profile design, a single representative value of D_{max} was adopted, taken as the mean of the three directional values:

$$D_{max} = \frac{3.706 + 3.772 + 4.215}{3} = 3.898 \text{ m}$$

This result compares favourably with the maximum nacelle diameter of the A350 family, which uses the same Rolls-Royce Trent XWB engine and has a measured D_{max} of 3.94 m [11]. Additionally, an industrial practitioner consulted during this project indicated that a common industry rule of thumb is to size the nacelle wall thickness at approximately 25% of the fan radius, giving:

$$D_{max} \approx D_{fan} + 2 \times 0.25 \times \frac{D_{fan}}{2} = 3.0 + 2 \times 0.375 = 3.75 \text{ m}$$

The value obtained through Mattingly's correlations is therefore consistent with both the A350 reference and this simplified industrial estimate, providing confidence in the sizing approach adopted here. All inlet dimensions are summarised in Table 5-1:

Table 5-1 : Key dimensions of the forebody

Scarf angle [°]	5.00				
Characteristic	TOP	SIDE	BOTTOM	Mean	Unit
D_{th}	2.740	2.740	2.740	2.740	m
D_{hl}	3.069	3.124	3.178	3.124	m
L_f	1.650	1.784	1.922	1.785	m
D_{max}	3.706	3.772	4.215	3.898	m

5.2.2.5 Inlet total pressure recovery validation

A further validation of the inlet sizing can be obtained through the inlet total pressure recovery at zero flight speed, π_d . This parameter is of particular physical significance as it quantifies the ability of the inlet lip to guide the incoming air into

the throat without generating excessive flow separation or pressure losses under the most demanding low-speed conditions — namely take-off and go-around, where the aircraft is stationary or at very low speed and the fan is operating at high power. Under these conditions, the air is entirely aspirated by the fan rather than rammed in by the forward motion of the aircraft, and the Mach number at the throat is therefore governed solely by the fan mass flow demand. For the BW-25, this zero flight speed throat Mach number is $M_{th} = 0.50$, corresponding to the fan operating condition at take-off power.

The total pressure recovery is evaluated following Mattingly's formulation :

$$\pi_d = \frac{\left(1 + \frac{\gamma - 1}{2} M_{th}^2\right)^{\frac{\gamma}{\gamma - 1}}}{1 + \gamma M_{th}^2} + \left(1 - \frac{\left(1 + \frac{\gamma - 1}{2} M_{th}^2\right)^{\frac{\gamma}{\gamma - 1}}}{1 + \gamma M_{th}^2}\right) \exp\left(-\frac{1}{7} \cdot \frac{A_{th}}{A_b}\right)$$

Equation 5.2 : Total pressure recovery at zero flight speed [39]

where A_b is the lip bellmouth area, defined as the annular cross-sectional area of the lip itself between the highlight and the throat, and physically represents the capacity of the lip to smoothly guide the flow without separation. It is computed from the mean highlight and throat diameters as follows:

$$\begin{aligned} A_{th} &= \frac{\pi}{4} \times D_{th}^2 = \frac{\pi}{4} \times 2.74^2 = 5.901 \text{ m}^2 \\ A_b &= \frac{\pi}{4} (D_{hl,mean}^2 - D_{th}^2) = \frac{\pi}{4} (3.124^2 - 2.74^2) = 1.768 \text{ m}^2 \\ \frac{A_{th}}{A_b} &= \frac{5.901}{1.768} = 3.34 \end{aligned}$$

Substituting $M_{th} = 0.50$, $\gamma = 1.4$ and $A_{th}/A_b = 3.34$ into Mattingly's expression yields:

$$\pi_d = 0.954$$

This value lies within the desired range indicated by Mattingly, confirming that the lip geometry resulting from the chosen highlight and throat diameters provides adequate total pressure recovery at zero flight speed. Physically, a π_d of 0.954 means that 4.6% of the total pressure available at the fan face is lost through lip flow

distortion and separation at take-off power — a level consistent with well-designed inlets of comparable high bypass ratio turbofan engines. The inlet sizing is therefore validated both geometrically through Mattingly's empirical correlations and aerodynamically through this pressure recovery criterion.

5.2.2.6 Forebody drag rise Mach number validation

A second aerodynamic validation of the inlet dimensions is provided by the forebody drag rise Mach number $M_{D,f}$, which represents the Mach number above which wave drag begins to develop on the nacelle forebody due to the acceleration of the local flow around the highlight and over the external nacelle surface. For the nacelle to operate without wave drag penalty at the design cruise condition, $M_{D,f}$ must exceed the cruise Mach number of 0.80. This parameter is governed by two geometric ratios: the highlight-to-maximum diameter ratio D_{hl}/D_{max} , which controls the local flow acceleration around the lip, and the forebody fineness ratio L_f/D_{max} , which determines how gradually the flow accelerates over the external nacelle surface — a finer nacelle pushing the drag rise to higher Mach numbers. Following the empirical correlation proposed in the lecture notes of Mr. Darrell Williams [43], derived from NACA 1-series intake performance data:

$$M_{D,f} = 1 - \frac{1}{8} \left[\frac{\sqrt{1 - \left(\frac{D_{hl}}{D_{max}} \right)^2}}{\frac{L_f}{D_{max}}} \right]$$

Equation 5.3 : Mach drag rise number correlation [43]

Substituting the mean inlet dimensions — $D_{hl} = 3.124$ m, $L_f = 1.785$ m and $D_{max} = 3.898$ m:

$$M_{D,f} = 1 - \frac{1}{8} \left[\frac{\sqrt{1 - \left(\frac{3.124}{3.898} \right)^2}}{\frac{1.785}{3.898}} \right] = 0.837$$

This yields a forebody drag rise margin of $0.837 - 0.800 = 0.037$ above the cruise Mach number, confirming that the nacelle forebody geometry does not generate wave drag at the design cruise condition. While this correlation was derived from NACA 1-series intake data and therefore represents a simplified estimate for a conceptual design study, the positive margin obtained provides confidence that the inlet sizing is aerodynamically consistent with the cruise flight envelope of the BW-25.

5.2.3 Inlet lip shape design

The inner lip contour defines the aerodynamic profile of the inlet between the highlight and the throat on the internal side of the nacelle. Its role is to guide the incoming air smoothly into the diffuser without generating flow separation, particularly under the critical conditions of high angle of attack and crosswind identified previously. The inner lip must therefore provide sufficient curvature to maintain flow attachment when the stagnation point migrates away from the highlight under these off-design conditions.



Figure 5-12 : Inlet lips

Following the criterion established by Albers [44] to achieve a minimum overall Mach number gradient throughout the inlet, the inner lip contour is defined by a super-ellipse of the form:

$$\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 1$$

Equation 5.4 : Elliptical geometry for the intake lips

where the parameters are defined as:

- $b_{inner} = \frac{D_{hl} - D_{th}}{2}$
- $a_{inner} = 2 \times b_{inner}$
- $n = 2$

The semi-minor axis b_{inner} corresponds physically to the lip thickness — the radial distance between the highlight and the throat — and therefore varies between the top, side and bottom sections in accordance with the highlight diameters determined previously. The resulting inner lip contour parameters are summarised in Table 5-2:

Table 5-2 : Inlet lip characteristics definition

Characteristic	Value			Unit
	Upper lip	Side lip	Lower lip	
$b_{inner} = (D_{hl} - D_{th})/2$	0.164	0.192	0.219	m
a_{inner}	0.329	0.384	0.438	m

The progressively larger inner lip dimensions from top to bottom are consistent with the aerodynamic drivers discussed previously: the bottom inner lip, being the most exposed to flow separation at high angles of attack, requires the greatest curvature to maintain attached flow under these critical conditions. By calculating each point, the author got these contours:

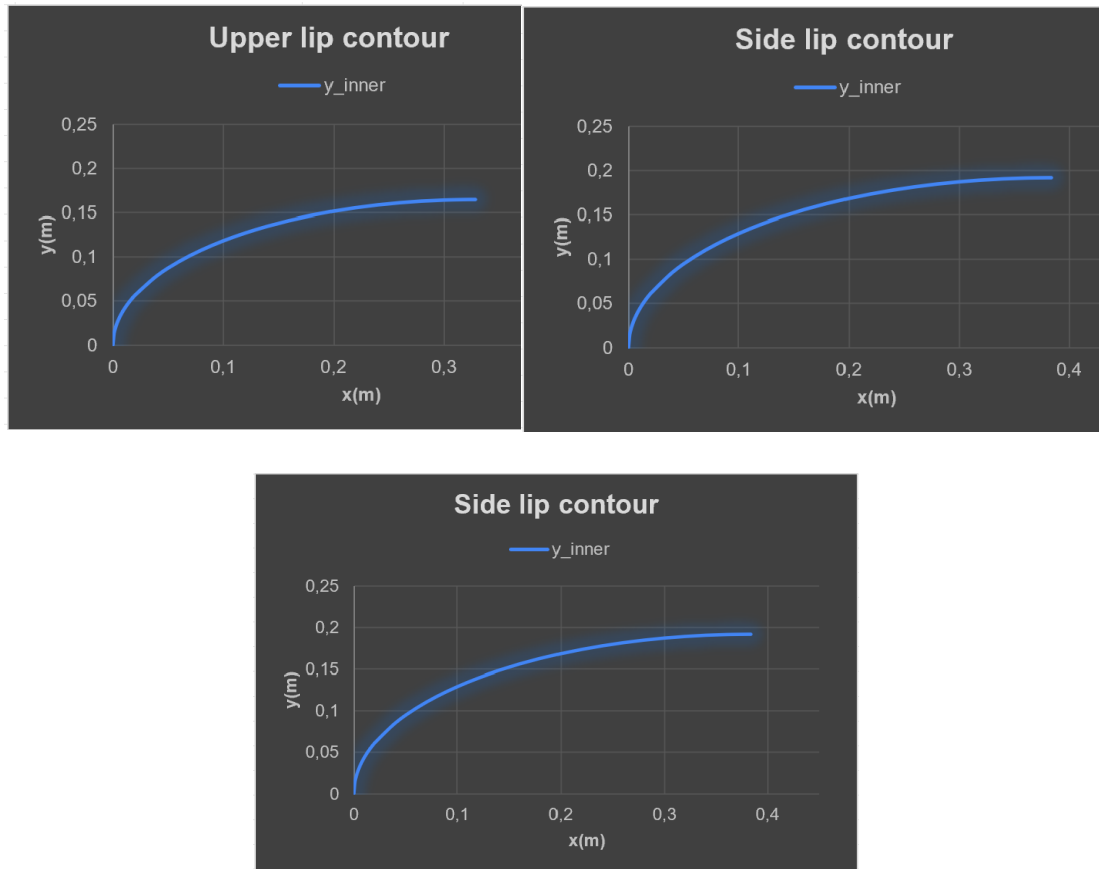


Figure 5-13 : Inlet lips profiles

5.2.4 Nacelle profile design

The external nacelle profile is divided into three sections: the forebody, which extends from the highlight to the maximum diameter; the midbody, a short cylindrical section at constant maximum diameter whose length has been greatly reduced in modern high bypass ratio engines compared to older designs; and the afterbody, which tapers from the maximum diameter down to the nozzle exit. The external contour must minimise skin friction and wave drag at cruise while ensuring that the forebody drag rise Mach number validated in the previous section is not exceeded.

5.2.4.1 Forebody design

The forebody external profile was generated using the NACA 1-Series analytical representation provided by ESDU Data Item 94013 [45]. The NACA 1-Series fore cowl profile is a well-established standard for axisymmetric nacelle forebody

design, originally developed for high critical speed inlets and subsequently widely adopted as a reference geometry for investigating flow characteristics of subsonic intakes. The ESDU document provides a smoothed analytical expression for the external profile coordinates that ensures regular variation of slopes and curvatures — a requirement for aerodynamically consistent geometry definition — as opposed to the original discrete coordinate table which exhibits irregularities in second divided differences that can introduce spurious pressure distribution features in flow computations. The smoothed profile is defined by:

$$\frac{y}{Y} = c \left(\frac{x}{X} \right)^{1/2} \left[\frac{1}{c} - \left(\frac{x}{X} - 1 \right) \frac{1}{2c} - \frac{1}{2c-1} \cdot \frac{1.5}{c} \left(\frac{x}{X} - 1 \right)^2 \right] + \left[\sum_{n=0}^7 (-1)^{n+1} A_n \left(\frac{x}{X} \right)^n \right] \left[\frac{\left(\frac{x}{X} \right)^{3/2} \left(1 - \frac{x}{X} \right)^2}{\left(b + \frac{x}{X} \right)^2} \right]$$

Equation 5.5 : Forebody geometry equation [45]

where the constants are $b = 0.05$ and $c = 1.044988$, and the coefficients A_n are given in Table 5-3. The normalising dimensions X and Y are defined as:

$$X = L_f ; Y = \frac{D_{max} - D_{hl}}{2}$$

Three distinct external profiles were generated, one for each section of the non-axisymmetric nacelle geometry, using the section-specific values of D_{hl} and L_f determined previously while keeping D_{max} constant at 3.898 m. The input parameters for each profile are summarised in Table 5-3:

Table 5-3 : External contour characteristics

Characteristic	Value			Unit	Constant	
	TOP	SIDE	BOTTOM		n	An
D_{max}	3.898	3.898	3.898	m	0	0.009466
D_{hl}	3.069	3.124	3.178	m	1	0.378874
$Y = (D_{max}-D_{hl})/2$	0.415	0.387	0.360	m	2	1.709298
$X = L_f$	1.650	1.784	1.922	m	3	7.731339
b	0.05	0.05	0.05	-	4	22.79108
c	1.044988	1.044988	1.044988	-	5	40.64622
					6	38.05716
					7	14.23322

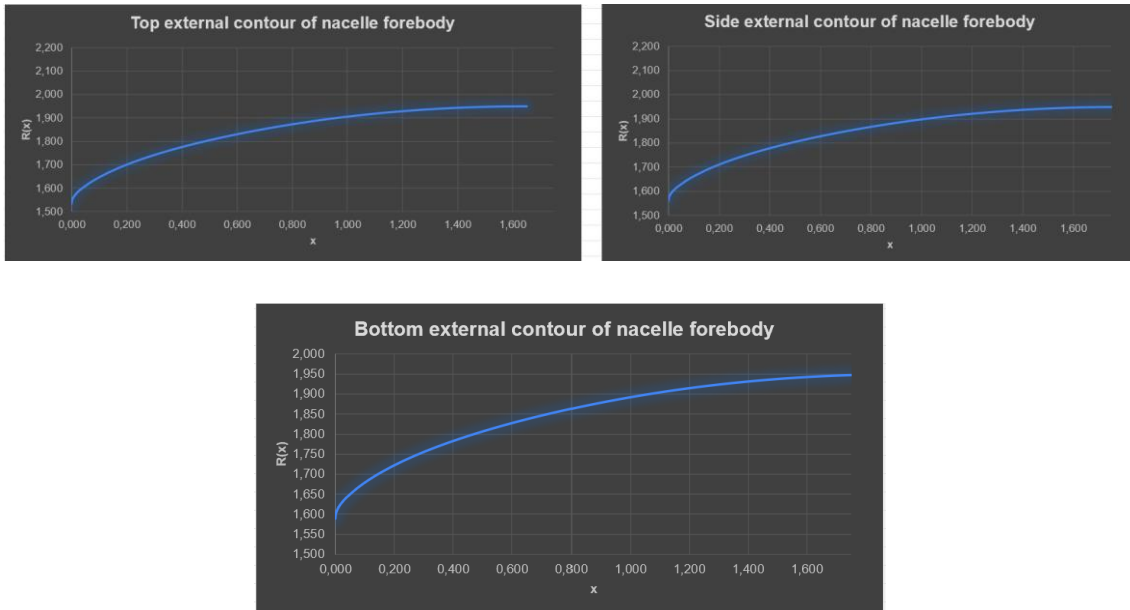


Figure 5-14 : External contour profiles

The three profiles show the expected behaviour of the NACA 1-Series geometry: the external radius increases rapidly near the highlight where the curvature is greatest, then transitions progressively toward $D_{max}/2$ at the aft end of the forebody. The

bottom profile, having the largest L_f and the smallest Y , exhibits the most gradual curvature transition, which is consistent with the lower drag rise requirement at the bottom section where the nacelle-fuselage proximity is greatest.

5.2.4.2 Afterbody design

The afterbody connects the maximum nacelle diameter to the bypass nozzle exit, tapering the external profile from D_{max} down to the nozzle diameter D_n . Unlike the forebody and inlet, the afterbody cross-section is circular and uniform in all directions, as it is not subject to the asymmetric incidence and crosswind constraints that govern the non-axisymmetric inlet geometry. The key geometric parameters of the afterbody are the boat-tail radius R_a , the afterbody length L_a , and the chord angle β_c , illustrated in Figure 5-15:

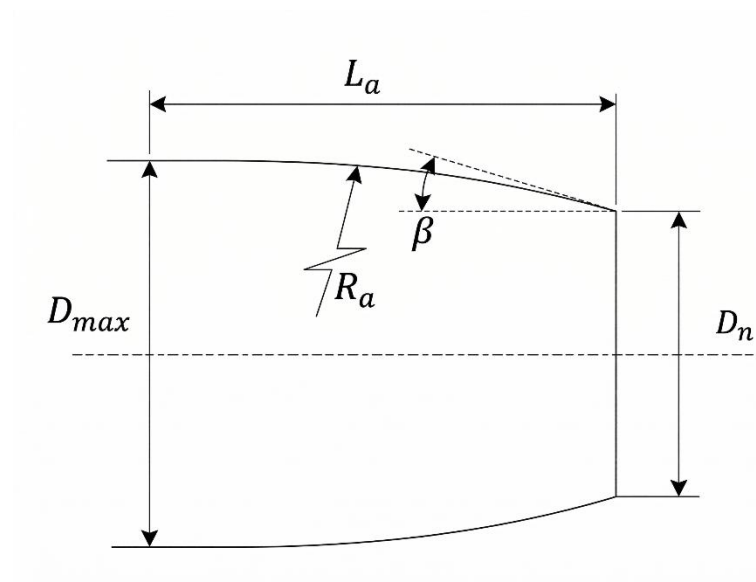


Figure 5-15 : Afterbody key parameters

The boat-tail radius governs the afterbody drag rise Mach number and must be selected carefully. A large R_a delays drag rise and reduces the risk of flow separation in the recompression region near the nozzle exit, but increases the wetted area and associated skin friction drag [43]. Conversely, a small R_a lowers the drag rise Mach number and promotes boundary layer growth and separation along the afterbody. The chord angle β_c — defined as half the boat-tail angle — must remain below 8° to

avoid excessive flow separation, with conventional afterbody designs targeting values between 6° and 8°.

The nozzle exit diameter $D_n = 2.887$ m was determined by reference to the measured bypass nozzle diameter of the A350 family, which uses the same Rolls-Royce Trent XWB engine. The afterbody length L_a was then selected iteratively to achieve a chord angle within the recommended range. The geometric parameters are defined as follows:

$$Y_a = \frac{D_{max} - D_n}{2} = \frac{3.898 - 2.887}{2} = 0.505 \text{ m}$$

$$\beta_c = \arctan \left(\frac{Y_a}{L_a} \right) = \arctan \left(\frac{0.505}{3.850} \right) = 7.48^\circ$$

The resulting chord angle of 7.48° falls within the recommended range of 6° to 8°, confirming that the selected afterbody length of $L_a = 3.850$ m is appropriate. The boat-tail radius is then derived geometrically from the afterbody length and chord angle:

$$R_a = \frac{L_a}{\sin(\beta_c)} = \frac{3.850}{\sin(7.48^\circ)} = 29.586 \text{ m}$$

The afterbody drag rise Mach number $M_{D,AB}$ is then evaluated from R_a as:

$$M_{D,AB} = 1 - \sqrt{\frac{0.04 \cdot D_{max}}{R_a}}$$

Equation 5.6 : Drag rise Mach number correlation for the afterbody [43]

$$M_{D,AB} = 1 - \sqrt{\frac{0.04 \times 3.898}{29.586}} = 0.927$$

This value significantly exceeds the cruise Mach number of 0.80, confirming that the afterbody geometry does not generate wave drag at the design cruise condition and provides a comfortable margin of 0.127 above the cruise point. The afterbody design parameters are summarised in Table 5-4, and since the geometry is axisymmetric, a single set of values applies uniformly to the top, side and bottom sections.

Table 5-4 : Afterbody key characteristics calculation

Characteristic	Relation	Value			Unit
		TOP	SIDE	BOTTOM	
D_{max}	-	3.898	3.898	3.898	m
D_n	-	2.887	2.887	2.887	m
Y_a	$Y_a = (D_{max} - D_n)/2$	0.505	0.505	0.505	m
β_c	$\beta_c = \arctan(Y_a / L_a)$	7.48	7.48	7.48	°
L_a	-	3.850	3.850	3.850	m
R_a	$R_a = L_a / \sin(\beta_c)$	29.586	29.586	29.586	m
M_{d,AB}	$M_{d,AB} = 1 - \text{SQRT}(0.04D_{max}/R_a)$	0.927	0.927	0.927	-

5.2.4.3 Midbody design

The midbody is a short cylindrical section at constant diameter D_{max} connecting the forebody to the afterbody. As discussed in the introduction to this section, modern high bypass ratio engines require only a minimal midbody length, in contrast to older lower bypass ratio designs where a long cylindrical mid-section was necessary to accommodate the structural and geometric constraints of the afterbody. For the BW-25, the total nacelle length was set to $L_{tot} = 6.185$ m by reference to the measured nacelle length of the A350 family, which uses the same Rolls-Royce Trent XWB engine. The midbody length is then derived as the remainder after subtracting the mean forebody and afterbody lengths:

$$L_m = L_{tot} - L_{f,mean} - L_a = 6.185 - 1.785 - 3.850 = 0.550 \text{ m}$$

This represents approximately 9% of the total nacelle length, consistent with the trend toward minimised midbody lengths in modern turbofan nacelle designs. The midbody contributes only skin friction drag to the overall nacelle drag budget, with no wave drag component given its constant diameter profile.

5.2.5 Primary nozzle and plug design

In addition to the external nacelle profile, two internal components were designed at a conceptual level to complete the nacelle geometry: the primary nozzle and the plug. These components were not sized through analytical calculations but were instead shaped by reference to existing turbofan engine architectures observed in the literature and in publicly available design documentation.

The primary nozzle surrounds the core exhaust flow and guides it toward the exit. It features a convergent-divergent internal profile, with an initial diameter of 1.60 m converging to a minimum diameter of 1.34 m before expanding to a final exit diameter of 1.80 m over a total length of 3.27 m. This hourglass geometry, visible in Figure 5-16, is typical of separate-flow turbofan primary nozzles and promotes controlled acceleration and expansion of the core exhaust gases before discharge into the atmosphere.

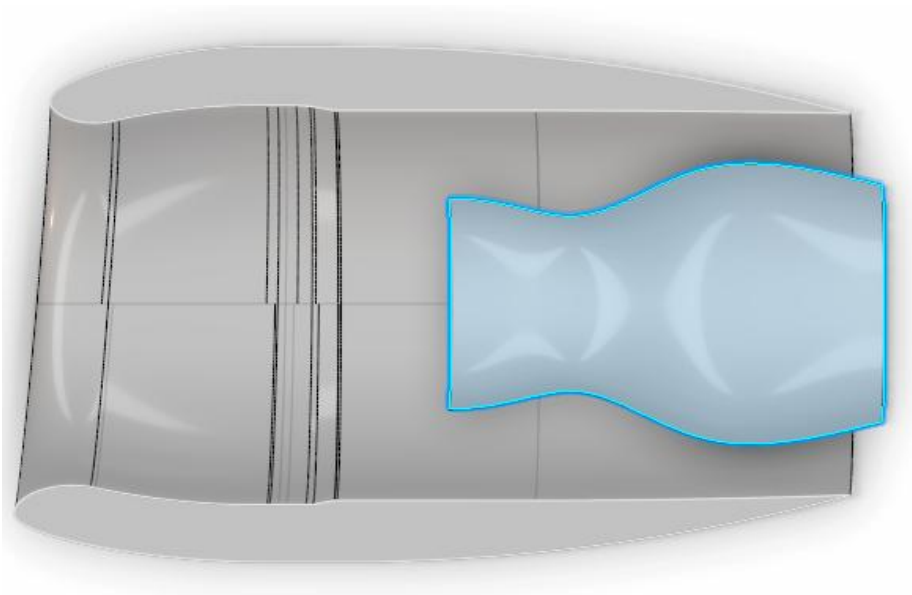


Figure 5-16 : Primary nozzle in blue

The plug is the conical centerbody positioned at the core nozzle exit. It serves to smoothly redirect the core exhaust flow outward and prevent flow separation at the nozzle centreline, reducing base drag and improving propulsive efficiency at the nozzle exit. The plug adopted for the BW-25 features a conical geometry with a maximum diameter of 1.72 m and a total length of 2.25 m, tapering to a point at the

aft end as shown in Figure 5-17. This geometry is consistent with plug designs observed on high bypass ratio turbofan engines of comparable size.

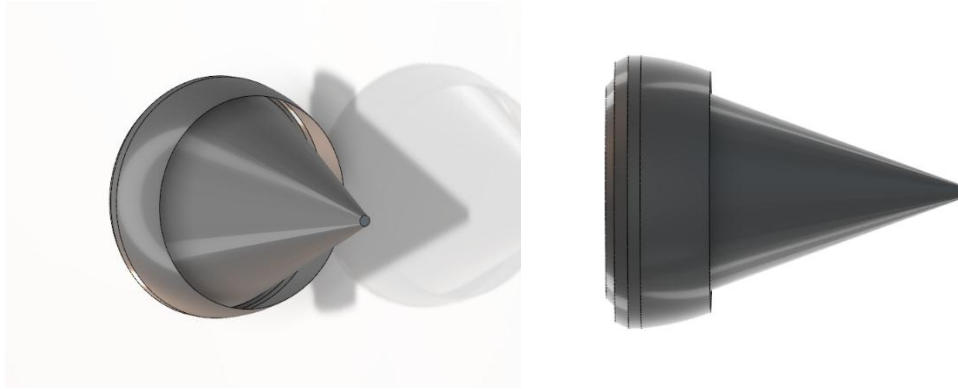
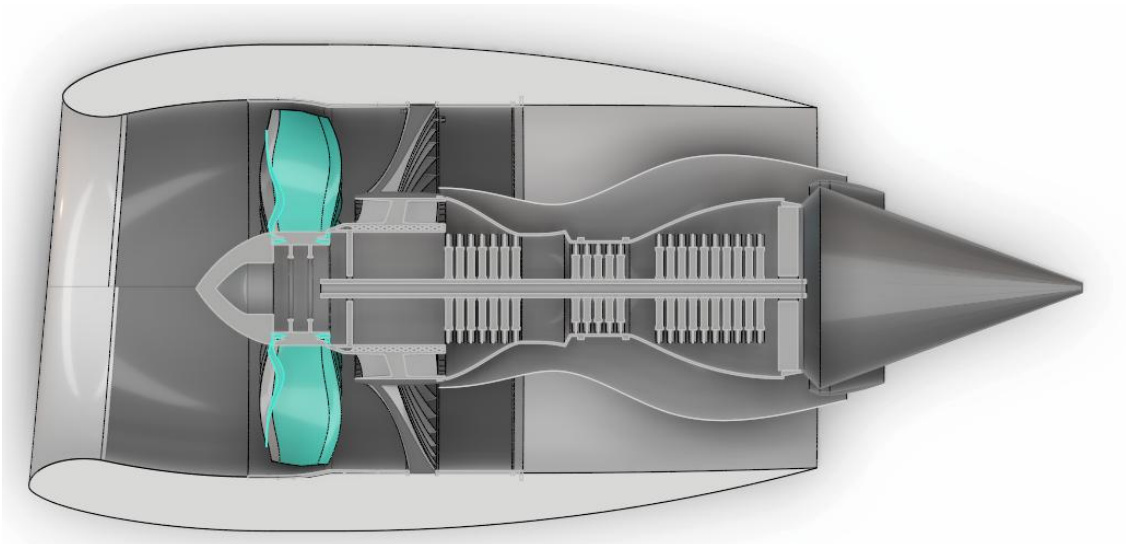
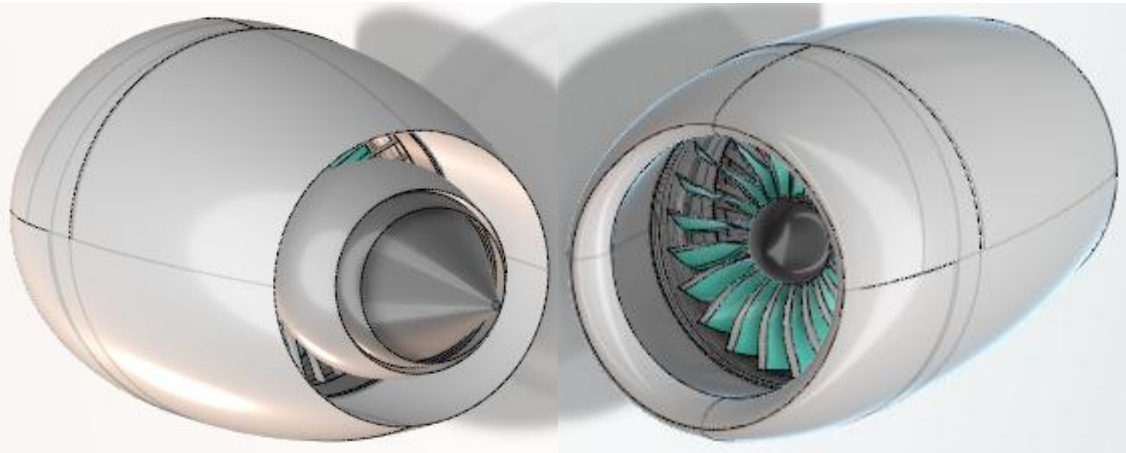
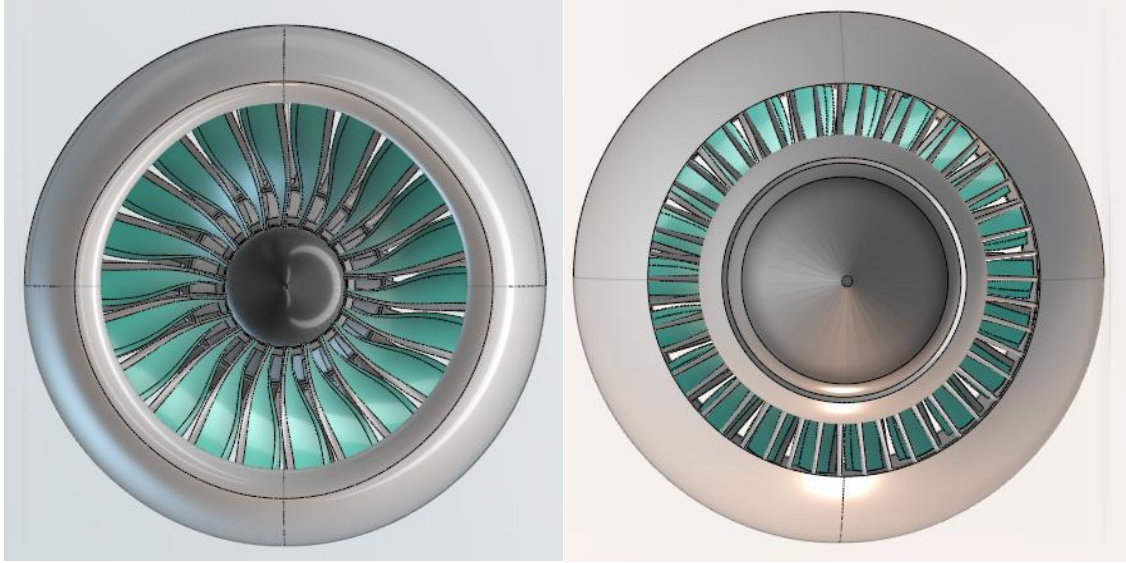


Figure 5-17 : Conical exhaust plug

5.2.6 Final design of the nacelle

The nacelle design presented in this section represents a conceptual-level integration of the propulsion system for the BW-25, combining empirical sizing methodologies with design choices informed by existing high bypass ratio turbofan installations. The overall geometry — a non-axisymmetric separate-flow nacelle with a negatively scarfed inlet, a NACA 1-Series forebody profile, a short midbody, and a circular arc afterbody — reflects a coherent set of trade-offs between aerodynamic performance, acoustic requirements, and the specific constraints of the upper-surface mounting configuration.



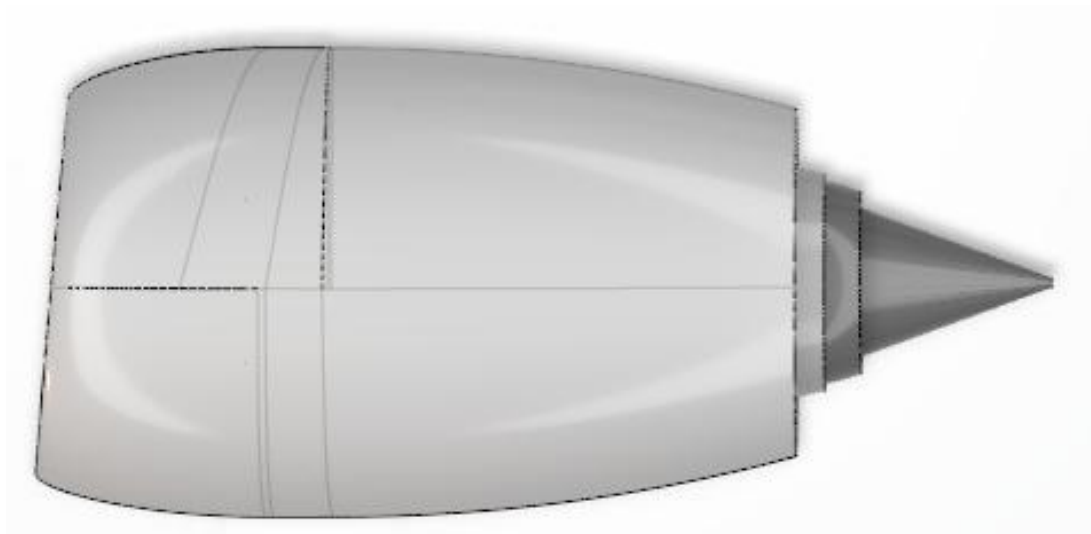


Figure 5-18 : Engine nacelle CAD model

The sizing approach based on Mattingly's empirical correlations, validated through the forebody drag rise Mach number criterion and the inlet total pressure recovery assessment, provides confidence that the principal dimensions are consistent with the aerodynamic requirements of the cruise design point. The comparison with the A350 nacelle dimensions, shown in Figure 5-19, further supports the overall coherence of the design, with the BW-25 nacelle exhibiting similar proportions to the reference aircraft despite the fundamentally different installation geometry.

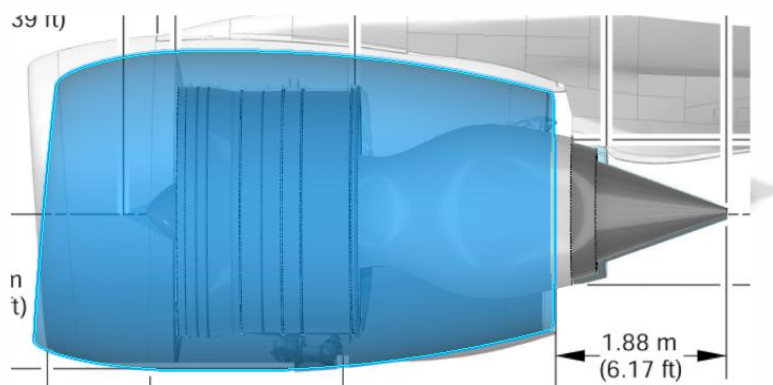


Figure 5-19 : Comparison between the nacelle designed and the A350's nacelle

It should be acknowledged, however, that the present design remains at a conceptual level and a number of aspects would require more detailed analysis in subsequent design phases. In particular, the aerodynamic optimisation of the

forebody and inlet profiles through high-fidelity CFD simulations would be necessary to validate the pressure distributions and refine the lip geometry for each section of the non-axisymmetric nacelle. The interaction between the nacelle external flow and the upper fuselage surface, identified in the literature as a significant source of interference drag for upper-surface mounted configurations, has not been addressed in the present work and represents a key area for future investigation. Similarly, the internal flow quality at the fan face — particularly the total pressure distortion at high angles of attack and under crosswind conditions — would need to be assessed through dedicated inlet flow analysis before the design could be considered mature.

5.3 Drag assessment

The nacelle drag was estimated under on-design cruise conditions following the methodology established by Williams [43], which models each section of the nacelle as an equivalent flat plate under the same Reynolds number and Mach number, with a fully turbulent boundary layer assumption. This approach uses empirical form factors to account for the geometric complexity of each nacelle section, avoiding the need for CFD simulations while providing a reliable first-order drag estimate. The ambient conditions at the cruise design point are summarised in Table 5-5.

Table 5-5 : Drag calculation parameters

Characteristic	Value	Unit
Altitude	45000	ft
Mach	0.8	-
R (gas constant)	287	J/kg*K
Gamma	1.4	-
ρ	2.22E-01	kg/m ³
μ (dynamic viscosity)	1.52E-05	Pa.s
ν (kinematic viscosity)	6.84E-05	m ² .s
Static Temperature	231.65	K
Total Temperature	261.3012	K
Static Pressure	14748	Pa
Total Pressure	22481	Pa

Mass flow	363.447	kg/s
MFP	0.038930121	-
A_{∞}	6.712913565	m ²
L_f	1.785	m
$D_{\max}$	3.898	m
$A_{\max}$	11.934	m ²
$S_{\text{wet,FB}} = \pi * D_{\max} * L_f$	21.85898177	m ²
a (sound of speed)	305.11	m/s
v	259.3435	m/s
Y (forebody thickness)	0.387	m
q_0 (dynamic pressure)	7.46E+03	Pa
Net thrust	69.43	kN
L_m	0.550	m
$S_{\text{wet,MB}}$	6.735	m ²
L_a	3.850	m
$S_{\text{wet,MA}}$	47.147	m ²

5.3.1 Forebody drag

The operating MFR is a key input to the forebody drag calculation as it determines the degree of spillage — the fraction of the incoming stream tube that is diverted around the inlet rather than captured by the engine. The MFR was computed from Mattingly's isentropic mass flow parameter formulation [39], which provides a rigorous thermodynamic definition of the freestream capture area A_{∞} as a function of the cruise Mach number, total temperature, total pressure, and required engine mass flow. The MFR is then obtained as the ratio of A_{∞} to the mean highlight area A_{hl} .

The forebody drag is evaluated following Williams, modelling the forebody as an equivalent flat plate under the same aerodynamic conditions, with a fully turbulent boundary layer assumption. The skin friction coefficient C_f is first obtained from the Prandtl-Schlichting turbulent flat plate correlation:

$$C_f = \frac{0.455}{(\log(Re_{FB}))^{2.58}} \text{ where } Re_{FB} = \frac{\rho \cdot v \cdot L_f}{\mu}$$

Equation 5.7 : Prandtl-Schlichting turbulent flat plate correlation

The skin friction coefficient C_F , which scales C_f by the forebody wetted area relative to the maximum cross-sectional area, is then:

$$C_F = C_f \times \frac{S_{wet,FB}}{A_{max}} \text{ where } S_{wet,FB} = \pi \cdot D_{max} \cdot L_f$$

Equation 5.8 : Skin friction coefficient

The drag coefficient at the ideal condition MFR = 1, denoted $C_{D,0}$, incorporates a form factor accounting for the forebody thickness:

$$C_{D,0} = \left(1 + \frac{2Y}{3L_f}\right)^{5/3} \times C_F \text{ where } Y = \frac{D_{max} - D_{hl}}{2}$$

Equation 5.9 : Drag coefficient at the ideal condition

For MFR < 1, spillage drag adds to the forebody drag through the empirical relation:

$$C_D = C_{D,0} \times [1 + K(1 - MFR)]^{5/3}$$

Equation 5.10 : Spillage drag empirical relation

where K is a geometry-dependent factor correlating cowl and spillage drag to the nacelle geometry:

$$K = 1.75 \left[\left(\frac{D_{max}}{D_{hl}} + 1 \right) \left\{ \left(1 + \frac{3L_f}{D_{max}} \right) \frac{D_{max}}{D_{hl}} - 1 \right\} \right]^{-1}$$

The operating MFR is determined from Mattingly's isentropic mass flow parameter formulation. The Mass Flow Parameter (MFP) is a dimensionless quantity that relates the mass flow rate, total conditions, and flow area through the isentropic flow equations. It is evaluated at the freestream Mach number M_∞ as follows:

$$MFP = M_{\infty} \sqrt{\frac{\gamma}{R}} \left(1 + \frac{\gamma - 1}{2} M_{\infty}^2 \right)^{\frac{1+\gamma}{2(1-\gamma)}}$$

Equation 5.11 : Mass Flow Parameter definition

The freestream stream tube capture area A_{∞} — that is, the cross-sectional area of the air mass that will ultimately be ingested by the engine — is then obtained by applying the definition of the MFP to the freestream flow conditions:

$$A_{\infty} = \frac{\dot{m} \sqrt{T_t}}{P_t \times MFP}$$

Equation 5.12 : Freestream tube capture area

where $\dot{m}$ is the required engine mass flow, T_t and P_t are the freestream total temperature and total pressure respectively. The MFR is finally obtained as the ratio of the freestream capture area to the highlight area, which represents the physical inlet opening:

$$MFR = \frac{A_{\infty}}{A_{hl}}$$

Equation 5.13 : Mass Flow Ratio definition

A value of $MFR < 1$ indicates that the inlet is capturing less air than the approaching stream tube, causing the excess flow to spill around the lip and generating spillage drag in addition to skin friction.

The forebody drag force is finally:

$$D_{FB} = q_0 \times A_{max} \times C_D$$

Equation 5.14 : Forebody drag

The numerical results are summarised in Table 5-6. The drag coefficient was also evaluated at the critical MFR, which represents the operating condition at which the forebody drag reaches its maximum and below which flow separation is likely to occur. The critical MFR is determined from the following geometric correlation from Williams, which relates it directly to the forebody proportions:

$$MFR_{crit} = \left[1 - 4 \times \frac{\left(1 - \frac{D_{hl}}{D_{max}}\right)^2}{\frac{L_f}{D_{max}}} \right]^{5/2}$$

Equation 5.15 : Critical Mass Flow Ratio

As shown in Table 5-6, the drag at $MFR_{crit} = 0.348$ reaches 958.8 N, significantly above the design operating drag of 707.7 N at MFR = 0.876. The BW-25 cruise MFR of 0.876 exceeds the 0.70–0.80 range cited in Section 3.3.2 for conventional inlets, reflecting the higher mass flow demand of the large UHBR engine at altitude. The design MFR is therefore comfortably above the critical value, confirming that the nacelle operates well within the efficient drag regime and is not at risk of flow separation on the forebody. It is also worth noting that drag is minimised at MFR = 1, corresponding to the ideal condition where the entire incoming stream tube is captured by the inlet with no spillage, giving a minimum forebody drag of 653.3 N.

Table 5-6 : Forebody drag calculation

Drag force at MFR = 1				CD0 calculation		
Characteristic	Representation	Value	Unit	Characteristic	Value	Unit
Reynold's number	$Re = (\rho v L_f) / \mu$	6.77E+06	-	D_{max}	3.898	m
Prandtl-Schlichting low speed friction coefficient	$C_f = 0.455 / (\log(Re))^{2.58}$	3.20E-03	-	D_{hl}	3.124	m
Wetted surface/max area ratio	S_{wet} / A_{max}	1.831708568	-	L_f	1.785	m
Skin friction coefficient	$C_F = C_f * (S_{wet} / A_{max})$	5.86E-03	-	$K = 1.75 * [(D_{max}/D_{hl} + 1) * \{(1 + 3 L_f / D_{max}) * (D_{max} / D_{hl}) - 1\}]^{-1}$	0.396729804	-
Drag coefficient for Forebody at MFR= 1	$C_{D0} = (1 + 2Y / 3 L_f)^{5/3} * C_F$	7.34E-03	-			
Drag force at MFR = 1	$D = q_0 * A_{max} * C_{D0}$	653.31	N			
Final forebody drag						
Characteristic	Representation	MFR	Value	Drag	Unit	
C_D	$C_D = C_{D0} * (1 + K * (1 - MFR))^{5/3}$	0.1	0.01221	1086.72344	N	
		0.2	0.01162	1034.29129	N	
		0.3	0.01104	982.90141	N	
		0.4	0.01048	932.56448	N	
		0.5	0.00992	883.29163	N	
		0.6	0.00938	835.09448	N	
		0.7	0.00885	787.98517	N	
		0.8	0.00834	741.97641	N	
		0.9	0.00783	697.08153	N	
		1	0.00734	653.31453	N	
C_D at MFR		0.876012585	0.00795	707.74837	N	
C_D at MFR _{crit}		0.347543541	0.01077	958.83748	N	

5.3.2 Midbody drag

The midbody is a cylinder at constant diameter D_{max} and therefore requires no form factor to establish equivalence with a flat plate — the drag coefficient reduces directly to the skin friction coefficient. However, since the boundary layer that develops over the midbody is a continuation of that initiated at the highlight, the Reynolds number must account for the cumulative length from the highlight:

$$Re_{MB} = \frac{\rho \cdot v \cdot (L_f + L_m)}{\mu}$$

The incremental skin friction coefficient of the midbody is then obtained by subtracting the forebody contribution from the combined forebody-midbody friction coefficient:

$$C_{F,MB} = C_f \left[\frac{S_{wet,FB}}{A_{max}} + \frac{S_{wet,MB}}{A_{max}} \right] - C_{F,FB} \text{ where } S_{wet,MB} = \pi \cdot D_{max} \cdot L_m$$

Equation 5.16 : Skin friction coefficient of the midbody

Since no form factor is required, $C_D = C_{F,MB}$, and the midbody drag force is:

$$D_{MB} = q_0 \times A_{max} \times C_{F,MB}$$

Equation 5.17 : Midbody drag

The numerical results are summarised in Table 5-7 alongside the forebody and afterbody contributions.

Table 5-7 : Midbody drag calculation

Characteristic	Representation	Value	Unit
Reynold's number	$Re = (\rho v (L_f + L_m)) / \mu$	8.86E+06	-
Prandtl-Schlichting low speed friction coefficient	$C_f = 0.455 / (\log(Re))^{2.58}$	3.06E-03	-
Wetted surface/max area ratio	S_{wet} / A_{max}	0.564391996	-
Skin friction coefficient	$C_F = C_f^* [(S_{wet} / A_{max})_{FB} + (S_{wet} / A_{max})_{MB}] - C_{F,FB}$	1.48E-03	-
Drag coefficient for Midbody	$C_D = C_F$	1.48E-03	-
Drag force	$D = q_0 * A_{max} * C_D$	131.56	N

5.3.3 Afterbody drag

The afterbody drag follows the same cumulative boundary layer approach as the midbody, with the Reynolds number now accounting for the full nacelle length:

$$Re_{AB} = \frac{\rho \cdot v \cdot (L_f + L_m + L_a)}{\mu}$$

The incremental skin friction coefficient of the afterbody is obtained by subtracting both the forebody and midbody contributions from the combined friction coefficient over the full nacelle length:

$$C_{F,AB} = C_f \left[\frac{S_{wet,FB}}{A_{max}} + \frac{S_{wet,MB}}{A_{max}} + \frac{S_{wet,AB}}{A_{max}} \right] - C_{F,FB} - C_{F,MB}$$

Equation 5.18 : Skin friction coefficient of the afterbody

Unlike the midbody, the conical taper of the afterbody introduces additional pressure drag associated with the boat-tail geometry, which cannot be captured by a flat plate analogy alone. A form factor is therefore applied in the form of a C_D/C_F ratio, determined from the empirical chart provided by Williams [43], which plots this ratio against the boat-tail chord angle β_c for a range of freestream Mach numbers. At the cruise Mach number of 0.80 and with $\beta_c = 7.48^\circ$ as determined in the afterbody sizing section, the chart yields $C_D/C_F = 1.2$.

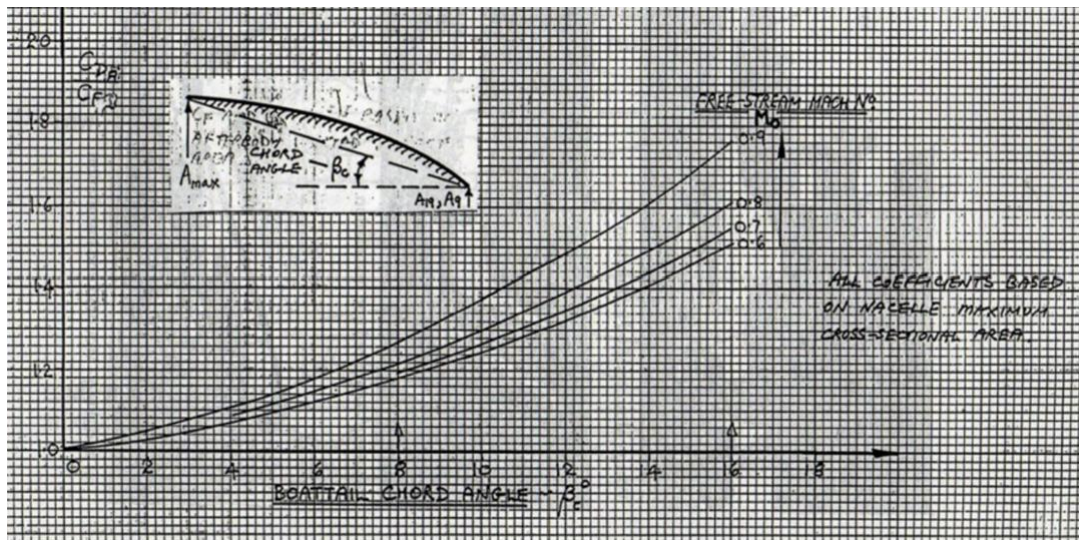


Figure 5-20 : Subsonic afterbody drag relation [43]

The afterbody drag force is therefore:

$$D_{AB} = q_0 \times A_{max} \times 1.2 \times C_{F,AB}$$

Equation 5.19 : Afterbody drag

Table 5-8 : Afterbody drag calculation

Characteristic	Representation	Value	Unit
Reynold's number	$Re = (\rho v(L_f + L_m + L_a)) / \mu$	2.35E+07	-
Prandtl-Schlichting low speed friction coefficient	$C_f = 0.455 / (\log(Re))^{2.58}$	2.63E-03	-
Wetted surface/max area ratio	S_{wet} / A_{max}	3.950743971	-
Skin friction coefficient	$C_F = C_f * [(S_{wet} / A_{max})_{FB} + (S_{wet} / A_{max})_{MB} + (S_{wet} / A_{max})_{AB}] - C_{F,FB} - C_{F,MD}$	9.35E-03	-
Drag coefficient for Afterbody	$C_D = 1.2 * C_F$	1.12E-02	-
Drag force	$D = q_0 * A_{max} * C_D$	998.70	N

5.3.4 Drag estimation results

The drag was calculated at the cruise design point of Mach 0.80 at 45,000 ft, representing the most demanding aerodynamic condition for the nacelle and therefore the appropriate sizing point. The results for one and two engines are summarised in Table 5-9.

Table 5-9 : Total nacelle drag

Characteristic	Drag	Total	Unit
One engine	Forebody	707.75	N
	Midbody	131.56	N
	Afterbody	998.70	N
Two engines	Forebody	1415.50	N
	Midbody	263.11	N
	Afterbody	1997.40	N

The afterbody is the dominant drag contributor, driven by the combination of skin friction over a large wetted area and the additional pressure drag penalty associated with the boat-tail geometry captured by the C_D/C_F form factor. The forebody represents the second largest contribution, reflecting the combined effect of skin friction and spillage drag at the operating MFR. The midbody contributes only marginally to the total, consistent with its short length of 0.550 m — a direct consequence of the modern high bypass ratio engine architecture that minimises the need for a long cylindrical mid-section.

A final validation of the overall nacelle drag level is provided by comparing the total drag to the net thrust. With a net thrust of 138.86 kN, the total nacelle drag of 3,676 N represents approximately 2.65% of the net thrust — a percentage similar to that found, for example, in Williams’s lecture notes, confirming that the nacelle sizing is aerodynamically consistent with the performance requirements of the BW-25. The drag was calculated in the same way for different altitudes for the flight performance responsible, Stephen Kam Pong Law (Law K. P., 2026, unpublished thesis: “Flight Path Performance – En-Route Performance and Emissions Analysis of an Advanced Blended Wing Body, Liquid Hydrogen Fuelled Airliner (BW-25)”; unreferenced). The mass airflow for each altitude from the engine performance responsible (Fleury G., 2026, unpublished thesis: “Engine Performance Design for an Advanced Blended Wing Body, Liquid Hydrogen Fuelled Airliner (BW-25)”; unreferenced) and the python script are given in the Appendix H and Appendix I.

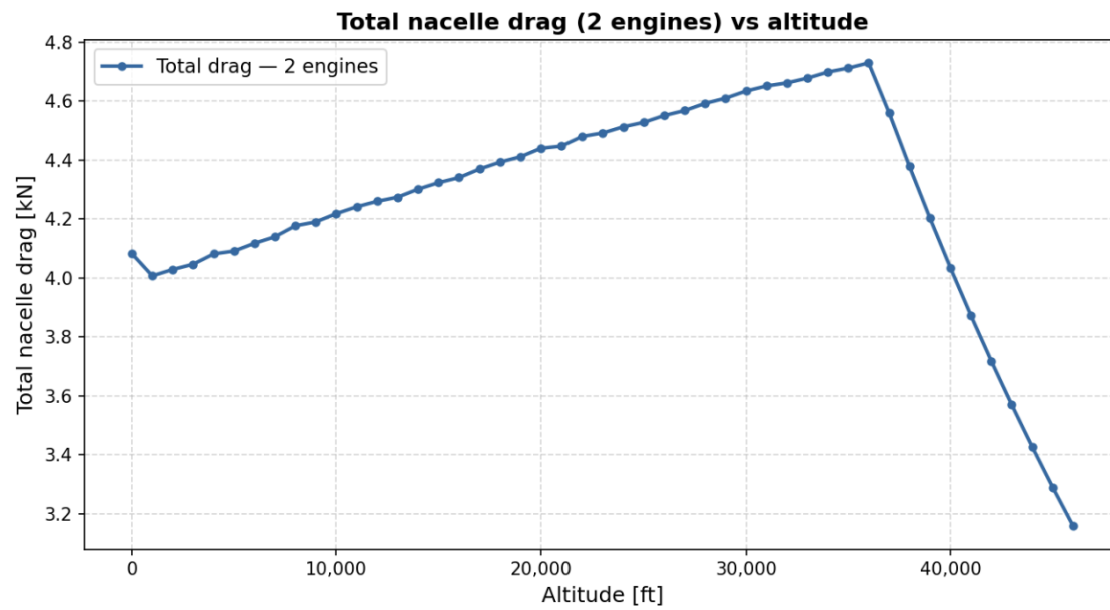


Table 5-10 : Total nacelle drag for every altitude

6 Thrust reverser system

Thrust reversers are a near-universal feature of modern commercial aircraft, and their role in aviation safety and operations has made them a standard expectation among airlines, pilots, and regulatory bodies alike. By redirecting engine exhaust gases forward upon landing, they provide an additional deceleration force that supplements the wheel brakes and spoilers, helping bring the aircraft to a stop within the available runway length.

Despite their widespread use, thrust reversers are not mandated by airworthiness regulations. CS-25 requires that an aircraft demonstrate safe landing performance relying solely on its primary braking systems, meaning that thrust reversers must be treated as a supplementary system rather than a certified means of stopping the aircraft. This distinction is important: the decision to include them is therefore driven not by regulatory necessity, but by operational and economic considerations.

In practice, the benefits they provide are significant enough that virtually all commercial operators have come to rely on them. Their primary contribution lies in reducing the load placed on the wheel brakes during the early, high-speed phase of the landing roll, where braking energy is at its highest. By absorbing part of the deceleration demand during this phase, thrust reversers reduce brake wear and thermal stress, extending brake service life and lowering maintenance costs. They also contribute to faster turnaround times by shortening the landing roll and allowing the aircraft to vacate the runway earlier — an important operational advantage for aircraft performing multiple rotations per day [46].

Finally, thrust reversers provide an additional safety margin in adverse conditions such as wet, icy, or contaminated runways, or in the presence of a tailwind, where the extra deceleration force can make a meaningful practical difference even if it is not formally accounted for in the certification process.

6.1 Feasibility and Interest Study

6.1.1 Landing Performance

Landing distance requirements were to be able to land on the runways of commercial airports. The following landing distances were calculated without thrust reversers by Rafic Sidani, the field performance responsible, in his thesis (Sidani R., 2026, unpublished thesis: "BW-25 VEGA: Airfield Performance Analysis And Flight Control Actuation System Design"; unreferenced):

Table 6-1 : Landing distances for the BW-25

Condition	LDR
Dry runway	968 m
Wet runway	1,196 m
Dry runway + 10-knot tailwind	1,207 m
Wet runway + 10-knot tailwind	1,436 m

Even in the most adverse conditions — wet runway combined with a 10-knot tailwind — the landing distance of 1,436 m remains well within the vast majority of runway distance in the world for large transport aircraft [47]. This confirms that the aircraft meets its landing performance requirements without any reliance on thrust reversers, and that their inclusion is not driven by certification necessity.

6.1.2 Safety

While thrust reversers are not required by CS-25 for normal operations, they provide a meaningful additional safety margin in degraded conditions. On wet, icy or contaminated runways, or in the presence of a tailwind, the wheel brakes alone may experience reduced effectiveness due to lower friction coefficients or increased thermal load. In these situations, the additional deceleration force provided by the reversers acts as a buffer, reducing the risk of a runway excursion.

It should however be noted that thrust reversers also introduce their own safety risks. In-flight deployment due to locking system failure, though rare, has been responsible for fatal accidents in the past. Foreign object ingestion risk is also

increased during reverser operation, as debris may be lifted from the runway surface and drawn into the engine intake [26].

6.1.3 Reliability

The mechanical complexity of thrust reverser systems — involving moving sleeves, blocker doors, actuators and locking mechanisms — has historically been a source of maintenance concerns. Common failure modes include actuator blockage, hydraulic fluid leakage, indicator failures, and difficulties in deployment or stowage. While the probability of a serious in-flight event remains low, the cumulative maintenance burden over the aircraft's service life is non-negligible.

However, the choice of an electrical actuation system, presented in 6.3, significantly mitigates these concerns by eliminating hydraulic components and reducing the number of moving parts, thereby improving overall system reliability.

6.1.4 Weight and Economics

Each thrust reverser system is estimated to weigh approximately 600 kg (roughly 30% of the nacelle weight which is 1535 kg) [48], representing a significant addition to the overall nacelle weight. This weight penalty directly impacts fuel consumption and operating costs throughout the aircraft's service life. However, this must be balanced against the operational savings generated by the system. By absorbing a portion of the braking energy during the high-speed phase of the landing roll, thrust reversers reduce brake wear and thermal stress, lowering the frequency of brake replacements and associated maintenance costs. For an aircraft performing multiple rotations per day, these savings accumulate rapidly and can offset a significant portion of the weight and acquisition cost penalty over the aircraft's lifetime.

Additionally, by shortening the landing roll, thrust reversers allow the aircraft to vacate the runway earlier, reducing runway occupancy time and enabling faster taxi to the gate. In high-traffic airport environments, this improvement in turnaround time has direct commercial value for the operating airlines.

6.1.5 Human Factors

Pilot reliance on thrust reversers is well documented in the literature. Standard airline training programmes incorporate reverser use as a routine part of the landing procedure, and pilots have come to regard them as a familiar and trusted tool, particularly in adverse weather conditions. Removing them entirely would require a significant adaptation of training programmes and operational procedures, and could increase pilot workload and perceived stress during demanding landings.

While it is technically feasible to operate safely without thrust reversers — as confirmed by the landing distance analysis — the human factors dimension represents a practical consideration that supports their inclusion, particularly for an aircraft intended for commercial airline operations where crew confidence and standardisation of procedures are important.

6.1.6 Conclusions

The feasibility study confirms that thrust reversers are not required for certification compliance, as the aircraft meets its landing distance requirements with a substantial margin even in the most adverse conditions. However, their inclusion is justified on operational, economic, and human factors grounds. The reduction in brake wear and temperatures, the improvement in turnaround times, the additional safety margin in degraded conditions, and the alignment with standard airline operating procedures all support the decision to integrate thrust reversers into the nacelle design. The weight penalty of approximately 600 kg per unit is considered acceptable given these benefits.

It should also be noted that thrust reversers have an average deployment time of approximately 3 seconds from activation to full operation. At typical touchdown speeds of around 70 m/s, this means the aircraft covers approximately 210 m before the full reverse thrust force is available. This delay slightly reduces the overall effectiveness of the system and further underlines that thrust reversers are a supplementary deceleration aid rather than a primary braking system. Nevertheless, once deployed, the force levels estimated above are sustained

throughout the high-speed phase of the landing roll down to the 60-knot cutoff, where their contribution is most significant.

6.2 Design choices

6.2.1 Reverser type

The cascade-type thrust reverser was selected for this design without detailed comparison with alternative concepts such as clamshell or bucket-type reversers. This choice is straightforward given the engine configuration: cascade reversers are the established standard for high bypass ratio turbofan engines with short-cowl nacelles, which precisely matches the propulsion system used here. The cascade arrangement redirects the bypass flow through a series of vanes placed circumferentially around the nacelle, providing effective deceleration while remaining compatible with the nacelle geometry and the cold-stream bypass flow that dominates thrust generation in high bypass ratio engines.

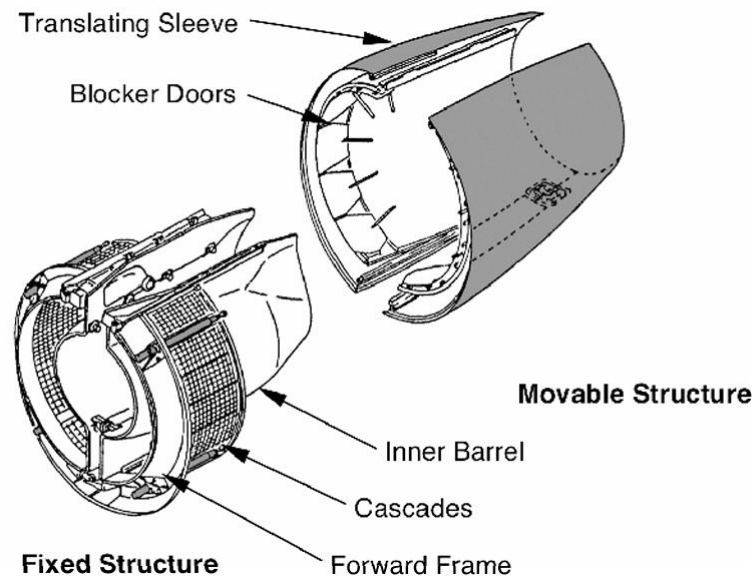


Figure 6-1 : Cascade thrust reverser structure [49]

6.2.2 Deflection angle

A forward velocity angle of $\beta = 60^\circ$ was selected for the cascade vanes. This value is drawn from a CFD study by Yu Gang et al. [18] on the feasibility of cascade thrust reversers applied to a Blended Wing Body aircraft — a configuration directly

comparable to the present design given the over-fuselage engine mounting arrangement. Their results show that at $\beta = 60^\circ$, the reversed flow remains confined to the region behind the reverser cascade with no re-ingestion occurring at the nacelle inlet for Mach numbers above 0.1, which corresponds to approximately 60 knots and covers the entire operational range of the thrust reverser. This allows the system to remain active down to the standard cutoff speed of 60 knots, maximising the deceleration contribution over the full landing roll. A more aggressive angle of $\beta = 45^\circ$ was also evaluated in the same study and shown to increase reverser efficiency marginally, but at the cost of significantly greater re-ingestion risk at low speeds, which could cause engine inlet distortion and instability, and would therefore require an earlier cutoff speed, reducing the overall effectiveness of the system. The 60° angle therefore represents a well-justified balance between reverser effectiveness, operational duration, and engine safety.

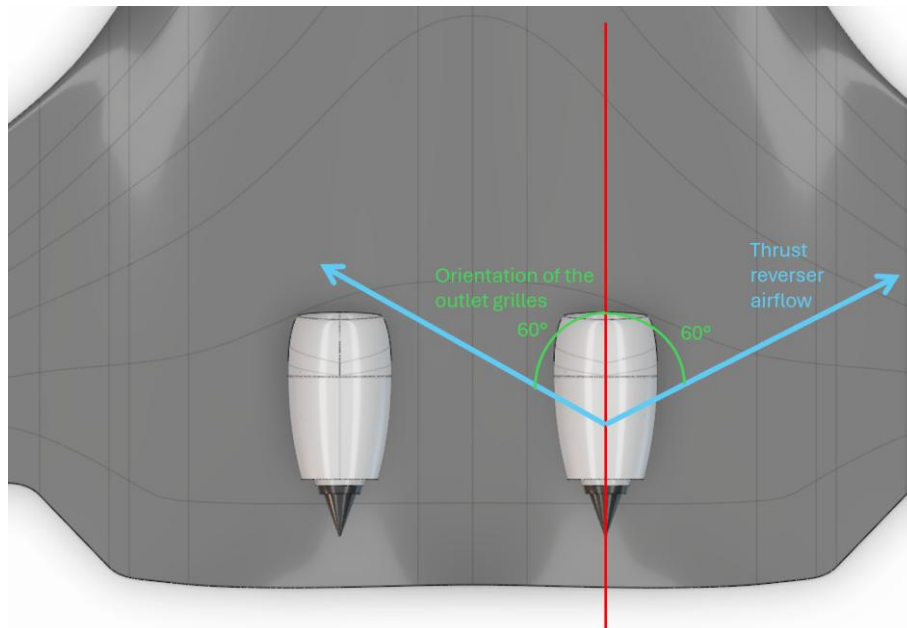


Figure 6-2 : Thrust reversers deflection angle

6.2.3 Blocking half configuration

Rather than deploying the reverser cascade over the full circumference of the nacelle, a blocking half configuration was adopted, in which only the lower half of the bypass flow is redirected. This design choice draws inspiration from the approach used on the C-17 Globemaster III, where a similar arrangement was

implemented to prevent reversed exhaust from lifting runway debris into the engine inlets due to the low engine mounting position. In the present case, the motivation is different but equally important: with the engines mounted above the fuselage in a BWB configuration, deploying the full circumference of the reverser would direct reversed exhaust gases onto the fuselage surface. This would expose structural panels to elevated thermal and aerodynamic loads, risk disturbing the flow over control surfaces, and potentially cause damage to surface-mounted components. Restricting the reverser to the lower half of the nacelle circumference avoids these issues while still providing meaningful deceleration force during the landing roll.

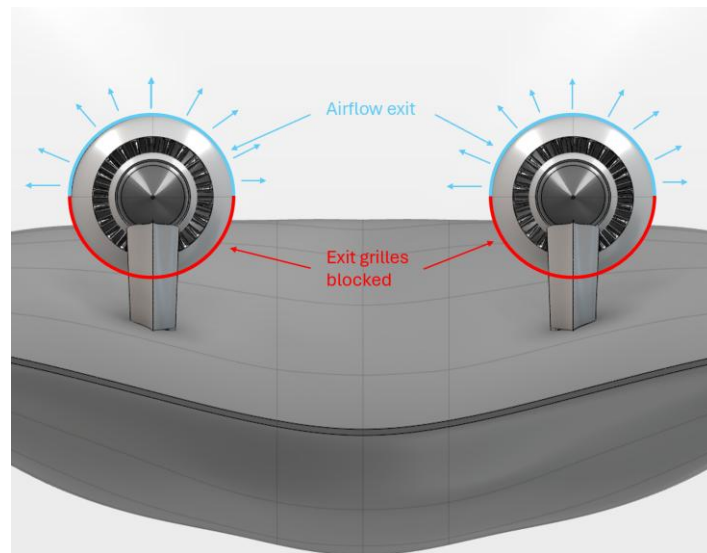


Figure 6-3 : Blocking half configuration illustration



Figure 6-4 : Block half thrust reversers on the C-17 [50]

6.2.4 Reverse Thrust Force Estimation

A preliminary estimation of the reverse thrust force was carried out to quantify the deceleration contribution of the system. The following assumptions were made:

- Each engine has a maximum thrust of 500 kN
- Thrust reversers are activated at 75% of maximum thrust, giving a throttle thrust of 375 kN per engine
- The reverser efficiency η is taken from Yu Gang et al.: $\eta = 0.36$ at $Ma = 0.2$ (activation speed) and $\eta = 0.32$ at $Ma = 0.1$ (approximately 60 knots, the cutoff speed)
- Two cases are considered for the blocking half: a conservative case applying a 0.5 coefficient to account for only half the bypass flow being redirected, and an optimistic case without this coefficient, acknowledging that the actual redirected mass flow depends on the nacelle geometry and flow distribution

The results are summarised in the following table:

Table 6-2 : Thrust reversers force

Condition	η	Blocking coefficient	Reverse thrust per engine	Total reverse thrust (2 engines)
Activation ($Ma = 0.2$) — optimistic	0.36	1.0	135 kN	270 kN
Activation ($Ma = 0.2$) — conservative	0.36	0.5	67.5 kN	135 kN
Cutoff ($Ma = 0.1$) — optimistic	0.32	1.0	120 kN	240 kN
Cutoff ($Ma = 0.1$) — conservative	0.32	0.5	60 kN	120 kN

The total reverse thrust therefore ranges between 120 kN and 270 kN depending on the assumptions made. Even in the most conservative case, this represents a meaningful deceleration force — approximately 1% of the maximum total engine

thrust — applied during the high-speed phase of the landing roll where it is most effective. It should be noted that these figures constitute a preliminary estimate only. The blocking half coefficient of 0.5 is a simplifying assumption, and the actual redirected mass flow would need to be determined through detailed aerodynamic analysis of the nacelle geometry. Nevertheless, these results confirm that the thrust reverser system, as designed, is capable of providing a significant contribution to landing deceleration.

6.3 Actuation System — E-TRAS

The thrust reverser actuation system selected for this design is the Electrical Thrust Reverser Actuation System (E-TRAS), developed by Safran and already proven in service on the Airbus A380, where it powers the two inboard engine reversers. This choice aligns with the More Electric Aircraft philosophy, which seeks to replace traditional hydraulic and pneumatic systems with electrically powered equivalents wherever possible, reducing the number of fluid systems on board and their associated weight, maintenance burden, and failure modes.

In conventional thrust reverser designs, the translating sleeves and blocker doors are driven by hydraulic actuators fed by the aircraft's central hydraulic network. This approach requires dedicated hydraulic lines routed to the nacelle, pressurised fluid circuits, and associated sealing and pumping components — all of which add weight, complexity, and potential failure points. The E-TRAS replaces these with Electro-Hydrostatic Actuators (EHAs), which are self-contained units powered directly by electrical wiring. Each actuator contains its own local hydraulic circuit, eliminating the need for centrally routed hydraulic lines to the nacelle while retaining the force output characteristics of conventional hydraulic actuation.

The key advantages of this approach for the present design are the following. First, the elimination of hydraulic fluid within the nacelle removes the risk of corrosive or flammable fluid leaks in a thermally demanding environment. Second, EHA-based systems have demonstrated improved reliability compared to conventional hydraulic actuation, with reduced common-mode failure risks and simplified maintenance procedures — no bleeding or disassembly of actuators is required

during routine servicing. Third, the speed of electrical signal transmission makes the system more responsive, which is particularly relevant given the 3-second deployment time constraint discussed previously. Fourth, the system is fully lockable at any position, which addresses one of the main safety concerns associated with thrust reversers, namely the risk of inadvertent in-flight deployment.

The main limitations of E-TRAS are its dependency on the reliability of the electrical power supply, the need for additional cooling to manage heat dissipation from the power electronics, and the potential for common-mode failures with other electrical systems on board — though the probability of such events is considered low. High operating temperatures in the nacelle environment also require careful thermal management of the actuator units.

Overall, the E-TRAS system represents a mature, flight-proven solution that is well-suited to the operational profile and design philosophy of this project. Its adoption on the A380 — one of the largest and most complex commercial aircraft ever built — provides strong evidence of its reliability and scalability, and supports its selection here with confidence.

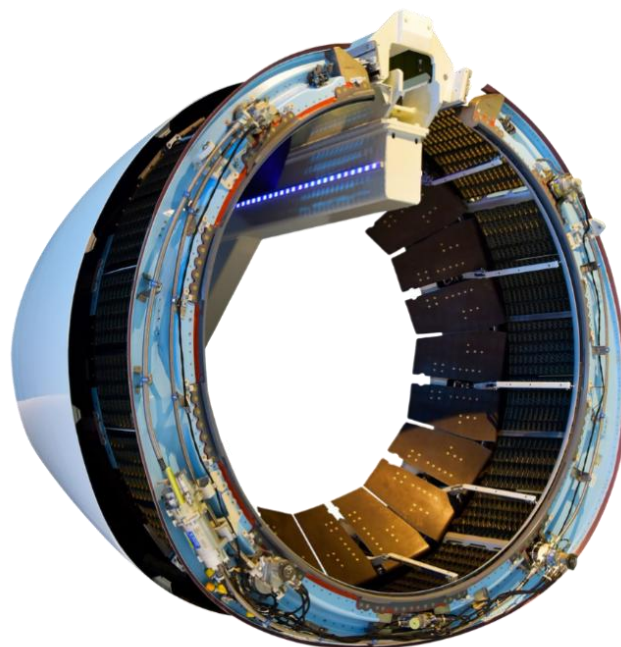


Figure 6-5 : E-TRAS thrust reverser [51]

7 Noise Mitigation

Aircraft noise certification is governed by ICAO Annex 16 [19], which defines strict limits at three reference measurement points around the airport — sideline, flyover, and approach — and whose requirements are expected to tighten significantly in the coming decades as aviation activity continues to grow. For a BWB aircraft with upper-surface mounted engines, the nacelle design plays a central role in meeting these requirements, as it sits at the intersection of the two dominant noise sources in a turbofan installation: fan noise, which propagates both forward through the inlet and rearward through the bypass duct and dominates during approach, and jet mixing noise, generated by the shear between the exhaust flow and the surrounding air, which dominates during takeoff. A layered approach to noise mitigation was therefore adopted, combining passive architectural advantages with active acoustic treatment integrated into the nacelle structure.

7.1 Acoustic Liners

Acoustic liners were selected as the primary active noise mitigation technology and are integrated into both the inlet cowl and the fan duct of the nacelle. These honeycomb-structured panels consist of a porous facing sheet over a cellular core backed by a solid sheet [21]. They absorb acoustic energy through viscous losses within the cells and behave as Helmholtz resonators, attenuating noise within a frequency range determined by the cell geometry and depth. When integrated into the inlet, they address fan noise propagating forward through the intake; when integrated into the fan duct, they attenuate the rearward fan noise component that dominates during approach. Their integration into the nacelle is straightforward, they add minimal complexity, and they represent the most mature and widely validated noise mitigation technology available for turbofan installations [52].

To broaden the absorption frequency range beyond what a single-layer liner can achieve, multi-degree-of-freedom configurations — consisting of superimposed layers with different cell depths — can be used to address both the tonal and broadband components of fan noise simultaneously. This is particularly relevant for a high bypass ratio engine, where the fan tonal noise and broadband noise

contributions both represent significant certification concerns. The impedance of the liners must be carefully tuned to the specific engine and operating conditions, and their optimal distribution within the inlet and fan duct requires dedicated acoustic design work that falls beyond the scope of this project. Nevertheless, their inclusion as a design feature is well-justified and consistent with standard industry practice on all modern turbofan nacelles.

One specific consideration for the BWB upper-surface installation is that the acoustic propagation environment within the inlet differs from conventional underwing geometries, due to the modified refraction effects of the boundary layer flow over the BWB surface. Whether existing liner impedance design methodologies remain fully valid under these conditions is not clearly established in the literature, and this represents a design uncertainty that would need to be addressed at a more detailed stage of the project through dedicated acoustic analysis.

7.2 Chevrons — Not Retained

Chevron nozzles were evaluated as a potential complement to the liner treatment. Chevrons are serrated or triangular-toothed features at the nozzle exit that function by inducing streamwise vortices into the jet shear layer, breaking down the large-scale turbulent structures responsible for low-frequency jet noise and promoting faster mixing between the bypass flow, core flow, and surrounding air. Computational studies have demonstrated noise reductions of up to approximately 2–3% in acoustic power levels depending on chevron geometry, with round-edged designs showing the best performance among those evaluated [15].

However, the decision not to retain chevrons on this design was made on the basis of two main considerations. First, chevrons introduce a non-negligible aerodynamic drag penalty due to the streamwise vortices they generate, which directly impacts specific fuel consumption — a particularly significant drawback for an aircraft whose design philosophy is oriented toward efficiency and environmental performance. Second, and more fundamentally, the combination of passive airframe shielding, scarfed inlet geometry, and acoustic liner treatment in both the inlet and fan duct was assumed to provide sufficient noise attenuation to meet certification

objectives without requiring the additional complexity and performance penalty of chevrons. This reasoning is consistent with the approach taken on the A320NEO, where the jet noise level was considered sufficiently reduced by other means for chevrons to not justify their aerodynamic cost [52]. Chevrons therefore remain an option that could be revisited at a later design stage if more detailed acoustic analysis were to reveal a compliance gap, but they are not part of the baseline nacelle design.

8 Engine accessories

Beyond the core propulsion components, a number of auxiliary systems must be accommodated within the nacelle envelope to ensure correct engine operation and to supply power to the aircraft. Although their detailed design falls outside the scope of this project, understanding their function and power requirements is necessary to make informed nacelle sizing decisions and to provide the secondary power responsible accurate loading estimates.

8.1 Architecture Overview

A more electric architecture was adopted for the accessory systems, consistent with the overall design philosophy of the project. No bleed air is extracted from the engine core — all systems that would conventionally rely on pneumatic power are instead powered electrically. This avoids the efficiency penalty associated with diverting high-pressure, high-temperature air from the compression stages and contributes to reduced specific fuel consumption. Engine starting and electrical power generation are both handled by Variable Frequency Starter-Generators (VFSGs), an architecture already demonstrated in service on the Boeing 787. However the oil pumps are driven directly with the shaft power from the gearbox. The accessory gearbox, driven by the engine shaft, connects the VFSGs and any remaining shaft-driven accessories, and its casing must be designed with robust sealing to prevent oil leakage within the nacelle fire zone [25].

8.2 Engine Starting

During engine start, electrical power supplied by the APU drives the VFSGs in motor mode, spinning the HP spool via the accessory gearbox until the engine reaches self-sustaining speed. The power required for this operation was estimated by modelling the HP spool as a hollow cylinder and calculating the power needed to accelerate it from rest to 20% of its maximum speed in 30 seconds — a typical target for engine light-off.

The moment of inertia of a hollow cylinder is given by:

$$I = \frac{m}{2} (R_{ext}^2 + R_{int}^2)$$

Equation 8.1 : Moment of inertia of a hollow cylinder

With a spool mass of 600 kg, an outer radius of 0.50 m and an inner radius of 0.35 m:

$$I = 300 \times (0.50^2 + 0.35^2) = 111.75 \text{ kg.m}^2$$

The angular acceleration required to reach a target speed $\omega_{target} = 0.2 \times N_{max}$ in a time t_{start} is:

$$\alpha = \frac{\omega_{target}}{t_{start}} = \frac{0.2 \times N_{max} \times 2\pi}{t_{start} \times 60}$$

Equation 8.2 : Angular acceleration for the engine start

With a maximum HP spool speed of $N_{max} = 12,600$ rpm and $t_{start} = 30$ s:

$$\alpha = \frac{0.2 \times 12600 \times 2\pi}{30 \times 60} = 8.8 \text{ rad.s}^{-2}$$

The mean angular velocity during the acceleration phase, assuming constant angular acceleration from rest, is:

$$\bar{\omega} = \frac{\omega_{target}}{2} = \frac{0.2 \times N_{max} \times \pi}{60}$$

Equation 8.3 : Mean angular velocity during the acceleration phase

$$\bar{\omega} = \frac{0.2 \times 12600 \times \pi}{60} = 132 \text{ rad.s}^{-1}$$

The mechanical power required to sustain this acceleration is the product of the torque and the mean angular velocity:

$$P_{start} = I \cdot \alpha \cdot \bar{\omega}$$

Equation 8.4 : Power required for the engine start

$$P_{start} = 111.75 \times 8.8 \times 132 \approx 130 \text{ kW per engine}$$

For both engines starting simultaneously, the total starting power demand is approximately 260 kW, which sets the minimum power requirement for the APU and the VFSG sizing.

8.3 Oil System

The engine oil system operates as a closed recirculatory loop in which Mobil Jet Oil II — a synthetic polyol ester oil qualified to MIL-PRF-23699 — is continuously circulated through the engine to provide lubrication and cooling to all rotating components including bearings and gears [53]. Its key properties are summarised in Table 8-1:

Table 8-1 : Properties of the Mobil Jet Oil II

Property	Value
Density	1000 kg/m ³
Specific heat capacity	2100 J/kg·K
Operating temperature range	-54°C to +204°C

Each engine is equipped with one shaft-driven feed pump and seven shaft-driven scavenge pumps, consistent with the distribution of oil collection points across the engine: front fan bearings, IPC bearings, HPC bearings, HPT bearings, IPT bearings, LPT bearings, and the accessory gearbox. Both pumps are mechanically driven by the accessory gearbox, which is itself driven by the HP spool. As a consequence, the oil flow rate is directly proportional to engine speed, which leads to over-pumping at high power settings. A bypass regulation valve is therefore required to return excess flow to the tank and maintain the feed pressure within acceptable limits across the flight envelope. The scavenge flow rate is taken as three times the feed flow rate to account for the air/oil mist that forms in the bearing cavities and significantly increases the volume to be evacuated relative to the supplied oil volume.

The shaft power extracted by the oil pumps was estimated from the mechanical heat dissipation in the bearings and gearbox. The total engine thermal power at each

flight phase was first derived from the fuel mass flow rate and the lower heating value of liquid hydrogen (LHV = 120 MJ/kg):

$$P_{thermal} = \dot{m}_{fuel} \times LHV \text{ where } \dot{m}_{fuel} = SFC \times F$$

The thermal power was computed from the propulsion performance table where data is available: the SLS condition (F = 500.31 kN, SFC = 3.0943 g/kN/s) was used for take-off, FL200 (F = 193.83 kN, SFC = 5.0246 g/kN/s) as a representative climb point, and FL450 (F = 69.43 kN, SFC = 5.9688 g/kN/s) for cruise. For the ground, approach, and landing phases, performance data at low thrust settings is not available in the table provided. The SLS condition was therefore used as a conservative upper bound for these phases — the actual thrust at ground idle, approach, and landing being significantly lower than maximum take-off thrust, the true pump power at these phases will be lower than the values presented here.

Bearing and gearbox losses were taken as 0.75% of total engine thermal power as a conservative upper bound applicable across all flight phases. The heat to be evacuated by the oil circuit is therefore:

$$Q = 0.75\% \times P_{thermal}$$

The required oil mass flow rate was then derived from the thermal balance equation:

$$\dot{m} = \frac{Q}{C_p \cdot \Delta T}$$

Equation 8.5 : Required oil mass flow for the oil pumps

Where Q is the heat to be evacuated, $C_p = 2100\text{J/kg}\cdot\text{K}$, and $\Delta T = 100\text{K}$ (oil inlet temperature of 60°C, outlet temperature of 160°C). The pump power was then calculated as:

$$P_{pump} = \frac{\dot{m} \cdot \Delta P}{\rho \cdot \eta}$$

Equation 8.6 : Pump power

With an operating pressure of 4 bar and a pump efficiency of $\eta = 0.65$. Since each engine is equipped with one feed pump and seven scavenge pumps, the total pump power per engine is obtained by summing the contribution of each pump individually:

$$P_{engine} = P_{feed} + 7 \times P_{scavenge,pump}$$

The scavenge pump power is calculated using the same expression as Equation 8.6, applied to the flow rate per scavenge pump. Since the total scavenge flow rate is three times the feed flow rate, the flow rate handled by each of the seven scavenge pumps is:

$$\dot{m} = \frac{3 \times \dot{m}_{feed}}{7}$$

The results across all flight phases are summarised in Table 8-2:

Table 8-2 : Oil pump system characteristics

Flight phase	P _{thermal} 2 engines (MW)	Source	Q 2 engines (kW)	$\dot{m}_{feed}$ pump (kg/s)	$\dot{m}_{scavenge}$ pump (kg/s)	P total 2 engines (kW)
Ground	185.9	SLS — upper bound	1394	6.638	2.845	32.7
Take-off	185.9	SLS — table	1394	6.638	2.845	32.7
Climb	116.8	FL200 — table	876	4.171	1.787	20.6
Cruise	49.7	FL450 — table	373	1.776	0.761	8.75
Approach	185.9	SLS — upper bound	1394	6.638	2.845	32.7
Landing	185.9	SLS — upper bound	1394	6.638	2.845	32.7

The dimensioning case is take-off, where the total shaft power demand for both engines reaches approximately 32.7 kW. The use of SLS conditions as an upper

bound for ground, approach, and landing phases ensures conservative pump sizing across the full flight envelope, while the true power at these low-thrust conditions will be significantly lower.

8.4 Cooling via Liquid Hydrogen

The cryogenic properties of liquid hydrogen are exploited as a heat sink for the thermal management of the engine. Rather than using dedicated cooling systems as in conventional kerosene-fuelled turbofans, the LH₂ absorbs heat from various engine components before reaching the combustor. The detailed design of this cooling architecture — covering the fuel feed circuit, the ECS, the exhaust ring, the turbines, and the combustion chamber thermal interfaces — falls outside the scope of this thesis and is addressed in the work of D. Barnett, responsible for the fuel feed system design within this project (Barnett D., 2026, unpublished thesis: " Fuel System Transfer and Feed Design for an Advanced Blended Wing Body, Liquid Hydrogen Fuelled Airliner (BW-25)"; unreferenced).

8.5 Power Loads Summary

Beyond the mechanical systems, a number of electrical systems within the nacelle contribute to the overall power offtake from the engines. Since the oil pumps are shaft-driven, three categories of purely electrical nacelle loads were identified and estimated: the E-TRAS thrust reverser actuation system, the engine sensor network, and the FADEC.

The E-TRAS system employs Double-sided Linear Induction Motor (DLIM) actuators, selected on the basis of their superior thrust-to-weight ratio of 132.3 lbf/lbm and thrust-to-power ratio of 174.5 lbf/kW compared to alternative linear motor designs [54]. Each reverser is equipped with three actuators, each rated at 47 kW during active deployment. The peak power demand during thrust reverser deployment is therefore:

$$P_{eTRAS} = 3 \times 47 \text{ kW} \times 2 \text{ nacelle} = 282 \text{ kW}$$

This is a transient demand occurring only during the landing roll for a duration of a few seconds. In standby mode, the power consumption drops to approximately 100 W for both nacelles combined.

The engine sensor network covers pressure, temperature, rotational speed, vibration, and flow measurements across the engine. Based on an estimated 75 sensors per engine with an average consumption of 3 W per sensor [55]:

$$P_{sensors} = 75 \times 3 \text{ W} \times 2 \text{ engines} = 450 \text{ W}$$

The FADEC manages all engine control functions in real time through two fully redundant channels. Its power consumption covers the processing units, communication interfaces, and command actuators including fuel metering valves and variable geometry systems. The estimated consumption per engine is 250 W, giving [56]:

$$P_{FADEC} = 250 \text{ W} \times 2 \text{ engines} = 500 \text{ W}$$

The complete nacelle power load budget across all flight phases is presented in Table 8-3:

Table 8-3 : Nacelle power requirements

Flight phase	Shaft power	Electrical power				Total (kW)
	Oil pumps (kW)	Engine starting (kW)	E-TRAS (kW)	Sensors (kW)	FADEC (kW)	
Ground	32.7*	260	0.10	0.45	0.50	293.75
Take-off	32.7	0	0.10	0.45	0.50	33.75
Climb	20.6	0	0.10	0.45	0.50	21.65
Cruise	8.75	0	0.10	0.45	0.50	9.80
Approach	32.7*	0	0.10	0.45	0.50	33.75
Landing	32.7*	0	282.10	0.45	0.50	315.75

*SLS upper bound — actual shaft power at this flight phase will be lower.

Two important observations follow from this budget. First, the normal operating load ranges from approximately 9.8 kW at cruise to 33.8 kW at take-off, dominated in all phases by the shaft power extracted by the oil pumps. The purely electrical load outside of transient events remains negligible at approximately 1.05 kW across all flight phases. Second, the two peak demands are non-concurrent: the engine starting peak of 260 kW occurs on the ground before departure, while the E-TRAS deployment peak of 282 kW occurs only during the landing roll for a few seconds, bringing the total nacelle load to a maximum of 315.75 kW at that instant — a conservative upper bound given that the true oil pump power at landing is lower than the SLS value used here. The fact that these two worst-case transient loads do not overlap significantly simplifies the dimensioning of the APU, the VFSGs, and the aircraft power distribution network. The nacelle power load budget presented here was shared with Ramón Máiquez Pérez, responsible for secondary power, for the sizing of the electrical generation system (Máiquez Pérez R., 2026, unpublished thesis: "BW-25 VEGA, Design of Secondary Power System and APU"; unreferenced).

9 CAD Team

9.1 Summary

The CAD work was carried out by a dedicated team of eight responsible for the global aircraft model throughout the project. The team's primary deliverables were the development of the aircraft surface model through successive iterations, the progressive integration of subsystem assemblies into a common CAD environment, and the maintenance of a CAD Rules Book defining modelling guidelines, naming conventions, colour coding standards, and assembly procedures for all project contributors. The CAD team acted as the backbone of the project's geometric definition, ensuring that the work of all discipline teams could be assembled into a coherent and consistent global model.

9.2 CAD Rules and Standards

To ensure consistency across a large and diverse team of contributors, a common set of CAD rules was defined and maintained throughout the project. A naming convention was established in which each part name includes the project name, a JASC code identifying the system or structure to which the part belongs, and a short description of the part. This convention allowed any team member to immediately identify the origin and function of any component in the assembly.

A colour coding system was also defined, assigning a specific colour to each aircraft system and structure. This allowed rapid visual identification of components within the global model and significantly improved readability during integration reviews and interface checks. Both the naming convention and the colour system were documented in the CAD Rules Book, which was distributed to all project teams and updated as the project evolved.

9.3 Integration Process

The integration workflow followed a structured process designed to manage the contributions of multiple teams working in parallel. Each contributor modelled their parts in their own collaborative space before submitting them to the CAD team for integration into the common assembly. Upon reception, the CAD team performed a

series of checks covering compliance with the naming convention, correct colour coding, geometric consistency with adjacent parts, and adherence to the modelling standards defined in the Rules Book.

When an interface issue was identified — for example a clash between two adjacent components, a gap at a structural junction, or a dimensional inconsistency with the specification booklet — it was logged using a dedicated interface issue form and communicated to the relevant team for correction. The CAD team tracked these issues through to resolution before validating the part into the assembly and releasing a new working version. This interface management function became increasingly important as the number of submitted parts grew, and represented a significant coordination effort throughout the project.

9.4 Guidance and Support to Project Teams

Beyond the management of the common assembly, the CAD team provided active support to other project members to help them correctly dimension and position their components. Many contributors were working in CAD for the first time in a collaborative multi-team environment, and the CAD team played a mentoring role — advising on modelling best practices, helping teams interpret the geometric constraints from the specification booklet, and assisting in the correct placement of components relative to the aircraft reference frame. This support was particularly important for teams whose components had tight spatial constraints, such as systems located within the fuselage or in close proximity to structural members.

9.5 Surface Modelling and Engine Integration

The surface model of the aircraft was developed through five successive iterations, each refining the geometry of the fuselage, wing blending, and integration areas based on feedback from the aerodynamics and structures teams. The challenge of interpolating between spanwise airfoil sections and longitudinal curves from the specification booklet — while maintaining surface continuity and minimising tangency and curvature discontinuities — drove much of the iterative work. A NACA airfoil fitting algorithm was developed to assist this process, using cubic

interpolation to find the closest NACA 4-digit profile to a set of intersection points. While this approach was ultimately not retained for the fuselage — where manual interpolation more closely followed the specification sections — it provided useful geometric insight during the early iterations.

10 Integrated propulsion system architecture

The integrated propulsion system architecture combines the engine core, fuel, lubrication, power, and control subsystems into a unified framework, as detailed in the architecture diagram provided in Appendix K. Liquid hydrogen is drawn from the tanks by a cryogenic pump and progressively warmed through three heat exchangers — cooling the turbines, the ECS supply, and the exhaust gases in sequence — before reaching the combustor as gaseous hydrogen. The oil circuit is cooled by a dedicated GH_2 -oil heat exchanger, closing the thermal loop between the fuel and lubrication systems. A Variable Frequency Starter-Generator, driven through the accessory gearbox, handles both engine starting and in-flight electrical power generation. The cascade thrust reverser is deployed by an AC electric motor via ball-screw actuators, and the FADEC supervises the entire system through distributed spool speed, temperature, and pressure sensor networks.

11 Conclusion and Further Work

11.1 Conclusion

This thesis has addressed the conceptual integration of a liquid hydrogen turbofan propulsion system into a Blended Wing Body aircraft, covering engine mounting configuration selection, nacelle aerodynamic design, safety assessment, thrust reverser system design, noise mitigation strategy, engine accessories sizing, and CAD integration.

The safety assessment captured 50 propulsion integration requirements from EASA CS-25, CS-E, and the project green book. A Functional Hazard Analysis covering 37 failure conditions informed design priorities, and two Fault Tree Analyses confirmed the adequacy of the safety architecture for uncontained rotor failure and uncommanded thrust reverser deployment. The rotor burst analysis confirmed that all fuel tanks are positioned outside the hazard zones across all fragment categories. The probabilistic assessment demonstrated that the 10-metre engine spacing keeps the dual-engine loss probability below 10^{-9} per flight hour across all flight phases, providing a quantitative validation of the engine layout.

The pylon-mounted configuration was selected and the pylon height set at 45 cm based on an analytically estimated boundary layer thickness of 39 cm at take-off rotation. The nacelle was designed as a non-axisymmetric separate-flow configuration with a 5° scarfed inlet, validated through inlet total pressure recovery of $\pi_d = 0.954$ and a forebody drag rise margin of 0.037 above cruise Mach number. Nacelle drag of 1,838 N per engine represents 2.65% of net thrust in cruise condition.

The thrust reverser feasibility study has been conducted, and a cascade-type E-TRAS system with 60° deflection angle and blocking half configuration was designed and sized, yielding between 120 kN and 270 kN of reverse thrust. The noise mitigation strategy combined BWB passive shielding, scarfed inlet geometry, and acoustic liner treatment. Engine accessories sizing yielded a starting power demand of 260 kW, an oil system dimensioning case of 33.8 kW at take-off, and a nacelle electrical load

budget peaking at 316 kW during thrust reverser deployment — non-concurrent with the starting peak.

Taken together, these results demonstrate that the integration of a large high-bypass ratio LH₂ turbofan on an upper-surface BWB configuration is conceptually feasible and compliant with the applicable regulatory requirements, providing a sound basis for progression to a preliminary design phase.

11.2 Further Work

Several areas require more detailed investigation before the design can progress to preliminary form.

The pylon height determination relied on a flat plate boundary layer correlation that neglects pressure gradients and three-dimensional effects. Higher-fidelity CFD analysis is recommended to confirm the 45 cm margin and to quantify inlet distortion at the fan face under take-off rotation conditions. The nacelle aerodynamic design should similarly be validated through CFD to confirm pressure distributions, refine the non-axisymmetric lip geometry, and quantify the nacelle-airframe interference drag not addressed in the present work.

The noise mitigation strategy was assessed qualitatively and should be verified quantitatively against ICAO Annex 16 Chapter 14 limits, including characterisation of BWB shielding effectiveness and acoustic liner impedance optimisation. The fire protection system, not designed in detail, represents a regulatory requirement requiring dedicated work — particularly given the challenges specific to hydrogen combustion, including invisible flames and the absence of a validated Halon replacement for aviation nacelle environments.

The engine accessories sizing should be refined with detailed engine thermal data, and the thermal interactions between cryogenic LH₂ lines and accessory gearbox and VFSG components require dedicated investigation. The safety analyses should be extended using more realistic rotor burst energy assumptions, and structural integration with the pylon and engine mount design requires close coordination with the structures team to define load paths and dynamic load cases.

Finally, variable pitch fan technology represents a compelling alternative to the cascade thrust reverser selected in this thesis and warrants dedicated investigation in the context of the BW-25. By reversing the blade pitch angle, the fan itself generates reverse thrust without any translating mechanical system, eliminating the structural complexity and weight of cascade sleeves and blocker doors. This approach is particularly relevant for an upper-surface BWB installation, where the risk of reversed exhaust impinging on the fuselage structure — which motivated the blocking half configuration adopted here — would be entirely avoided. The implications for engine stall margin, boundary layer interaction over the BWB upper surface, and the certification pathway under CS-E for such an architecture in a commercial application all represent areas requiring further analysis before a meaningful comparison with the cascade solution could be made.

References

- [1] ACI World, "Global Air Travel Forecasted to Reach 9.8 Billion Passengers in 2025, Nearing the Historic 10 Billion Milestone," Sep. 30, 2025. [Online]. Available: <https://aci.aero/2025/09/30/global-air-travel-forecasted-to-reach-9-8-billion-passengers-in-2025-nearing-the-historic-10-billion-milestone/> [Accessed: Apr. 09, 2026].
- [2] Airbus S.A.S., "Future Aircraft," Airbus. [Online]. Available: <https://www.airbus.com/en/innovation/future-aircraft> [Accessed: Apr. 09, 2026].
- [3] "Blended Wing Body," *Wikipedia*, Mar. 23, 2026. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Blended_wing_body&oldid=1344905523 [Accessed: Apr. 09, 2026].
- [4] JetZero Inc., "JetZero." [Online]. Available: <https://www.jetzero.aero> [Accessed: Apr. 09, 2026].
- [5] "Green Book Specifications BW-25," Sep. 2025.
- [6] D. Keller, "Aerodynamic Investigation of Propulsion Integration for a Low Noise Hybrid Wing-Body with Podded UHBR Turbofan Engines," *J. Phys. Conf. Ser.*, vol. 2716, no. 1, p. 012008, Mar. 2024, doi: 10.1088/1742-6596/2716/1/012008.
- [7] K. Abu Salem, G. Palaia, P. D. Bravo-Mosquera, and A. A. Quarta, "A Review of Novel and Non-Conventional Propulsion Integrations for Next-Generation Aircraft," *Designs*, vol. 8, no. 2, p. 20, Feb. 2024, doi: 10.3390/designs8020020.
- [8] G. Wang, D. Li, P. Li, and B. Zhang, "Momentum- and Energy-Based Analyses of the Aerodynamic Effects of Boundary Layer Ingestion and Propulsion–Airframe Integration on a Blended Wing Body–Turbofan Configuration," *Aerospace*, vol. 12, no. 9, p. 846, Sep. 2025, doi: 10.3390/aerospace12090846.
- [9] A. Gangoli Rao, A. Sharma, and R. Van Dijk, "A CFD Based Parametric Analysis of S-Shaped Inlet for a Novel Blended Wing Body Aircraft," *SSRN Electron. J.*, 2017, doi: 10.2139/ssrn.3101299.
- [10] Rolls-Royce plc, "Trent XWB (Airbus A350-800/900) — Line and Base Maintenance," Issue 3, Jun. 2017.
- [11] Airbus S.A.S., "A350 Aircraft Characteristics — Airport and Maintenance Planning," Nov. 2016.
- [12] P. Okonkwo and H. Smith, "Review of Evolving Trends in Blended Wing Body Aircraft Design," *Prog. Aerosp. Sci.*, vol. 82, pp. 1–23, Apr. 2016, doi: 10.1016/j.paerosci.2015.12.002.
- [13] T. Silva, "Aerodynamic Design and Experimental Investigation of Short Nacelles for Future Turbofan Engines," Ph.D. thesis, TU Delft, Delft, Netherlands, 2020.

- [14] Z. Xin *et al.*, "Nacelle-Airframe Integration Design Method for Blended-Wing-Body Transport with Podded Engines," *Chin. J. Aeronaut.*, vol. 32, no. 8, pp. 1860–1868, Aug. 2019, doi: 10.1016/j.cja.2019.06.009.
- [15] S. Bagy, "Design and Optimization of Aircraft Engine Nozzles in Under-Wing Configuration," Ph.D. thesis, Université de Montpellier, Montpellier, France, 2020. [Online]. Available: <https://theses.hal.science/tel-03155091>
- [16] M. B. Malone and C. C. Peavey, "Installed Transonic 2D Nozzle Nacelle Boattail Drag Study," NASA Technical Report No. 20000044629, NASA Langley Research Center, Hampton, VA, 2000. [Online]. Available: <https://ntrs.nasa.gov/citations/20000044629>
- [17] D. J. Bentley, "Reverse Thrust Aerodynamics of Ultra-High Bypass Variable-Pitch Turbofans," Ph.D. thesis, School of Aerospace, Transport and Manufacturing, Cranfield University, Cranfield, UK, 2018. [Online]. Available: <https://dspace.lib.cranfield.ac.uk/handle/1826/18970>
- [18] Y. Gang, L. Dong, C. Zhengli, and Z. Zeyu, "Blended Wing Body Thrust Reverser Cascade Feasibility Evaluation Through CFD," *IEEE Access*, vol. 7, pp. 155184–155193, 2019, doi: 10.1109/ACCESS.2019.2949062.
- [19] International Civil Aviation Organization, *Annex 16 to the Convention on International Civil Aviation — Environmental Protection, Volume I: Aircraft Noise*, 8th ed. Montreal, QC, Canada: ICAO, Jul. 2017.
- [20] C. Villena Muñoz, "Aircraft Power Plant Installation," Lecture 2, School of Aerospace, Transport and Manufacturing, Cranfield University, Cranfield, UK, 2026.
- [21] Rolls-Royce Limited and Rolls-Royce plc, Eds., *The Jet Engine*, 6th ed. London, UK: Rolls-Royce plc, 2005.
- [22] S. H. S. and K. M. C. Rao, "Study of Different Designs of Chevron for Effective Noise Reduction in Jet Engines," presented at AeroCON 2024, Bangalore, India, Jun. 2024, pp. 2024-26-0408, doi: 10.4271/2024-26-0408.
- [23] O. Cleynen, "The Left Engine (A General Electric GENx-1B) on an Air India Boeing 787," Wikimedia Commons, 2013. [Online]. Available: [https://commons.wikimedia.org/wiki/File:GENx-1B_on_Air_India_B787_\(2\).jpg](https://commons.wikimedia.org/wiki/File:GENx-1B_on_Air_India_B787_(2).jpg) [Accessed: Apr. 09, 2026].
- [24] X. Liu, D. Zhao, D. Guan, S. Becker, D. Sun, and X. Sun, "Development and Progress in Aeroacoustic Noise Reduction on Turbofan Aeroengines," *Prog. Aerosp. Sci.*, vol. 130, p. 100796, Apr. 2022, doi: 10.1016/j.paerosci.2021.100796.
- [25] "787 Variable Frequency Starter Generator," Scribd. [Online]. Available: <https://www.scribd.com/document/875278743/B787-Electrical-Power-System> [Accessed: Apr. 09, 2026].
- [26] European Union Aviation Safety Agency, "CS-25 — Certification Specifications for Large Aeroplanes," Amendment 28. EASA, Cologne, Germany.

- [27] European Union Aviation Safety Agency, "CS-E — Certification Specifications for Engines." EASA, Cologne, Germany.
- [28] European Union Aviation Safety Agency, "AMC-20 — General Acceptable Means of Compliance for Airworthiness of Products, Parts and Appliances." EASA, Cologne, Germany.
- [29] SKYbrary, "Aircraft Systems Handbook," SKYbrary Aviation Safety. [Online]. Available: <https://skybrary.aero/sites/default/files/bookshelf/33388.pdf> [Accessed: Apr. 10, 2026].
- [30] NASA Office of Safety and Mission Assurance, "Fault Tree Handbook with Aerospace Applications," NASA/SP-2002-1580, NASA, Washington, DC, 2002.
- [31] B. Makiela, "How Safe Are Turbine Engine Rotor Life Limits?," MAK Aviation Engineering. [Online]. Available: <https://www.mak-aviation.com/post/how-safe-are-turbine-engine-rotor-life-limits> [Accessed: Apr. 10, 2026].
- [32] Airbus S.A.S., "A350 Family: The Long Range Leader — Facts & Figures," Airbus, Toulouse, France, Mar. 2026. [Online]. Available: <https://www.airbus.com/newsroom.html> [Accessed: Apr. 13, 2026].
- [33] R. T. Kawai, D. M. Friedman, and L. Serrano, "Blended Wing Body (BWB) Boundary Layer Ingestion (BLI) Inlet Configuration and System Studies," NASA Contractor Report NASA/CR-2006-214534, Boeing Phantom Works, Huntington Beach, CA, Dec. 2006. [Online]. Available: <https://ntrs.nasa.gov/citations/20070006754>
- [34] N. Khan, "School of Aerospace, Transport and Manufacturing", MSc Thesis, Cranfield University.
- [35] R. L. Barnola, "Interaction Between Aerodynamics, Structure and Packaging in a Blended Wing Body Configuration Conceptual Design", MSc Thesis, Cranfield University.
- [36] M. L. Olano, "CFD Aerodynamic Study of the Cranfield BWB-11 Eagle Ray Aircraft with Pod Mounted Engines", MSc Thesis, Cranfield University.
- [37] S. Prince, D. Di Pasquale, K. Garry, and C. Nuzzo, "A Rapid Aerodynamic Prediction Method for Unconventional Transonic Aircraft Configurations", MSc Thesis, Cranfield University.
- [38] ANSYS, Inc., "Turbulent Boundary Layers," in *Basics of Turbulent Flows*, Lesson 6, ANSYS Training Material, 2020.
- [39] J. D. Mattingly, K. M. Boyer, B. A. Haven, W. H. Heiser, and D. T. Pratt, *Aircraft Engine Design*, 3rd ed., in AIAA Education Series. Reston, VA: American Institute of Aeronautics and Astronautics, 2018.
- [40] I. Sadrehaghighi, "Aircraft Nacelle Design & Optimization," ResearchGate, 2025, doi: 10.13140/RG.2.2.13712.12801/4.

- [41] S. L. Heath *et al.*, "NASA Hybrid Wing Aircraft Aeroacoustic Test Documentation Report," NASA Technical Memorandum NASA/TM-2016-219185, NASA Langley Research Center, Hampton, VA, 2016. [Online]. Available: <https://ntrs.nasa.gov/citations/20160006330>
- [42] DCube s.r.l., "Digital Design Development," DCube. [Online]. Available: <http://www.d-cube.it/en/aeronautica.php> [Accessed: Apr. 10, 2026].
- [43] M. D. Williams, "Propulsion System Performance and Integration," Lecture Notes, MSc Thermal Power, School of Aerospace, Transport and Manufacturing, Cranfield University, Cranfield, UK.
- [44] J. A. Albers, "Effect of Subsonic Inlet Lip Geometry on Predicted Surface and Flow Mach Number Distributions," NASA Technical Note NASA TN D-7520, NASA Lewis Research Center, Cleveland, OH, 1973. [Online]. Available: <https://ntrs.nasa.gov/citations/19740003699>
- [45] ESDU, "NACA 1-Series Geometry Representation for Computational Fluid Dynamics," ESDU Data Item 94013, ESDU International, London, UK, 1994.
- [46] T. Mahmood, A. Jackson, V. Sethi, B. Khanal, and F. Ali, "Computational Fluid Dynamics Investigation of a Core-Mounted Target-Type Thrust Reverser — Part 1: Reverser Stowed Configuration," *J. Eng. Gas Turbines Power*, vol. 140, no. 9, p. 091204, Sep. 2018, doi: 10.1115/1.4038816.
- [47] "Runway," *Wikipedia*, Apr. 08, 2026. [Online]. Available: <https://en.wikipedia.org/w/index.php?title=Runway&oldid=1347780315> [Accessed: Apr. 10, 2026].
- [48] J. A. Yetter, "Why Do Airlines Want and Use Thrust Reversers? A Compilation of Airline Industry Responses to a Survey Regarding the Use of Thrust Reversers on Commercial Transport Airplanes," NASA Technical Memorandum NASA/TM-109158, NASA Langley Research Center, Hampton, VA, Jan. 1995. [Online]. Available: <https://ntrs.nasa.gov/citations/19950014289>
- [49] C. Liu, J. Deng, and S. Su, "Design of Loading System for Thrust Reverser Ground Simulation Test," *IOP Conf. Ser. Earth Environ. Sci.*, vol. 170, p. 042093, Jul. 2018, doi: 10.1088/1755-1315/170/4/042093.
- [50] K. Boyer, "Practical Aero Spotlight: John Norton and C-17 Operational Test and Evaluation," *Practical Aeronautics*, May 2024. [Online]. Available: <https://practicalaero.com/2024/05/pai-spotlight-john-norton-and-c-17-operational-test-and-evaluation/> [Accessed: Apr. 10, 2026].
- [51] Safran Group, "eTRAS — Electrical Thrust Reverser Actuation System," Safran. [Online]. Available: <https://www.safran-group.com/products-services/etras-electrical-thrust-reverser-actuation-system> [Accessed: Apr. 10, 2026].
- [52] L. Leylekian, M. Lebrun, and P. Lempereur, "An Overview of Aircraft Noise Reduction Technologies," *AerospaceLab J.*, no. 7, pp. 1–15, 2014, doi: 10.12762/2014.AL07-01.

- [53] ExxonMobil Aviation, "Mobil Jet™ Oil II," Product Data Sheet, ExxonMobil. [Online]. Available: <https://www.mobil.com/en-sd/aviation/pds/gl-xx-mobil-jet-oil-ii> [Accessed: Apr. 10, 2026].
- [54] E. D. Bol, "Investigation of Linear Motors as Electric Actuators for Aircraft," *Boeing Technical Journal*, Boeing, 2018.
- [55] Sensata Technologies, "Aerospace Products Overview," Sensata Technologies, Inc., Attleboro, MA, USA, 2025.
- [56] Crane Aerospace & Electronics, "Aeronautical FADEC," Rev. 3, Crane Co., Lynnwood, WA, USA.

Appendices

Appendix A Capture requirements

The main goal of this project is to get a type certificate, which means following all the rules set by the certifying agency. Airworthiness requirements depend on the type of aircraft and system. At the start of the project, the main requirements from EASA regulations were collected. The author summarized them, keeping only what is directly relevant to the thesis and related components. This list was also used as a checklist during the design to make sure the project was on track.

Table_Apx 1 : Propulsion system integration requirements capture

Requirement Serial	Requirement Derived From	Reference	Strategic Requirement	Interfaces	Importance	Plan to meet objective
ENG-INT-001	Standards	CS-E 20(a)/AMC E20	Ensure a single, controlled and traceable engine type design baseline is available for all integration, analysis and certification activities.	-	High	Maintain a complete type design list and configuration rules; ensure all integration artefacts reference it.
ENG-INT-002	Standards	CS-E 20(b), 30(a)	Ensure engine assumptions remain compatible with the aircraft certification basis to avoid gaps in compliance demonstration.	-	Medium	Document assumed aircraft certification standards and update as programme decisions change.

ENG-INT-003	Standards	CS-E 20(c)	Ensure all non-Type Certified equipment mounted on, or driven by, the engine is accounted for so that their effects on engine safety, loads and performance are controlled.	-	High	Identify all equipment, define mechanical, electrical, and pneumatic interfaces, and document approvals.
ENG-INT-004	Standards	CS-E 20(d)	Ensure the aircraft integrator has a complete and unambiguous definition of engine-aircraft interfaces and operating limitations to design a safe installation.	-	High	Develop and maintain installation and operating manuals covering all interfaces and operational modes, including limitations.
ENG-INT-005	Standards	CS-E 20(e)	Ensure aircraft performance, handling qualities and structural sizing are based on accurate and conservative engine performance data.	Propulsion system - Performance	High	Provide minimum and maximum performance datasets to the airframer, validated by test and analysis.
ENG-INT-006	Standards	CS-E 20(f)	Ensure the aircraft can verify in-service One Engine Inoperative capability and margins using robust and traceable power assurance methods.	Propulsion system - Performance	High	Provide performance data for one-engine-inoperative conditions and recommend power assurance procedures.
ENG-INT-007	Standards	CS-E 25(a)	Ensure continued airworthiness and safe operation of the installed engine through clearly defined and controlled maintenance tasks and limitations.	Reliability / Maintainability	High	Produce instructions for continued airworthiness including limitations and maintenance tasks.

ENG-INT-008	Standards	CS-E 30(a)	Ensure all key assumptions used in engine certification are visible to the airframer so the installation can be shown compatible or differences can be addressed.	-	High	Collect and publish structured installation assumptions in manuals and certification documents.
ENG-INT-009	Standards	CS-E 30(b)	Ensure the engine's safety and performance analyses correctly reflect the behaviour and reliability of non-engine components it depends on.	-	High	Document interface conditions, environmental ratings, and reliability targets for non-engine components.
ENG-INT-010	Standards	CS-E 50(a)	Ensure the Electronic Engine Control System maintains safe engine operation across all declared operating and installation conditions.	Avionics	High	Validate engine control laws and protection functions through analysis and testing over the operating envelope.
ENG-INT-011	Standards	CS-E 50(b)	Ensure transitions between control modes/channels do not create hazardous or unacceptable engine response in the aircraft environment.	Avionics	High	Define and verify control mode and channel transition logic, and document required notifications and procedures.
ENG-INT-012	Standards	CS-E 50(c)(3), 50(d)	Ensure that failures in the Electronic Engine Control System do not lead to hazardous engine effects or loss of safe controllability in the intended installation.	Avionics	High	Perform a safety assessment including engine and installation interfaces, and reference limitations in manuals.

ENG-INT-013	Standards	CS-E 50(g)	Ensure that loss/corruption of aircraft-supplied data is safely managed by the engine control without hazardous effects.	Avionics	High	Define data quality, validity, latency, and fallback strategies; verify through analysis and testing.
ENG-INT-014	Standards	CS-E 50(h)	Ensure the Electronic Engine Control System can tolerate loss/interruption of aircraft power and that the aircraft can provide power within declared limits.	Avionics	High	Specify nominal and transient power quality; define dedicated power sources and verify performance.
ENG-INT-015	Standards	CS-E 50(i)	Ensure air signal lines for Electronic Engine Control System do not introduce unsafe behaviour due to blockage, leakage or icing in the installation.	Avionics	Medium	Define routing, environmental, filtration, and heating requirements for signal lines; verify operation.
ENG-INT-016	Standards	CS-E 50(l)	Ensure Electronic Engine Control System is protected against unauthorised access or malicious interaction that could compromise engine safety.	Avionics	High	Identify safety-relevant interfaces and define cybersecurity requirements; document in manuals.
ENG-INT-017	Standards	CS-E 60	Ensure the aircraft provides the engine indications and recordings needed for safe operation, crew awareness and post-flight monitoring.	Avionics	High	Define required engine parameters, ranges, accuracy, sampling, and recording requirements; provide guidance.

ENG-INT-018	Standards	CS-E 80	Ensure engine-mounted equipment and their attachments do not jeopardise engine or aircraft safety in normal and failed conditions.	Structures	High	Define strength, stiffness, and containment requirements; validate through analysis and tests.
ENG-INT-019	Standards	CS-E 100, 520	Ensure aircraft and engine structures are sized against all relevant engine-induced loads during normal and abnormal operation.	Structures	High	Provide load envelopes and definitions for structural sizing and verification.
ENG-INT-020	Standards	CS-E 520(c)(2)	Ensure the aircraft installation can withstand engine interface loads and dynamic behaviour following blade loss/unbalance events.	Structures	High	Analyse and test rotor unbalance and blade loss events; provide interface forces to the airframer.
ENG-INT-021	Standards	CS-E 130	Ensure the engine installation supports adequate fire protection to prevent or limit hazardous engine fire effects.	Structures	High	Classify parts for fire safety, define fire zones, and document reliance on firewalls and safety systems.
ENG-INT-022	Standards	CS-E 135	Ensure engine and aircraft bonding arrangements provide adequate paths for lightning and fault currents without compromising safety.	-	High	Define bonding points, maximum resistance, and inspection criteria; provide guidance on installation and maintenance.

ENG-INT-023	Standards	CS-E 170	Ensure engine systems, including controls, remain operational or safely fail under environmental, HIRF and lightning conditions of the intended installation.	-	High	Qualify systems and equipment to environmental, electromagnetic interference, and lightning standards; publish limitations.
ENG-INT-024	Standards	CS-E 210	Ensure engine failures, including failures in the control system, are analysed considering a typical installation so that unsafe conditions are prevented or mitigated.	-	High	Perform failure mode and effects analysis covering engine failures and installation-sensitive effects; record assumptions.
ENG-INT-025	Standards	CS-E 510	Ensure hazardous engine effects are identified and that any reliance on aircraft features to mitigate them is clearly documented.	-	High	Identify hazardous engine effects and provide supporting safety analysis.
ENG-INT-026	Standards	CS-E 540	Ensure the engine and installation provide acceptable behaviour when subject to likely FOD events, including multi-engine exposure where relevant.	Propulsion system – Performance	High	Perform foreign object damage tests and analyses; provide guidance and limitations.
ENG-INT-027	Standards	CS-E 790	Ensure engine performance, stall margin and operability remain acceptable in the rain and hail conditions relevant to the intended installation.	Propulsion system – Performance	Medium	Provide rain and hail ingestion test or analysis results; document limitations affecting the intake and nacelle.

ENG-INT-028	Standards	CS-E 800	Ensure the engine and aircraft installation can tolerate defined bird strikes/ingestion without leading to hazardous effects.	Propulsion system – Performance	High	Conduct bird ingestion tests; provide results including damage locations and residual performance.
ENG-INT-029	Standards	CS-E 560	Ensure the engine receives fuel of appropriate quality, pressure and temperature to operate safely across the envelope.	Fuel systems – Tanks and gauging	High	Declare approved fuel types, additives, pressure and temperature ranges, filtration, and water separation requirements.
ENG-INT-030	Standards	CS-E 570	Ensure engine oil specification, consumption and return/venting arrangements are compatible with aircraft systems and fire safety.	Fuel systems – Transfer and feed	Medium	Provide approved oil list, maximum consumption, and constraints on venting and monitoring.
ENG-INT-031	Standards	CS-E 580/590	Ensure bleed and starter system demands and failure behaviours are compatible with aircraft pneumatic and structural capabilities.	Secondary power and APU	Medium	Define bleed extraction envelopes, starter torque and speed, and any weak-link provisions; document limitations.
ENG-INT-032	Standards	CS-E 650	Ensure engine vibration characteristics are compatible with the installation and do not cause unacceptable stress levels or resonances.	Structures	High	Perform vibration surveys; provide significant modes, frequencies, and damping guidance.

ENG-INT-033	Standards	CS-E 740	Ensure fatigue capability of critical components under resonant vibration is demonstrated for the intended usage.	Structures	Medium	Integrate dwell testing at resonance peaks into endurance testing and report results.
ENG-INT-034	Standards	CS-E 700	Ensure engine protection systems and limits prevent or mitigate harmful excess operating conditions that could impact aircraft safety.	Propulsion system – Performance	Medium	Define operating limits for temperatures, speeds, and pressures; validate and document procedures.
ENG-INT-035	Standards	CS-E 720	Ensure continuous ignition capability supports safe relight and operation in conditions requiring it (e.g. severe turbulence, icing, precipitation).	Propulsion system – Performance	Medium	Define when continuous ignition is required and specify electrical and control requirements; verify performance.
ENG-INT-036	Standards	CS-E 745	Ensure engine response characteristics support aircraft handling, protection and performance requirements.	Propulsion system – Performance	High	Characterise thrust and power response; provide data and constraints for control law and performance.
ENG-INT-037	Standards	CS-E 750	Ensure engine starting characteristics are compatible with aircraft/APU/ground start systems and cold-soak conditions.	Secondary power and APU	Medium	Provide start envelopes, torque, and speed requirements; verify low temperature and worst-case starting conditions.

ENG-INT-038	Standards	CS-E 780	Ensure the engine and installation provide acceptable performance and durability in icing environments covered by the aircraft CS.	IPS	High	Perform icing tests and analyses; provide design precautions and operational limitations.
ENG-INT-39	Standards	CS-E 810	Ensure blade failures do not result in uncontrolled debris hazards for the aircraft structure and systems.	-	High	Provide blade failure and containment test or analysis results, including energy and containment angles.
ENG-INT-040	Standards	CS-E 840/850	Ensure rotor and shaft systems remain structurally adequate under installation-induced loads and abnormal events.	-	Medium	Perform analyses and tests of shafts and rotors under normal and abnormal loadings; document constraints.
ENG-INT-041	Standards	CS-E 870	Ensure Exhaust Gas Temperature overtemperature events are understood and managed so that engine and aircraft structures are not exposed to unsafe thermal conditions.	Propulsion system – Performance	High	Define exhaust gas temperature limits, overtemperature behaviours, and required crew actions; provide thermal data.
ENG-INT-042	Standards	CS-E 890	Ensure any thrust reverser used with the engine operates safely and compatibly with the aircraft systems and structures.	Actuation systems & Field Performance	High	Define approved reverser systems and interface requirements; provide results to the airframer.

ENG-INT-043	Standards	CS-E 930	Ensure the Integrated Maintenance Procedure test configuration realistically reflects installed engine hardware and loads, so maintenance intervals remain valid in service.	Reliability / Maintainability	High	Plan and conduct maintenance tests with representative accessories and load profiles; correlate results with maintenance tasks.
ENG-INT-044	Standards	CS-E 1000	Ensure engine design supports aircraft environmental and operational requirements (e.g. altitude, temperature, special operations).	Flight path performance – En-route performance and emissions	Medium	Provide engine environmental and operational envelopes; relate them to aircraft mission and operational needs.
ENG-INT-045	Standards	CS-E 1010/1020	Ensure fuel venting and exhaust emissions characteristics are available to support aircraft environmental compliance and design.	Fuel systems – Transfer and feed	Medium	Provide measured or analysed fuel venting rates and emissions data; document operational restrictions.
ENG-INT-046	Standards	CS-E 1030	Ensure Time Limited Dispatch use of control system faults remains within safe exposure limits compatible with aircraft operations.	Reliability / Maintainability	High	Summarise time-limited dispatch eligible faults, exposure times, and maintenance approaches; communicate in manuals.
ENG-INT-047	Standards	CS-E 1040	Ensure engine in-flight shutdown performance supports requested ETOPS diversion times and operational approvals.	Flight path performance – En-route performance and emissions	High	Provide in-flight shutdown rate targets, monitoring plans, and in-service data; support reliability analyses.

ENG-INT-048	Standards	CS-E 1050	Ensure susceptibility of the engine to volcanic ash is understood and operators receive adequate guidance for safe operation and inspection.	-	Medium	Perform studies and use experience to define exposure effects; provide operator guidance for avoidance, inspection, and maintenance.
ENG-INT-049	Green Book	2.6	Anticipated requirements appropriate for year 2050	Structures / Propulsion system - Performance	Medium	Perform studies to reduce engine thrust requirements and determine cowl insulation methods.
ENG-INT-50	Green Book	4.4	Consideration should be given to uncontained engine rotor failures	Structures / Fuel systems	High	Determine possible rotor failure trajectories and define areas to be armoured.

Appendix B Functional Hazard Analysis

A functional hazard analysis is essential to understand the consequences of possible failures within the designed system. As part of the propulsion system work, a system-level functional hazard analysis was therefore performed. This analysis covers the integration of the propulsion system and its performance-related functions, and evaluates their effects at aircraft level.

Table_Apx 2 : Functional Hazard Analysis of propulsion systems

ID	Function	Failure Condition	Flight Phase	Effect on Aeroplane/Crew/Occupants	Probability		Classification
					Qualitative	Quantitative	
1.1	Provide thrust generation	Loss of thrust on one engine	Take-Off, Climb, Cruise, Approach	Asymmetric thrust and increased drag leading to yaw, reduced climb performance and possible drift-down; high crew workload and noticeable discomfort for occupants.	Extremely Remote	1,00E-07	Hazardous
1.2		Loss of thrust on both engines (dual failure)	Take-Off, Climb, Cruise, Descent	Total loss of climb capability resulting in a glide and potential forced landing; extreme crew workload and high risk of serious or fatal injuries.	Extremely Improbable	1,00E-09	Catastrophic
1.3		Large uncommanded thrust asymmetry	Take-Off, Climb, Approach	Strong yaw or roll causing loss of directional control during take-off or landing; very high crew	Extremely Improbable	1,00E-09	Hazardous-Catastrophic

				workload and significant injury risk if control is lost.			
1.4		Degraded / slow spool-up (thrust response lag)	Take-Off, Climb, Approach	Reduced thrust response or acceleration affecting take-off, climb or go-around margins; increased crew workload and moderate discomfort for occupants.	Extremely Remote	1,00E-09	Major-Hazardous
1.5		Incorrectly low thrust setting (wrong flex/derate)	Take-Off, Climb	Insufficient thrust for aircraft weight or environmental conditions, leading to rejected take-off or marginal climb performance; risk of injuries during abrupt manoeuvres or overrun.	Extremely Remote	1,00E-07	Hazardous
1.6		Higher-than-necessary thrust used	All	Increased engine stress and fuel consumption without loss of control; low crew workload and no significant effect on occupants.	Probable	1,00E-03	Minor
1.7		Inability to restart failed engine	Climb, Cruise, Approach	Prolonged one-engine-inoperative operation requiring drift-down and diversion; high crew workload and mild passenger discomfort due to longer flight and descent.	Remote	1,00E-05	Major

1.8		Thrust limiter / protection failure (engine exceeds max limits)	Take-Off, Climb	Engine operation beyond certified limits with risk of damage or asymmetric thrust; high crew workload and limited direct impact on occupants unless abrupt changes occur.	Extremely Remote	1,00E-07	Hazardous
2.1	Provide engine control / FADEC / indications	Loss of all primary engine indications	Take-Off, Climb, Cruise, Approach	Loss or corruption of engine indications leading to incorrect crew actions or delayed response; high crew workload and potential passenger anxiety.	Extremely Remote	1,00E-07	Hazardous
2.2		Misleading engine indications	Take-Off, Climb, Approach, Taxi and cruise	Uncommanded or unstable thrust behaviour due to control system faults; very high crew workload and noticeable aircraft motions causing discomfort or fear.	Remote	1,00E-05	Major
2.3		Sensor drift/failure to FADEC	All except Taxi	Combined failures resulting in severe thrust asymmetry or instability, with high risk of loss of control; extreme crew workload and high injury potential.	Remote	1,00E-05	Major
2.4		FADEC total loss / uncommanded thrust	Take-Off, Climb, Cruise, Approach	Engine fire within the nacelle or surrounding structure, possibly leading to structural damage; extreme crew workload and high risk of smoke, fire and serious injuries.	Extremely Remote	1,00E-07	Hazardous

2.5		Combination failure (sensor + FADEC + latent)	Take-Off, Climb, Approach	False warnings causing unnecessary engine shutdown and asymmetric flight; high crew workload and moderate passenger discomfort.	Extremely Improbable	1,00E-09	Catastrophic
3.1	Fire / nacelle / bleed / thermal	Failure to detect actual engine fire	Take-Off, Climb, Cruise	Contaminated bleed air affecting crew and cabin air quality; moderate health effects requiring diversion, with discomfort to occupants.	Extremely Improbable	1,00E-09	Catastrophic
3.2		Failure of extinguishing system	Take-Off, Climb, Cruise	Engine flameout leading to thrust loss and yaw at low altitude; high crew workload and short-term fear for occupants.	Extremely Improbable	1,00E-09	Catastrophic
3.3		Spurious engine fire warning	Take-Off, Climb, Approach	Dual engine flameout resulting in glide flight and potential off-airport landing; extreme crew workload and very high fatality risk.	Remote	1,00E-05	Major
3.4		Short circuit / chafing causing in-engine fire or FADEC loss	Take-Off, Climb, cruise	Rotor or blade failure causing severe vibration, thrust loss and possible uncontained debris; extreme crew workload and high risk of catastrophic structural damage and fatalities.	Extremely Remote	1,00E-07	Hazardous-Catastrophic

3.5		Nacelle overheat (non-fire)	Take-Off, Climb, Cruise	Progressive engine damage leading to vibration, performance loss and shutdown; high crew workload and noticeable discomfort for occupants.	Remote	1,00E-05	Major
3.6		Bleed air contamination (oil/hydraulic fumes)	Climb, Cruise, Approach	Ice, foreign object or bird ingestion causing flow distortion, surge or shutdown; high crew workload and short-duration passenger fear and discomfort.	Extremely Remote	1,00E-07	Hazardous
4.1	Fuel system (LH ₂ /GH ₂ & conventional failure modes)	Fuel supply interruption (one engine)	Take-Off, Climb, Cruise	Thrust reverser or engine attachment failure causing large aerodynamic disturbances and possible loss of control; extreme crew workload and very high risk of serious injuries or fatalities.	Extremely Remote	1,00E-07	Major-Hazardous
4.2		Common-cause fuel supply loss (both engines)	Take-Off, Climb, Cruise	Asymmetric thrust and increased drag leading to yaw, reduced climb performance and possible drift-down; high crew workload and noticeable discomfort for occupants.	Extremely Improbable	1,00E-09	Catastrophic
4.3		Fuel contamination / clogged filters	Take-Off, Climb, Cruise	Total loss of climb capability resulting in a glide and potential forced landing; extreme crew workload and high risk of serious or fatal injuries.	Remote	1,00E-05	Major

4.4		Fuel control / metering failure	Take-Off, Climb, Cruise	Strong yaw or roll causing loss of directional control during take-off or landing; very high crew workload and significant injury risk if control is lost.	Extremely Remote	1,00E-07	Hazardous
5.1	Mechanical rotating group	Fan blade fracture / liberation	Take-Off, Climb, Cruise, Approach, Landing	Reduced thrust response or acceleration affecting take-off, climb or go-around margins; increased crew workload and moderate discomfort for occupants.	Extremely Improbable	1,00E-09	Hazardous-Catastrophic
5.2		Fan blade-off (contained)	Take-Off, Climb, Cruise	Insufficient thrust for aircraft weight or environmental conditions, leading to rejected take-off or marginal climb performance; risk of injuries during abrupt manoeuvres or overrun.	Extremely Remote	1,00E-07	Major-Hazardous
5.3		LP/IP/HP shaft fracture or rotor imbalance	All	Increased engine stress and fuel consumption without loss of control; low crew workload and no significant effect on occupants.	Extremely Improbable	1,00E-09	Hazardous-Catastrophic
5.4		Turbine disk/blade uncontained release	Take-Off, Climb, Cruise	Prolonged one-engine-inoperative operation requiring drift-down and diversion; high crew workload and mild passenger discomfort due to longer flight and descent.	Extremely Improbable	1,00E-09	Catastrophic

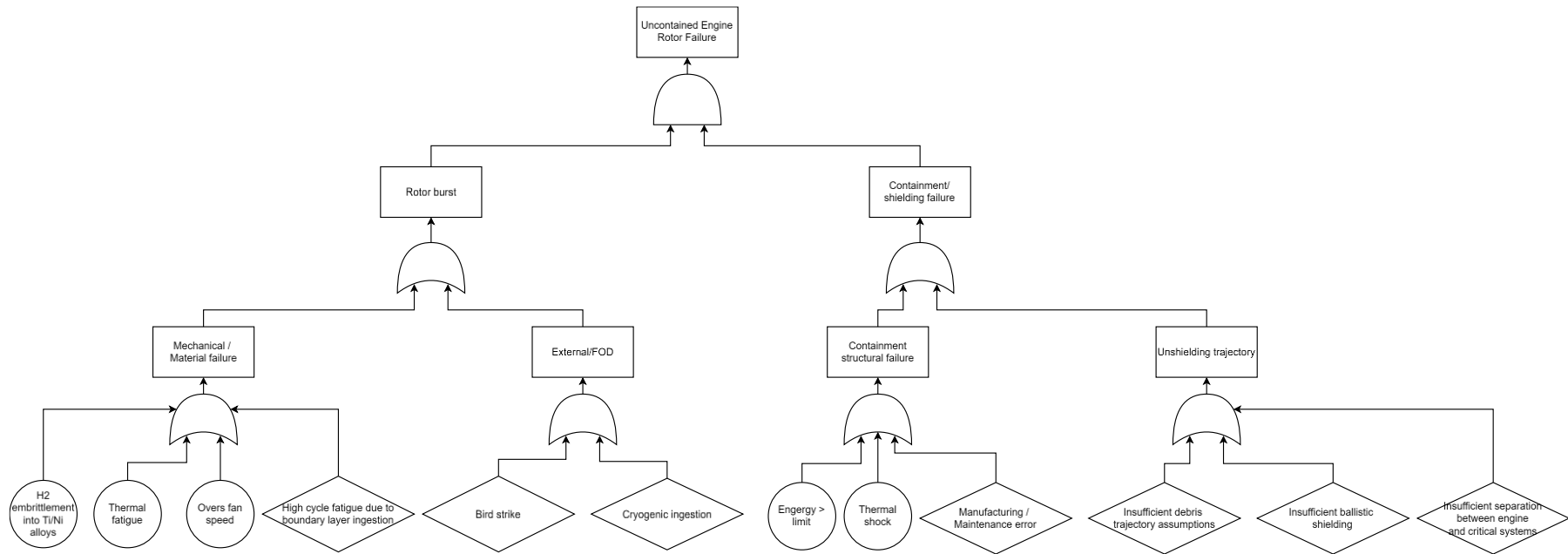
5.5		High-energy fragment impacting wing/fuel tank	Take-Off, Climb, Cruise, Landing	Engine operation beyond certified limits with risk of damage or asymmetric thrust; high crew workload and limited direct impact on occupants unless abrupt changes occur.	Extremely Improbable	1,00E-09	Catastrophic
5.6		False high vibration indication (unnecessary shutdown)	Take-Off, Climb, Approach	Loss or corruption of engine indications leading to incorrect crew actions or delayed response; high crew workload and potential passenger anxiety.	Remote	1,00E-05	Major-Hazardous
6.1	Accessory systems	Oil pump failure / starvation / contamination	Take-Off, Climb, Cruise	Uncommanded or unstable thrust behaviour due to control system faults; very high crew workload and noticeable aircraft motions causing discomfort or fear.	Remote	1,00E-05	Hazardous
6.2		Gearbox seizure / accessory drive loss	Take-Off, Climb	Combined failures resulting in severe thrust asymmetry or instability, with high risk of loss of control; extreme crew workload and high injury potential.	Remote	1,00E-05	Major
6.3		Loss of vibration monitoring	Cruise	Engine fire within the nacelle or surrounding structure, possibly leading to structural damage; extreme crew workload and high risk of smoke, fire and serious injuries.	Remote	1,00E-05	Major

7.1	LH ₂ thermal management / engine airflow control	Failure/degradation of charge-air intercooler	All	False warnings causing unnecessary engine shutdown and asymmetric flight; high crew workload and moderate passenger discomfort.	Remote	1,00E-05	Major
7.2		Loss/degradation of turbine cooling-air supply	Climb, Cruise	Contaminated bleed air affecting crew and cabin air quality; moderate health effects requiring diversion, with discomfort to occupants.	Extremely Remote	1,00E-07	Hazardous
7.3		Failure of hydrogen expander cycle	Take-Off, Climb, Cruise	Engine flameout leading to thrust loss and yaw at low altitude; high crew workload and short-term fear for occupants.	Remote	1,00E-05	Major-Hazardous
7.4		VSV / bleed-valve stuck (open/closed)	Take-Off, Climb	Dual engine flameout resulting in glide flight and potential off-airport landing; extreme crew workload and very high fatality risk.	Remote	1,00E-05	Hazardous
8.1	Environmental / inlet	Degraded inlet flow / distortion	Take-Off, Climb, Approach	Rotor or blade failure causing severe vibration, thrust loss and possible uncontained debris; extreme crew workload and high risk of catastrophic structural damage and fatalities.	Remote	1,00E-05	Major-Hazardous

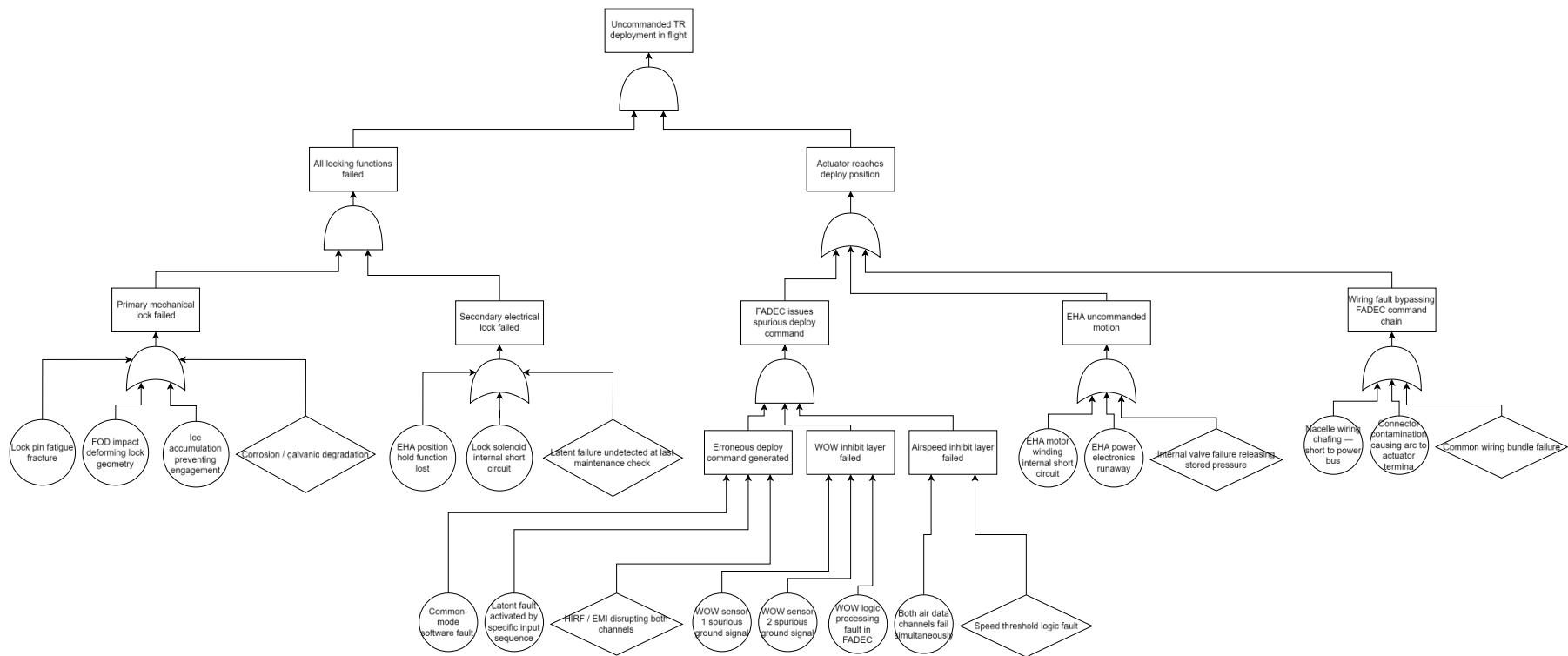
8.2		Inlet/nacelle icing / ice shedding	Climb, Cruise, Approach	Progressive engine damage leading to vibration, performance loss and shutdown; high crew workload and noticeable discomfort for occupants.	Remote	1,00E-05	Major
8.3		Bird strike ingestion	Take-Off, Climb, Approach	Ice, foreign object or bird ingestion causing flow distortion, surge or shutdown; high crew workload and short-duration passenger fear and discomfort.	Remote	1,00E-05	Hazardous
9.1	Thrust reverser	Loss of reverse thrust	Landing	Thrust reverser or engine attachment failure causing large aerodynamic disturbances and possible loss of control; extreme crew workload and very high risk of serious injuries or fatalities.	Remote	1,00E-05	Major

9.2		Uncommanded thrust reverser deployment (in flight)	Climb, Cruise	Asymmetric thrust and increased drag leading to yaw, reduced climb performance and possible drift-down; high crew workload and noticeable discomfort for occupants.	Extremely Improbable	1,00E-09	Catastrophic
10.1	Structural / integration	Engine mount / pylon structural failure (partial/full)	Take-Off, Climb, Cruise	Total loss of climb capability resulting in a glide and potential forced landing; extreme crew workload and high risk of serious or fatal injuries.	Extremely Improbable	1,00E-09	Catastrophic

Appendix C Fault tree analysis



Figure_Apx 1 : UERF Fault tree analysis



Figure_Apx 2 : Uncommanded thrust reversers deployment in flight Fault tree analysis

Appendix D Inlet lip coordinates

Table_Apx 3 : Intake lip coordinates

Upper lip		Side lip		Lower lip	
x	y_inner	x	y_inner	x	y_inner
0	0	0	0	0	0
0,001	0,012812104	0,001	0,013840159	0,001	0,014797
0,002	0,018105248	0,002	0,019560164	0,002	0,0209141
0,003	0,022157392	0,003	0,023940551	0,003	0,0255998
0,004	0,025565602	0,004	0,027626075	0,004	0,0295432
0,005	0,028561338	0,005	0,030866649	0,005	0,0330114
0,006	0,031263397	0,006	0,033790531	0,006	0,0361414
0,007	0,033742407	0,007	0,036473963	0,007	0,0390147
0,008	0,036044417	0,008	0,038966652	0,008	0,0416845
0,009	0,03820144	0,009	0,041303148	0,009	0,0441877
0,01	0,040236799	0,01	0,04350862	0,01	0,046551
0,011	0,042168116	0,011	0,045602083	0,011	0,048795
0,012	0,04400909	0,012	0,047598319	0,012	0,0509353
0,013	0,045770624	0,013	0,04950909	0,013	0,0529844
0,014	0,047461563	0,014	0,051343938	0,014	0,0549527
0,015	0,049089205	0,015	0,053110733	0,015	0,0568485
0,016	0,050659649	0,016	0,054816056	0,016	0,0586788
0,017	0,052178061	0,017	0,056465476	0,017	0,0604496
0,018	0,053648858	0,018	0,058063758	0,018	0,0621659
0,019	0,055075857	0,019	0,059615015	0,019	0,0638322
0,02	0,056462377	0,02	0,061122827	0,02	0,0654523
0,04	0,078587531	0,04	0,085276022	0,04	0,0914768
0,06	0,094678403	0,06	0,102995146	0,06	0,1106888
0,08	0,107480231	0,08	0,117234807	0,08	0,1262379
0,1	0,118067777	0,1	0,129151074	0,1	0,1393557
0,12	0,126996063	0,12	0,139341308	0,12	0,1506785
0,14	0,134595691	0,14	0,148162073	0,14	0,1605864
0,16	0,141081537	0,16	0,155846078	0,16	0,1693281
0,18	0,146601501	0,18	0,162554606	0,18	0,1770763
0,2	0,151261363	0,2	0,168404275	0,2	0,1839565
0,22	0,155138648	0,22	0,173481988	0,22	0,1900631
0,24	0,158290871	0,24	0,177853873	0,24	0,1954687
0,26	0,160760692	0,26	0,181570923	0,26	0,2002299
0,28	0,162579211	0,28	0,184672683	0,28	0,2043918
0,3	0,163768129	0,3	0,187189743	0,3	0,2079904
0,32	0,164341109	0,32	0,189145447	0,32	0,2110545
0,329	0,1644	0,34	0,190557078	0,34	0,2136071
		0,36	0,191436674	0,36	0,2156664
		0,38	0,191791554	0,38	0,2172464
		0,384	0,1918	0,4	0,2183575
				0,42	0,2190068
				0,438	0,2192

Appendix E Top external contour coordinates

Table_Apx 4 : Top external contour coordinates

x/X	y/Y	x	y
0,00	0	0,000	1,534
0,001	0,032948093	0,002	1,548
0,002	0,046482753	0,003	1,554
0,003	0,056818296	0,005	1,558
0,004	0,065507549	0,007	1,562
0,005	0,073155014	0,008	1,565
0,006	0,080071726	0,010	1,568
0,007	0,086442684	0,012	1,570
0,008	0,092388314	0,013	1,573
0,009	0,097992003	0,015	1,575
0,01	0,103314148	0,017	1,577
0,011	0,108400017	0,018	1,579
0,012	0,11328446	0,020	1,581
0,013	0,117994895	0,021	1,583
0,014	0,122553282	0,023	1,585
0,015	0,126977475	0,025	1,587
0,016	0,131282177	0,026	1,589
0,017	0,135479633	0,028	1,591
0,018	0,139580138	0,030	1,592
0,019	0,143592427	0,031	1,594
0,02	0,147523965	0,033	1,596
0,04	0,215221985	0,066	1,624
0,06	0,270734859	0,099	1,647
0,08	0,318813443	0,132	1,667
0,1	0,361505892	0,165	1,684
0,12	0,400073364	0,198	1,700
0,14	0,435377833	0,231	1,715
0,16	0,468033137	0,264	1,728
0,18	0,498487677	0,297	1,741
0,2	0,527076235	0,330	1,753
0,22	0,554054093	0,363	1,764
0,24	0,579619935	0,396	1,775
0,26	0,603931335	0,429	1,785
0,28	0,627115247	0,462	1,794
0,3	0,649275141	0,495	1,804
0,32	0,670495884	0,528	1,812
0,34	0,690847128	0,561	1,821
0,36	0,71038574	0,594	1,829
0,38	0,72915764	0,627	1,837
0,4	0,747199276	0,660	1,844
0,42	0,764538928	0,693	1,851
0,44	0,781197901	0,726	1,858
0,46	0,797191696	0,759	1,865
0,48	0,812531155	0,792	1,871
0,5	0,827223593	0,825	1,877

0,52	0,841273886	0,858	1,883
0,54	0,8546855	0,891	1,889
0,56	0,86746142	0,924	1,894
0,58	0,879604942	0,957	1,899
0,6	0,891120303	0,990	1,904
0,62	0,902013114	1,023	1,908
0,64	0,912290576	1,056	1,913
0,66	0,921961472	1,089	1,917
0,68	0,931035928	1,122	1,920
0,7	0,93952496	1,155	1,924
0,72	0,947439832	1,188	1,927
0,74	0,954791266	1,221	1,930
0,76	0,961588577	1,254	1,933
0,78	0,967838789	1,287	1,936
0,8	0,973545856	1,320	1,938
0,82	0,978710096	1,353	1,940
0,84	0,983327977	1,386	1,942
0,86	0,987392432	1,419	1,944
0,88	0,990893888	1,452	1,945
0,9	0,99382222	1,485	1,946
0,92	0,99616988	1,518	1,947
0,94	0,997936463	1,551	1,948
0,96	0,99913501	1,584	1,949
0,98	0,999800374	1,617	1,949
1	1	1,650	1,949

Appendix F Side external contour coordinates

Table_Apx 5 : Side external contour coordinates

x/X	y/Y	x	y
0,00	0	0,000	1,562
0,001	0,032948093	0,002	1,575
0,002	0,046482753	0,004	1,580
0,003	0,056818296	0,005	1,584
0,004	0,065507549	0,007	1,587
0,005	0,073155014	0,009	1,590
0,006	0,080071726	0,011	1,593
0,007	0,086442684	0,012	1,595
0,008	0,092388314	0,014	1,598
0,009	0,097992003	0,016	1,600
0,01	0,103314148	0,018	1,602
0,011	0,108400017	0,020	1,604
0,012	0,11328446	0,021	1,606
0,013	0,117994895	0,023	1,607
0,014	0,122553282	0,025	1,609
0,015	0,126977475	0,027	1,611
0,016	0,131282177	0,029	1,613
0,017	0,135479633	0,030	1,614
0,018	0,139580138	0,032	1,616
0,019	0,143592427	0,034	1,617
0,02	0,147523965	0,036	1,619
0,04	0,215221985	0,071	1,645
0,06	0,270734859	0,107	1,667
0,08	0,318813443	0,143	1,685
0,1	0,361505892	0,178	1,702
0,12	0,400073364	0,214	1,717
0,14	0,435377833	0,250	1,730
0,16	0,468033137	0,285	1,743
0,18	0,498487677	0,321	1,755
0,2	0,527076235	0,357	1,766
0,22	0,554054093	0,392	1,776
0,24	0,579619935	0,428	1,786
0,26	0,603931335	0,464	1,796
0,28	0,627115247	0,499	1,805
0,3	0,649275141	0,535	1,813
0,32	0,670495884	0,571	1,821
0,34	0,690847128	0,606	1,829
0,36	0,71038574	0,642	1,837
0,38	0,72915764	0,678	1,844
0,4	0,747199276	0,713	1,851
0,42	0,764538928	0,749	1,858
0,44	0,781197901	0,785	1,864
0,46	0,797191696	0,821	1,870
0,48	0,812531155	0,856	1,876
0,5	0,827223593	0,892	1,882

0,52	0,841273886	0,928	1,888
0,54	0,8546855	0,963	1,893
0,56	0,86746142	0,999	1,898
0,58	0,879604942	1,035	1,902
0,6	0,891120303	1,070	1,907
0,62	0,902013114	1,106	1,911
0,64	0,912290576	1,142	1,915
0,66	0,921961472	1,177	1,919
0,68	0,931035928	1,213	1,922
0,7	0,93952496	1,249	1,926
0,72	0,947439832	1,284	1,929
0,74	0,954791266	1,320	1,932
0,76	0,961588577	1,356	1,934
0,78	0,967838789	1,391	1,937
0,8	0,973545856	1,427	1,939
0,82	0,978710096	1,463	1,941
0,84	0,983327977	1,498	1,943
0,86	0,987392432	1,534	1,944
0,88	0,990893888	1,570	1,945
0,9	0,99382222	1,605	1,947
0,92	0,99616988	1,641	1,948
0,94	0,997936463	1,677	1,948
0,96	0,99913501	1,712	1,949
0,98	0,999800374	1,748	1,949
1	1	1,784	1,949

Appendix G Bottom external contour coordinates

Table_Apx 6 : Bottom external contour coordinates

x/X	y/Y	x	y
0,00	0	0,000	1,589
0,001	0,03294809	0,002	1,601
0,002	0,04648275	0,004	1,606
0,003	0,0568183	0,006	1,610
0,004	0,06550755	0,008	1,613
0,005	0,07315501	0,010	1,616
0,006	0,08007173	0,012	1,618
0,007	0,08644268	0,013	1,620
0,008	0,09238831	0,015	1,622
0,009	0,097992	0,017	1,624
0,01	0,10331415	0,019	1,626
0,011	0,10840002	0,021	1,628
0,012	0,11328446	0,023	1,630
0,013	0,1179949	0,025	1,632
0,014	0,12255328	0,027	1,633
0,015	0,12697748	0,029	1,635
0,016	0,13128218	0,031	1,636
0,017	0,13547963	0,033	1,638
0,018	0,13958014	0,035	1,639
0,019	0,14359243	0,037	1,641
0,02	0,14752396	0,038	1,642
0,04	0,21522199	0,077	1,667
0,06	0,27073486	0,115	1,687
0,08	0,31881344	0,154	1,704
0,1	0,36150589	0,192	1,719
0,12	0,40007336	0,231	1,733
0,14	0,43537783	0,269	1,746
0,16	0,46803314	0,308	1,758
0,18	0,49848768	0,346	1,769
0,2	0,52707624	0,384	1,779
0,22	0,55405409	0,423	1,789
0,24	0,57961994	0,461	1,798
0,26	0,60393133	0,500	1,807
0,28	0,62711525	0,538	1,815
0,3	0,64927514	0,577	1,823
0,32	0,67049588	0,615	1,830
0,34	0,69084713	0,654	1,838
0,36	0,71038574	0,692	1,845
0,38	0,72915764	0,730	1,852
0,4	0,74719928	0,769	1,858
0,42	0,76453893	0,807	1,864
0,44	0,7811979	0,846	1,870
0,46	0,7971917	0,884	1,876
0,48	0,81253116	0,923	1,882
0,5	0,82722359	0,961	1,887

0,52	0,84127389	1,000	1,892
0,54	0,8546855	1,038	1,897
0,56	0,86746142	1,076	1,901
0,58	0,87960494	1,115	1,906
0,6	0,8911203	1,153	1,910
0,62	0,90201311	1,192	1,914
0,64	0,91229058	1,230	1,917
0,66	0,92196147	1,269	1,921
0,68	0,93103593	1,307	1,924
0,7	0,93952496	1,346	1,927
0,72	0,94743983	1,384	1,930
0,74	0,95479127	1,422	1,933
0,76	0,96158858	1,461	1,935
0,78	0,96783879	1,499	1,937
0,8	0,97354586	1,538	1,940
0,82	0,9787101	1,576	1,941
0,84	0,98332798	1,615	1,943
0,86	0,98739243	1,653	1,944
0,88	0,99089389	1,692	1,946
0,9	0,99382222	1,730	1,947
0,92	0,99616988	1,768	1,948
0,94	0,99793646	1,807	1,948
0,96	0,99913501	1,845	1,949
0,98	0,99980037	1,884	1,949
1	1	1,922	1,949

Appendix H Engine performances for each altitude

Table_Apx 7 : Engine performances

Mach Number	Altitude (ft)	Altitude (m)	Thrust (kN)	SFC	Mass air flow (kg/s)
0.378	0	0	500.31	3.0943	1442.92
0.385	1000	304.8	366.26	4.3932	1524.228
0.392	2000	609.6	355.53	4.4118	1479.371
0.399	3000	914.4	344.93	4.4319	1435.579
0.407	4000	1219.2	334.34	4.457	1393.489
0.414	5000	1524	324.07	4.4796	1351.756
0.422	6000	1828.8	313.87	4.5067	1311.667
0.430	7000	2133.6	303.88	4.5347	1272.544
0.439	8000	2438.4	294.02	4.5666	1234.992
0.447	9000	2743.2	284.5	4.5959	1197.73
0.456	10000	3048	275.11	4.6291	1161.988
0.465	11000	3352.8	265.97	4.6628	1127.116
0.474	12000	3657.6	257.07	4.6973	1093.096
0.483	13000	3962.4	248.41	4.7323	1059.91
0.493	14000	4267.2	239.93	4.7708	1028.12
0.503	15000	4572	231.69	4.8097	997.112
0.513	16000	4876.8	223.68	4.849	966.869
0.524	17000	5181.6	215.88	4.8917	937.932
0.535	18000	5486.4	208.31	4.9347	909.712
0.546	19000	5791.2	200.97	4.978	882.19
0.558	20000	6096	193.83	5.0246	855.89
0.569	21000	6400.8	186.94	5.0685	829.71
0.582	22000	6705.6	180.24	5.1185	805.231
0.594	23000	7010.4	173.77	5.1658	780.84
0.607	24000	7315.2	167.5	5.2162	757.567
0.620	25000	7620	161.44	5.2668	734.871
0.634	26000	7924.8	155.58	5.3204	713.239
0.648	27000	8229.6	149.92	5.3742	692.14
0.663	28000	8534.4	144.46	5.4309	672.053
0.678	29000	8839.2	139.19	5.4877	652.46
0.694	30000	9144	134.12	5.5475	633.828
0.710	31000	9448.8	129.24	5.6073	615.683
0.726	32000	9753.6	124.51	5.6673	597.913
0.743	33000	10058.4	119.97	5.7301	581.069
0.761	34000	10363.2	115.62	5.7958	565.092
0.779	35000	10668	111.42	5.8615	549.495

0.798	36000	10972.8	107.39	5.9301	534.72
0.800	37000	11277.6	102.43	5.9397	510.705
0.800	38000	11582.4	97.57	5.9427	486.739
0.800	39000	11887.2	92.95	5.946	463.898
0.800	40000	12192	88.55	5.9493	442.129
0.800	41000	12496.8	84.35	5.9529	421.381
0.800	42000	12801.6	80.34	5.9566	401.607
0.800	43000	13106.4	76.53	5.9604	382.761
0.800	44000	13411.2	72.89	5.9645	364.799
0.800	45000	13716	69.43	5.9688	347.68
0.800	46000	14020.8	66.12	5.9733	331.365

Appendix I Python script for the drag calculation at each altitude

```

"""
Nacelle Drag Estimation Script
=====
Calculates total nacelle drag (2 engines) for altitudes from 0 to 46,000
ft
using ISA atmosphere and drag estimation method from GDP.xlsx (nacelle
design sheet).

Mach number and mass air flow per altitude are taken directly from the
mission table provided. MFR is computed from the actual mass flow at each
altitude rather than using a fixed design-point value.

Drag components:
- Forebody drag : skin friction + MFR correction ( $CD = CD_0 * (1 + K * (1 - MFR))^{(5/3)}$ )
- Midbody drag : cumulative skin friction up to  $L_f + L_m$ 
- Afterbody drag : cumulative skin friction up to  $L_f + L_m + L_a$ , with
1.2 factor

References (from "Nacelle design" sheet of GDP.xlsx):
Prandtl-Schlichting:  $C_f = 0.455 / (\log_{10}(Re))^{2.58}$ 
Forebody:  $CD_0 = (1 + 2 * \gamma / (3 * L_f))^{(5/3)} * C_{f\_FB}$ 
            $K = 1.75 * [(D_{max}/D_{h1} + 1) * \{(1 + 3 * L_f / D_{max}) * (D_{max}/D_{h1}) - 1\}]^{-1}$ 
            $CD = CD_0 * (1 + K * (1 - MFR))^{(5/3)}$ 
MFR      :  $A_\infty = W_a * \sqrt{TAT} / (TAP * MFP)$  then  $MFR = A_\infty / A_{h1}$ 
Midbody:   $CF_{MB} = C_{f\_MB} * (Swet_{FB} / A_{max} + Swet_{MB} / A_{max}) - CF_{FB}$ 
Afterbody:  $CF_{AB} = C_{f\_AB} * (\sum Swet / A_{max}) - CF_{FB} - CF_{MB}$ 
            $CD_{AB} = 1.2 * CF_{AB}$ 
"""

import math
import csv
import os

# =====
# MISSION TABLE (Mach, altitude_ft, mass_air_flow_kg_s per engine)
# =====
MISSION_TABLE = [
    (0.378, 0, 1442.920),
    (0.385, 1000, 1524.228),
    (0.392, 2000, 1479.371),
    (0.399, 3000, 1435.579),
    (0.407, 4000, 1393.489),
    (0.414, 5000, 1351.756),

```

```

(0.422, 6000, 1311.667),
(0.430, 7000, 1272.544),
(0.439, 8000, 1234.992),
(0.447, 9000, 1197.730),
(0.456, 10000, 1161.988),
(0.465, 11000, 1127.116),
(0.474, 12000, 1093.096),
(0.483, 13000, 1059.910),
(0.493, 14000, 1028.120),
(0.503, 15000, 997.112),
(0.513, 16000, 966.869),
(0.524, 17000, 937.932),
(0.535, 18000, 909.712),
(0.546, 19000, 882.190),
(0.558, 20000, 855.890),
(0.569, 21000, 829.710),
(0.582, 22000, 805.231),
(0.594, 23000, 780.840),
(0.607, 24000, 757.567),
(0.620, 25000, 734.871),
(0.634, 26000, 713.239),
(0.648, 27000, 692.140),
(0.663, 28000, 672.053),
(0.678, 29000, 652.460),
(0.694, 30000, 633.828),
(0.710, 31000, 615.683),
(0.726, 32000, 597.913),
(0.743, 33000, 581.069),
(0.761, 34000, 565.092),
(0.779, 35000, 549.495),
(0.798, 36000, 534.720),
(0.800, 37000, 510.705),
(0.800, 38000, 486.739),
(0.800, 39000, 463.898),
(0.800, 40000, 442.129),
(0.800, 41000, 421.381),
(0.800, 42000, 401.607),
(0.800, 43000, 382.761),
(0.800, 44000, 364.799),
(0.800, 45000, 347.680),
(0.800, 46000, 331.365),
]

# =====
# NACELLE GEOMETRY (from "Nacelle design" sheet, GDP.xlsx)
# =====
Dmax = 3.898 # Max nacelle outer diameter [m]

```

```

Dh1  = 3.1236 # Highlight diameter (mean) [m]
Lf   = 1.785  # Forebody length [m]
Lm   = 0.55   # Midbody length [m]
La   = 3.85   # Afterbody length [m]
Y    = 0.3872 # Forebody half-thickness  $= (D_{max} - Dh1) / 2$ , side mean) [m]

A_h1  = math.pi * (Dh1 / 2) ** 2 # Highlight (intake) area [m²]
Amax  = math.pi * (Dmax / 2) ** 2 # Max cross-section area [m²]
Swet_FB = math.pi * Dmax * Lf      # Wetted area - forebody [m²]
Swet_MB = math.pi * Dmax * Lm      # Wetted area - midbody [m²]
Swet_AB = 47.146823                # Wetted area - afterbody [m²]
(GDP.xlsx row 504)

# Shape factor K for MFR correction (GDP.xlsx row 512)
K = 1.75 / ((Dmax / Dh1 + 1) * ((1 + 3 * Lf / Dmax) * (Dmax / Dh1) - 1))

# =====
# ISA ATMOSPHERE
# =====
R_gas  = 287.05 # J/(kg·K)
gamma  = 1.4
g0     = 9.80665 # m/s²
T0     = 288.15 # K
P0     = 101325 # Pa
lapse  = -0.0065 # K/m
h_trop = 11000.0 # m
mu0    = 1.716e-5 # Pa·s (Sutherland reference)
T_ref  = 273.15 # K
S_suth = 110.4 # K

def isa(altitude_ft):
    h = altitude_ft * 0.3048
    if h <= h_trop:
        T = T0 + lapse * h
        P = P0 * (T / T0) ** (-g0 / (lapse * R_gas))
    else:
        T_t = T0 + lapse * h_trop
        P_t = P0 * (T_t / T0) ** (-g0 / (lapse * R_gas))
        T = T_t
        P = P_t * math.exp(-g0 * (h - h_trop) / (R_gas * T_t))
    rho = P / (R_gas * T)
    a = math.sqrt(gamma * R_gas * T)
    mu = mu0 * (T / T_ref) ** 1.5 * (T_ref + S_suth) / (T + S_suth)
    nu = mu / rho
    return T, P, rho, a, mu, nu

```

```

def prandtl_schlichting(Re):
    return 0.455 / math.log10(Re) ** 2.58

# =====
# DRAG CALCULATION
# =====
def compute_drag(altitude_ft, Mach, Wa):
    """Wa : mass air flow for ONE engine [kg/s]."""
    T, P, rho, a, mu, nu = isa(altitude_ft)
    V = Mach * a
    q0 = 0.5 * rho * V ** 2

    # Mass Flow Ratio via stream tube area at infinity (GDP.xlsx, nacelle
    design sheet)
    #  $A_\infty = W_a * \sqrt{TAT} / (TAP * MFP)$  then  $MFR = A_\infty / A_{h1}$ 
    TAT = T * (1 + (gamma - 1) / 2 * Mach ** 2)
    TAP = P * (1 + (gamma - 1) / 2 * Mach ** 2) ** (gamma / (gamma -
1))
    MFP = (math.sqrt(gamma / R_gas) * Mach
            * (1 + (gamma - 1) / 2 * Mach ** 2) ** (-(gamma + 1) / (2 *
(gamma - 1))))
    A_inf = Wa * math.sqrt(TAT) / (TAP * MFP)
    MFR = A_inf / A_h1

    # --- FOREBODY ---
    Re_FB = rho * V * Lf / mu
    Cf_FB = prandtl_schlichting(Re_FB)
    CF_FB = Cf_FB * (Swet_FB / Amax)
    CD0 = (1 + 2 * Y / (3 * Lf)) ** (5 / 3) * CF_FB
    CD_FB = CD0 * (1 + K * (1 - MFR)) ** (5 / 3)
    D_FB = q0 * Amax * CD_FB

    # --- MIDBODY ---
    Re_MB = rho * V * (Lf + Lm) / mu
    Cf_MB = prandtl_schlichting(Re_MB)
    CF_MB = Cf_MB * (Swet_FB / Amax + Swet_MB / Amax) - CF_FB
    D_MB = q0 * Amax * CF_MB

    # --- AFTERBODY ---
    Re_AB = rho * V * (Lf + Lm + La) / mu
    Cf_AB = prandtl_schlichting(Re_AB)
    CF_AB = (Cf_AB * (Swet_FB / Amax + Swet_MB / Amax + Swet_AB / Amax)
            - CF_FB - CF_MB)
    CD_AB = 1.2 * CF_AB
    D_AB = q0 * Amax * CD_AB

```

```

D_one = D_FB + D_MB + D_AB
D_two = 2 * D_one

return {
    "altitude_ft" : altitude_ft,
    "Mach"        : Mach,
    "Wa_kg_s"     : Wa,
    "MFR"         : MFR,
    "T_K"         : T,
    "P_Pa"        : P,
    "rho_kgm3"    : rho,
    "a_ms"        : a,
    "mu_Pas"      : mu,
    "nu_m2s"      : nu,
    "TAS_ms"      : V,
    "q0_Pa"       : q0,
    "Re_FB"       : Re_FB,
    "Cf_FB"       : Cf_FB,
    "CF_FB"       : CF_FB,
    "CD0_FB"      : CD0,
    "CD_FB"       : CD_FB,
    "D_FB_N"      : D_FB,
    "Re_MB"       : Re_MB,
    "Cf_MB"       : Cf_MB,
    "CF_MB"       : CF_MB,
    "D_MB_N"      : D_MB,
    "Re_AB"       : Re_AB,
    "Cf_AB"       : Cf_AB,
    "CF_AB"       : CF_AB,
    "CD_AB"       : CD_AB,
    "D_AB_N"      : D_AB,
    "D_1eng_N"    : D_one,
    "D_2eng_N"    : D_two,
    "D_2eng_kN"   : D_two / 1000,
}

# =====
# RUN SWEEP
# =====
results = [compute_drag(alt, M, Wa) for M, alt, Wa in MISSION_TABLE]

# =====
# PRINT
# =====
header = (
    f"{'Alt [ft]':>10} {'Mach':>6} {'Wa [kg/s]':>10} {'MFR':>6} "

```

```

f"{ 'T [K]':>7} { 'P [Pa]':>9} { 'rho':>9} { 'a [m/s]':>8} "
f"{ 'mu [Pa.s]':>11} { 'nu [m²/s]':>11} { 'q0 [Pa]':>9} "
f"{ 'D_FB [N]':>10} { 'D_MB [N]':>10} { 'D_AB [N]':>10} "
f"{ 'D_1eng [N]':>11} { 'D_2eng [kN]':>12}"
)
sep = "=" * len(header)

print(sep)
print("NACELLE DRAG ESTIMATION - ISA atmosphere - 2 engines")
print("Geometry : Dmax=3.898 m Dh1=3.124 m Lf=1.785 m Lm=0.55
m La=3.85 m")
print("Speed & Wa : taken from mission table per altitude")
print("MFR via  $A_\infty = Wa \cdot \sqrt{TAT / (TAP \cdot MFP)}$ , then MFR =  $A_\infty / A_{h1}$  ( $A_{h1}=7.663$ 
m²)")
print(sep)
print(header)
print("-" * len(header))

for r in results:
    print(
        f"{r['altitude_ft']:>10} {r['Mach']:>6.3f} {r['Wa_kg_s']:>10.3f}
    } "
        f"{r['MFR']:>6.4f} {r['T_K']:>7.2f} {r['P_Pa']:>9.1f} "
        f"{r['rho_kgm3']:>9.5f} {r['a_ms']:>8.2f} "
        f"{r['mu_Pas']:>11.3e} {r['nu_m2s']:>11.3e} {r['q0_Pa']:>9.1f}
    "
        f"{r['D_FB_N']:>10.1f} {r['D_MB_N']:>10.1f} {r['D_AB_N']:>10.1f}
    } "
        f"{r['D_1eng_N']:>11.1f} {r['D_2eng_kN']:>12.4f}"
    )

print("-" * len(header))

drags = [r["D_2eng_kN"] for r in results]
i_max = drags.index(max(drags))
i_min = drags.index(min(drags))
print(f"\nMax total drag (2 eng): {max(drags):.4f} kN @
{results[i_max]['altitude_ft']} ft"
      f" (M={results[i_max]['Mach']:.3f},
Wa={results[i_max]['Wa_kg_s']:.1f} kg/s,"
      f" MFR={results[i_max]['MFR']:.4f})")
print(f"Min total drag (2 eng): {min(drags):.4f} kN @
{results[i_min]['altitude_ft']} ft"
      f" (M={results[i_min]['Mach']:.3f},
Wa={results[i_min]['Wa_kg_s']:.1f} kg/s,"
      f" MFR={results[i_min]['MFR']:.4f})")

```

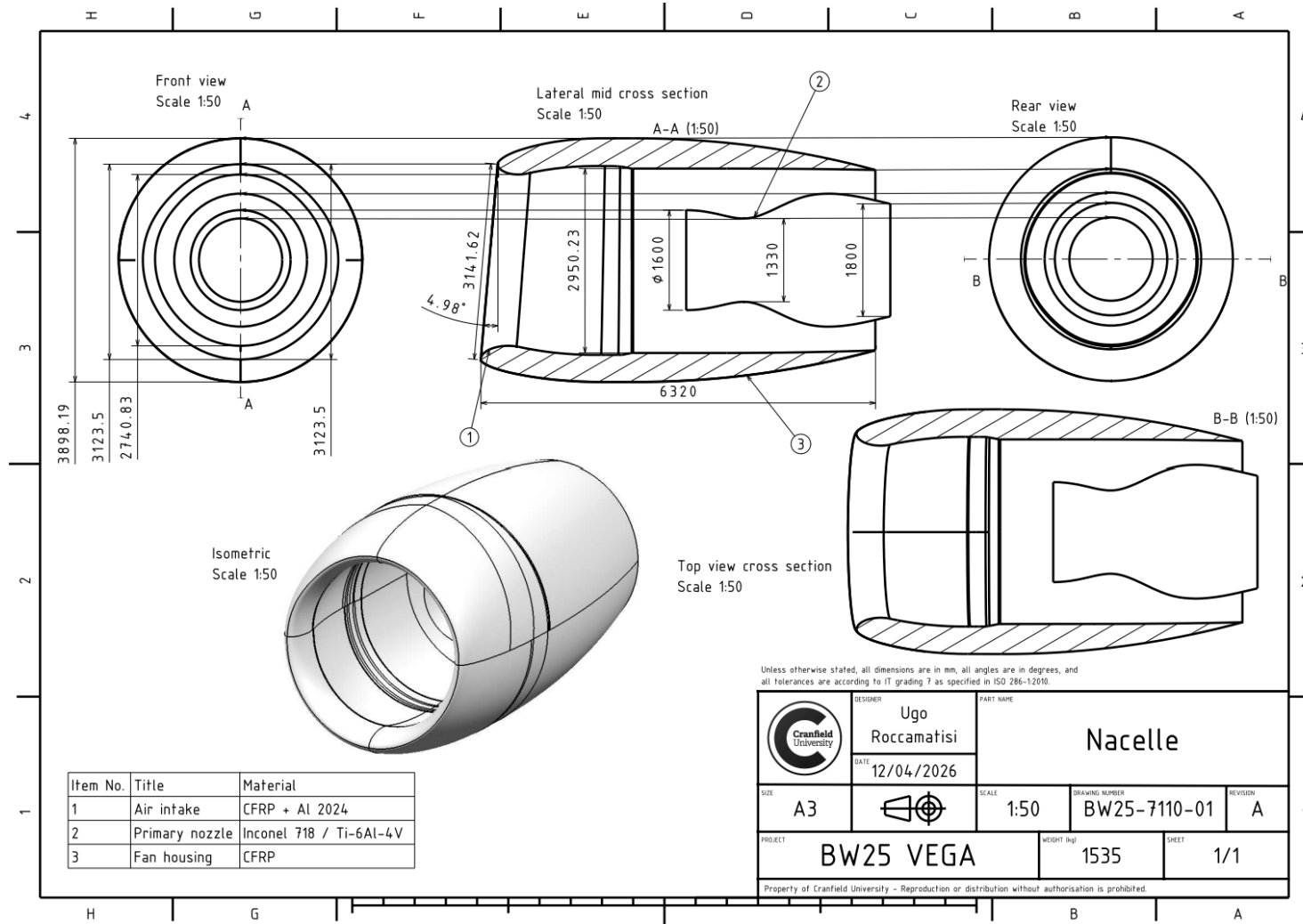
```

# =====
# CSV EXPORT
# =====
csv_path = os.path.join(os.path.dirname(os.path.abspath(__file__)),
                        "nacelle_drag_results.csv")
with open(csv_path, "w", newline="") as f:
    writer = csv.DictWriter(f, fieldnames=list(results[0].keys()))
    writer.writeheader()
    writer.writerows(results)
print(f"\nResults exported to: {csv_path}")

```

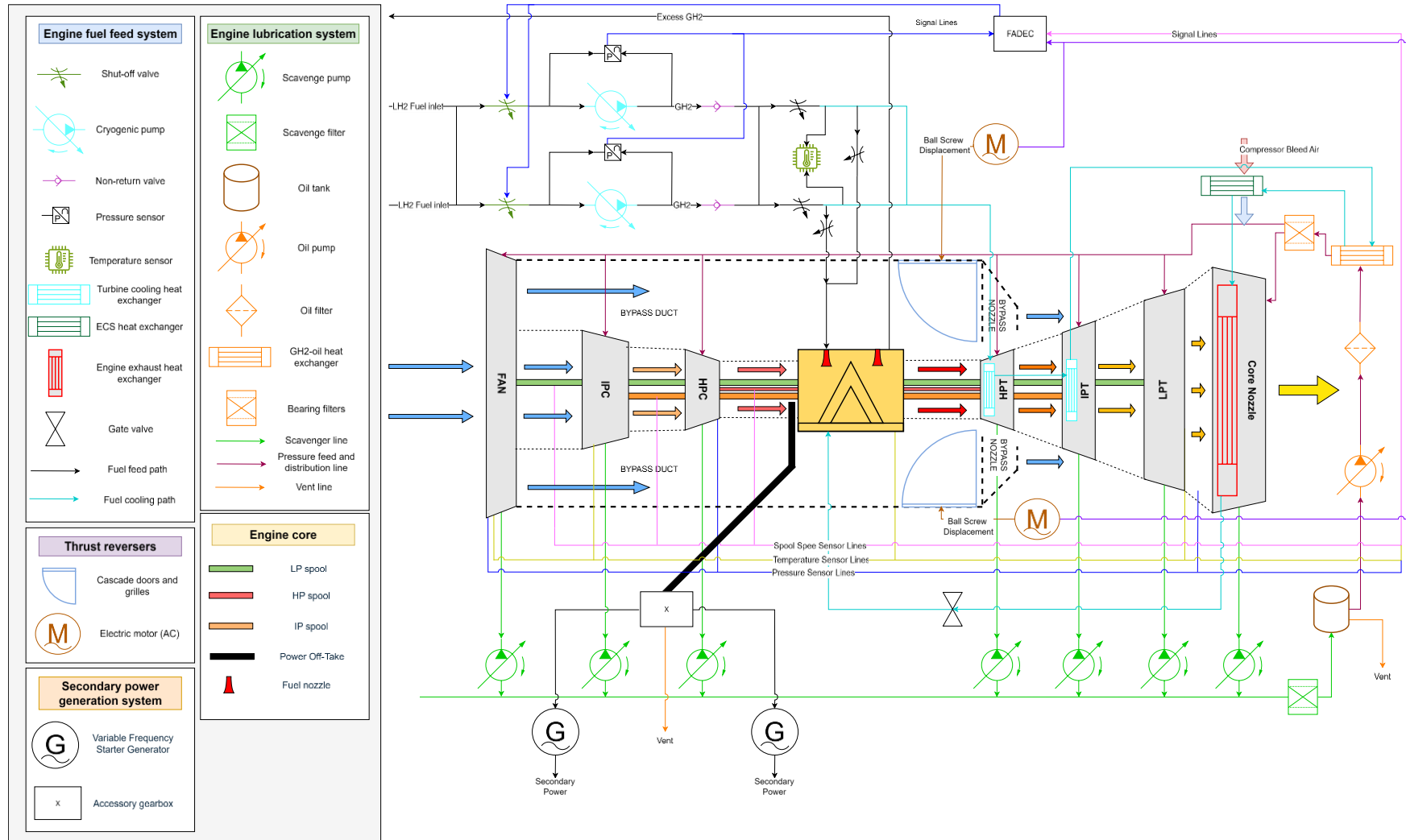
Figure_Apx 3 : Nacelle drag calculator python script

Appendix J CAD Drawings



Figure_Apx 4 : CAD drawings

Appendix K Integrated propulsion architecture diagram



Figure_Apx 5 : Final architecture

